Waymo_AV report

Waymo_AV — Robust Joint Motion Forecasting Under Missing and Noisy Observations

Generated 2026-11-03 18:41 from commit `d3076b4`. Simplified MotionLM-style two-agent joint predictor; local metrics (not official challenge scores).

0. Key results

Development set (500 scenes; checkpoint selection; NOT held-out) (per-method means over seeds; CIs and paired differences in the tables below)

method	clean joint minADE@6 (m)	core-suite joint minADE@6 (m)	clean overlap, own 6 outputs (%)	clean offroad (%)	n seeds
constant velocity	10.82	11.25	16.6	5.4	1
independent decoder	3.90 (seeds: 3.87, 4.02, 3.80)	7.87 (seeds: 8.10, 8.06, 7.46)	18.8 (seeds: 17.7, 18.4, 20.3)	5.8 (seeds: 6.0, 5.5, 5.9)	3
joint M0 (clean-trained)	3.74 (seeds: 3.78, 3.67, 3.77)	7.52 (seeds: 7.81, 7.31, 7.44)	15.0 (seeds: 14.6, 15.2, 15.4)	5.8 (seeds: 5.0, 5.8, 6.7)	3
joint M1 (corruption-trained)	3.51 (seeds: 3.56, 3.48, 3.50)	4.26 (seeds: 4.33, 4.23, 4.22)	14.9 (seeds: 15.8, 14.2, 14.7)	4.7 (seeds: 4.9, 4.7, 4.7)	3
M1 + best-6-of-32 reward selection	3.31 (seeds: 3.33, 3.27, 3.33)	4.06 (seeds: 4.09, 4.02, 4.08)	1.2 (seeds: 1.1, 0.6, 1.8)	1.6 (seeds: 1.6, 1.6, 1.6)	3
M1 + extra SFT	3.47 (seeds: 3.45, 3.47, 3.49)	4.25 (seeds: 4.30, 4.21, 4.25)	14.6 (seeds: 15.8, 13.6, 14.4)	4.8 (seeds: 5.0, 4.4, 4.9)	3

method	clean joint minADE@6 (m)	core-suite joint minADE@6 (m)	clean overlap, own 6 outputs (%)	clean offroad (%)	n seeds
M1 + reward-filtered SFT	3.55 (seeds: 3.61, 3.56, 3.49)	4.36 (seeds: 4.38, 4.39, 4.30)	14.4 (seeds: 15.2, 14.3, 13.7)	4.8 (seeds: 5.0, 4.6, 4.8)	3
M1 + bounded RL	4.35 (seeds: 4.50, 4.41, 4.14)	5.64 (seeds: 5.91, 5.75, 5.26)	5.3 (seeds: 5.3, 4.3, 6.2)	3.5 (seeds: 3.7, 3.3, 3.5)	3

- M1 vs M0 core-suite joint minADE@6 (seed of reference): -44.6 % [-47.3, -41.9] (group-level bootstrap). Per-seed verdicts against the preregistered thresholds: see the comparison section.

FINAL held-out set (1,000 locked scenes; one pass per frozen method) (per-method means over seeds; CIs and paired differences in the tables below)

method	clean joint minADE@6 (m)	core-suite joint minADE@6 (m)	clean overlap, own 6 outputs (%)	clean offroad (%)	n seeds
constant velocity	11.11	11.51	17.3	5.7	1
independent decoder	3.65 (seeds: 3.69, 3.58, 3.69)	7.62 (seeds: 7.87, 7.79, 7.22)	19.1 (seeds: 18.5, 18.3, 20.4)	5.7 (seeds: 5.9, 5.4, 5.9)	3
joint M0 (clean-trained)	3.49 (seeds: 3.52, 3.42, 3.54)	7.30 (seeds: 7.55, 7.18, 7.18)	16.0 (seeds: 15.0, 16.4, 16.5)	5.5 (seeds: 5.1, 5.8, 5.8)	3
joint M1 (corruption-trained)	3.20 (seeds: 3.20, 3.14, 3.25)	3.92 (seeds: 3.95, 3.89, 3.91)	15.8 (seeds: 14.7, 16.9, 15.9)	4.5 (seeds: 4.5, 4.5, 4.6)	3
M1 + best-6-of-32 reward selection	3.03 (seeds: 3.04, 2.99, 3.05)	3.71 (seeds: 3.72, 3.69, 3.70)	1.2 (seeds: 0.5, 1.7, 1.5)	1.3 (seeds: 1.3, 1.3, 1.4)	3
M1 + extra SFT	3.12 (seeds: 3.12, 3.08, 3.16)	3.90 (seeds: 3.94, 3.88, 3.87)	15.6 (seeds: 15.4, 15.8, 15.6)	4.7 (seeds: 4.7, 4.5, 5.0)	3
M1 + reward-filtered SFT	3.19 (seeds: 3.18, 3.18, 3.22)	3.98 (seeds: 3.97, 4.00, 3.99)	14.7 (seeds: 14.0, 14.7, 15.5)	4.4 (seeds: 4.2, 4.5, 4.7)	3
M1 + bounded RL	3.97 (seeds: 4.04, 4.06, 3.80)	5.27 (seeds: 5.46, 5.44, 4.92)	4.8 (seeds: 3.9, 4.5, 6.1)	3.1 (seeds: 3.2, 3.0, 3.2)	3

- M1 vs M0 core-suite joint minADE@6 (seed of reference): -47.7 % [-49.5, -45.9] (group-level bootstrap). Per-seed verdicts against the preregistered thresholds: see the comparison section.

RL outcome (D-031): recorded as mixed/negative under the frozen comparison rule — see section 3a (Q5/Q6) and DECISIONS.md.

1. Setup

- Dataset: { "dataset": "Waymo Open Motion Dataset", "version": "1.3.0", "raw_subdir": "womd_v1_3_0", "format": "scenario tfrecord", "bucket": "gs://waymo_open_dataset_motion_v_1_3_0", "listed": "2026-11-03", "listing_sha256": "d34a243222ae7bc2443da13cc8691847e2b34455f7dfb82a81798c091be3cfa1" }
- Calibration: { "n_scenes": 500, "tokenizer": { "saturation_rate": 0.0008159166509759618, "recon_err_mean_m": 0.09240365773439407, "recon_err_p99_m": 0.16335155069828033, "recon_err_max_m": 3.338541030883789, "vocab_used": 95, "top10_tokens": [[84, 3102], [71, 1884], [97, 1823], [85, 928], [83, 854], [58, 799], [110, 636], [70, 628], [96, 592], [72, 589]], "zero_token_frac": 0.19469026548672566 }, "d0_speed_mps": { "mean": 6.568544387817383, "p50": 5.209609031677246, "p99": 22.173171997070312, "max": 32.964210510253906 }, "history_valid_frac_targets": 0.9890802556818181, "future_valid_frac_targets": 0.9957275390625, "initial_overlap_rate_gt": 0.001953125, "future_overlap_rate_gt": 0.032421875, "transform_roundtrip_max_err_m": 5.684341886080802e-14, "d0_init": { "mean_err_m_vs_true_first_half_second": 0.24413703676310622, "p99_err_m": 0.84452919952413, "zero_velocity_err_m": 3.2363052368164062 }, "cv_baseline_clean": { "joint_minADE_8s": 10.822137399683946, "joint_minFDE_8s": 27.731932333955744 }, "target_types": { "1": 895, "2": 73, "3": 32 }, "offroad_oracle": { "sign_+1": 0.009799354392039928, "sign_-1": 0.99020064560796, "chosen_sign": 1, "oracle_offroad_frac_with_chosen_sign": 0.009799354392039928, "metric_valid": true } }
- Decisions: see DECISIONS.md; metric definitions: docs/metrics.md; preregistered thresholds: PLAN.md.

2. Runs

run	kind	status	step	params	best dev score
extra_sft_pilot_s0	train	completed	600	1787049	3.94908966326966
extra_sft_pilot_s1	train	completed	600	1787049	3.93531043102798
extra_sft_pilot_s2	train	completed	600	1787049	3.90064741090491

run	kind	status	step	params	best dev score
filtered_sft_pilot_s0	filtered_sft	completed	600		
filtered_sft_pilot_s1	filtered_sft	completed	600		
filtered_sft_pilot_s2	filtered_sft	completed	600		
m0_overfit	train	completed	600	1787049	0.09314770238688
m0_pilot_s0	train	completed	12000	1787049	3.82606548388915
m0_pilot_s1	train	completed	12000	1787049	3.74378330702582
m0_pilot_s2	train	completed	12000	1787049	3.59648358112898
m1_pilot_s0	train	completed	12000	1787049	3.98811603750914
m1_pilot_s1	train	completed	12000	1787049	3.84353757475528
m1_pilot_s2	train	completed	12000	1787049	3.88669393636379
mind_pilot_s0	train	completed	12000	1787049	3.84350423648226
mind_pilot_s1	train	completed	12000	1787049	4.05884199395848
mind_pilot_s2	train	completed	12000	1787049	3.83129117551994
rl_pilot_s0	rl	completed	600		
rl_pilot_s1	rl	completed	600		
rl_pilot_s2	rl	completed	600		

GPU-hours used (all runs): 2.27

2b. Metric qualification (before any post-training)

- Logged-future overlap audit v2 (D-028a; 500 dev scenes): reference rate on jointly valid intervals 1.0 % (5 scenes; classes {‘heading_inference’: 2, ‘annotation_contact’: 3}); the old unmasked full-horizon test flagged 15 scenes, 10 of them only on zero-filled invalid label steps. Superseded v1: manifests/overlap_floor_audit_v1_superseded.json.
- Generated-trajectory overlap inspection (D-028b; m0_pilot_s0, clean): 449/2994 (scene, sample) pairs flagged (15.0 %); verdicts {‘approximation_sensitive’: 56, ‘confirmed_geometric’: 324, ‘ambiguous’: 69}; by stratum {‘type:1-1’: {‘approximation_sensitive’: 40, ‘confirmed_geometric’: 252, ‘ambiguous’: 50}, ‘turning’: {‘approximation_sensitive’: 36, ‘ambiguous’: 26, ‘confirmed_geometric’: 134}, ‘straight_moving’: {‘confirmed_geometric’: 190, ‘ambiguous’: 32, ‘approximation_sensitive’: 20}, ‘stationary’: {‘ambiguous’: 13, ‘approximation_sensitive’: 2}, ‘type:1-2’: {‘approximation_sensitive’: 12, ‘ambiguous’: 17, ‘confirmed_geometric’: 48}, ‘type:1-3’: {‘approximation_sensitive’: 4, ‘confirmed_geometric’: 24, ‘ambiguous’: 2}}. The RL overlap reward is an approximate geometric-consistency objective, not a verified safety measure.
- Penetration depth does not separate logged-reference flags from model flags (p50 0.75 m vs 0.98 m); no depth margin is used.
- Offroad: sign convention +1 calibrated by log playback (manifests/metric_config.json); KD-tree nearest-segment implementation verified against brute force (tests).

3. Development-set results (checkpoint selection, sampling freeze)

Selection rules: M0/Mind checkpoints = lowest clean joint minADE@6 on the first 256 dev scenes; M1 = lowest mean of clean and core-suite joint minADE@6 on the same scenes (D-017). These selection scores are NOT comparable across models; every table below is a matched evaluation of the selected checkpoints on identical scenes, conditions and sampling.

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s	h
cp	clean	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [(0
cv	clean	500	10.822 [10.236, 11.404]	27.732 [26.190, 29.256]	10.817 [10.236, 11.402]	0 [(0
extra_sft_pilot_s0	clean	500	3.454 [3.200, 3.706]	8.442 [7.750, 9.133]	2.968 [2.730, 3.198]	0 [(0
extra_sft_pilot_s1	clean	500	3.470 [3.241, 3.691]	8.401 [7.767, 9.049]	2.956 [2.747, 3.162]	0 [(0
extra_sft_pilot_s2	clean	500	3.493 [3.238, 3.732]	8.375 [7.672, 9.053]	2.997 [2.763, 3.228]	0 [(0
filtered_sft_pilot_s0	clean	500	3.609 [3.334, 3.877]	8.667 [7.953, 9.391]	3.064 [2.809, 3.312]	0 [(0
filtered_sft_pilot_s1	clean	500	3.559 [3.321, 3.803]	8.677 [8.000, 9.368]	3.081 [2.856, 3.305]	0 [(0
filtered_sft_pilot_s2	clean	500	3.488 [3.234, 3.728]	8.411 [7.690, 9.084]	2.984 [2.757, 3.208]	0 [(0
m0_pilot_s0	clean	500	3.776 [3.522, 4.037]	9.195 [8.468, 9.916]	3.185 [2.947, 3.413]	0 [(0
m0_pilot_s1	clean	500	3.666 [3.417, 3.915]	8.672 [8.003, 9.338]	3.048 [2.816, 3.271]	0 [(0
m0_pilot_s2	clean	500	3.765 [3.491, 4.027]	9.125 [8.428, 9.842]	3.157 [2.910, 3.397]	0 [(0
m1_pilot_s0	clean	500	3.561 [3.310, 3.811]	8.418 [7.739, 9.089]	3.034 [2.806, 3.259]	0 [(0

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s	h
m1_pilot_s1	clean	500	3.481 [3.234, 3.724]	8.424 [7.734, 9.092]	2.993 [2.756, 3.223]	0 [0
m1_pilot_s2	clean	500	3.495 [3.246, 3.728]	8.360 [7.680, 9.019]	2.959 [2.733, 3.185]	0 [0
mind_pilot_s0	clean	500	3.871 [3.598, 4.137]	9.522 [8.764, 10.272]	3.270 [3.020, 3.528]	0 [0
mind_pilot_s1	clean	500	4.020 [3.755, 4.277]	9.724 [9.001, 10.440]	3.309 [3.074, 3.541]	0 [0
mind_pilot_s2	clean	500	3.796 [3.544, 4.053]	9.109 [8.437, 9.796]	3.147 [2.914, 3.380]	0 [0
oracle_log	clean	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	clean	500	4.504 [4.225, 4.755]	11.114 [10.349, 11.874]	3.984 [3.724, 4.223]	0 [0
rl_pilot_s1	clean	500	4.409 [4.147, 4.663]	10.856 [10.135, 11.545]	3.906 [3.646, 4.148]	0 [0
rl_pilot_s2	clean	500	4.139 [3.864, 4.404]	10.119 [9.368, 10.894]	3.603 [3.340, 3.855]	0 [0
select_m1_s0	clean	500	3.331 [3.081, 3.586]	7.818 [7.131, 8.509]	2.859 [2.622, 3.089]	0 [0
select_m1_s1	clean	500	3.273 [3.040, 3.509]	7.899 [7.269, 8.549]	2.823 [2.600, 3.047]	0 [0
select_m1_s2	clean	500	3.334 [3.102, 3.564]	8.073 [7.412, 8.728]	2.863 [2.640, 3.076]	0 [0
cp	compound_0	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	compound_0	500	11.868 [11.308, 12.458]	29.125 [27.641, 30.505]	11.861 [11.302, 12.454]	0 [0
extra_sft_pilot_s0	compound_0	500	5.347 [5.079, 5.613]	11.493 [10.758, 12.229]	4.602 [4.350, 4.859]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
extra_sft_pilot_s1	compound_0	500	5.193 [4.924, 5.469]	11.230 [10.493, 11.992]	4.440 [4.182, 4.708]	0 [0
extra_sft_pilot_s2	compound_0	500	5.321 [5.041, 5.600]	11.375 [10.655, 12.120]	4.487 [4.218, 4.758]	0 [0
filtered_sft_pilot_s0	compound_0	500	5.416 [5.131, 5.705]	11.363 [10.652, 12.112]	4.657 [4.391, 4.932]	0 [0
filtered_sft_pilot_s1	compound_0	500	5.623 [5.319, 5.928]	12.107 [11.329, 12.901]	4.842 [4.550, 5.131]	0 [0
filtered_sft_pilot_s2	compound_0	500	5.439 [5.141, 5.740]	11.959 [11.182, 12.755]	4.589 [4.321, 4.865]	0 [0
m0_pilot_s0	compound_0	500	10.052 [9.579, 10.533]	21.337 [20.169, 22.545]	8.657 [8.219, 9.097]	0 [0
m0_pilot_s1	compound_0	500	10.404 [9.896, 10.925]	22.592 [21.395, 23.819]	9.053 [8.583, 9.548]	0 [0
m0_pilot_s2	compound_0	500	10.272 [9.791, 10.745]	22.026 [20.900, 23.332]	9.086 [8.626, 9.541]	0 [0
m1_pilot_s0	compound_0	500	5.334 [5.067, 5.619]	11.194 [10.445, 11.961]	4.546 [4.301, 4.800]	0 [0
m1_pilot_s1	compound_0	500	5.264 [4.998, 5.542]	11.307 [10.592, 12.053]	4.495 [4.231, 4.762]	0 [0
m1_pilot_s2	compound_0	500	5.265 [4.987, 5.559]	11.351 [10.625, 12.091]	4.443 [4.190, 4.708]	0 [0
mind_pilot_s0	compound_0	500	10.355 [9.888, 10.814]	21.527 [20.460, 22.659]	9.177 [8.737, 9.599]	0 [0
mind_pilot_s1	compound_0	500	11.323 [10.793, 11.823]	24.499 [23.199, 25.774]	9.707 [9.221, 10.170]	0 [0
mind_pilot_s2	compound_0	500	9.809 [9.316, 10.296]	20.813 [19.687, 21.951]	8.628 [8.164, 9.079]	0 [0
oracle_log	compound_0	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	c 1 h
rl_pilot_s0	compound_0	500	7.717 [7.360, 8.088]	16.986 [16.048, 17.941]	6.986 [6.640, 7.355]	0 [0
rl_pilot_s1	compound_0	500	7.494 [7.123, 7.865]	16.806 [15.807, 17.810]	6.743 [6.380, 7.114]	0 [0
rl_pilot_s2	compound_0	500	6.767 [6.397, 7.126]	15.266 [14.333, 16.170]	5.965 [5.604, 6.308]	0 [0
select_m1_s0	compound_0	500	5.002 [4.738, 5.285]	10.655 [9.948, 11.382]	4.350 [4.103, 4.613]	0 [0
select_m1_s1	compound_0	500	5.114 [4.809, 5.419]	10.956 [10.172, 11.744]	4.412 [4.122, 4.710]	0 [0
select_m1_s2	compound_0	500	5.107 [4.817, 5.404]	11.034 [10.282, 11.871]	4.425 [4.137, 4.712]	0 [0
cp	missing_0.1	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	missing_0.1	500	10.814 [10.238, 11.395]	27.719 [26.177, 29.231]	10.808 [10.219, 11.387]	0 [0
extra_sft_pilot_s0	missing_0.1	500	3.440 [3.198, 3.691]	8.281 [7.576, 8.979]	2.869 [2.652, 3.090]	0 [0
extra_sft_pilot_s1	missing_0.1	500	3.511 [3.275, 3.754]	8.345 [7.706, 9.014]	2.931 [2.713, 3.141]	0 [0
extra_sft_pilot_s2	missing_0.1	500	3.612 [3.356, 3.861]	8.647 [7.916, 9.363]	3.068 [2.822, 3.287]	0 [0
filtered_sft_pilot_s0	missing_0.1	500	3.524 [3.262, 3.769]	8.429 [7.709, 9.093]	3.018 [2.780, 3.250]	0 [0
filtered_sft_pilot_s1	missing_0.1	500	3.590 [3.332, 3.833]	8.640 [7.959, 9.325]	3.091 [2.849, 3.332]	0 [0
filtered_sft_pilot_s2	missing_0.1	500	3.545 [3.297, 3.795]	8.582 [7.872, 9.284]	3.056 [2.813, 3.290]	0 [0
m0_pilot_s0	missing_0.1	500	3.941 [3.691, 4.195]	9.485 [8.754, 10.208]	3.348 [3.109, 3.581]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
m0_pilot_s1	missing_0.1	500	3.785 [3.536, 4.043]	9.044 [8.365, 9.746]	3.209 [2.981, 3.447]	0 [0
m0_pilot_s2	missing_0.1	500	3.949 [3.677, 4.211]	9.587 [8.829, 10.349]	3.393 [3.141, 3.639]	0 [0
m1_pilot_s0	missing_0.1	500	3.462 [3.213, 3.719]	8.382 [7.674, 9.084]	2.919 [2.687, 3.159]	0 [0
m1_pilot_s1	missing_0.1	500	3.533 [3.271, 3.784]	8.575 [7.902, 9.282]	2.966 [2.725, 3.201]	0 [0
m1_pilot_s2	missing_0.1	500	3.463 [3.197, 3.698]	8.386 [7.677, 9.062]	2.983 [2.737, 3.199]	0 [0
mind_pilot_s0	missing_0.1	500	4.039 [3.746, 4.314]	9.863 [9.058, 10.636]	3.395 [3.124, 3.653]	0 [0
mind_pilot_s1	missing_0.1	500	4.076 [3.808, 4.335]	9.719 [9.004, 10.430]	3.370 [3.140, 3.589]	0 [0
mind_pilot_s2	missing_0.1	500	4.173 [3.896, 4.467]	9.928 [9.160, 10.704]	3.505 [3.245, 3.770]	0 [0
oracle_log	missing_0.1	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	missing_0.1	500	4.485 [4.205, 4.760]	11.078 [10.296, 11.857]	3.994 [3.724, 4.250]	0 [0
rl_pilot_s1	missing_0.1	500	4.428 [4.163, 4.685]	10.935 [10.181, 11.660]	3.861 [3.603, 4.106]	0 [0
rl_pilot_s2	missing_0.1	500	4.197 [3.916, 4.464]	10.380 [9.593, 11.080]	3.652 [3.386, 3.900]	0 [0
select_m1_s0	missing_0.1	500	3.384 [3.134, 3.637]	8.064 [7.343, 8.802]	2.913 [2.681, 3.160]	0 [0
select_m1_s1	missing_0.1	500	3.290 [3.041, 3.531]	7.852 [7.243, 8.498]	2.843 [2.620, 3.068]	0 [0
select_m1_s2	missing_0.1	500	3.397 [3.136, 3.630]	8.081 [7.381, 8.767]	2.914 [2.669, 3.137]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	c 1 h
cp	missing_0.3	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [(0
cv	missing_0.3	500	10.768 [10.193, 11.348]	27.647 [26.120, 29.150]	10.763 [10.177, 11.347]	0 [(0
extra_sft_pilot_s0	missing_0.3	500	3.544 [3.301, 3.788]	8.470 [7.786, 9.129]	3.022 [2.784, 3.248]	0 [(0
extra_sft_pilot_s1	missing_0.3	500	3.512 [3.270, 3.756]	8.423 [7.794, 9.088]	2.991 [2.766, 3.220]	0 [(0
extra_sft_pilot_s2	missing_0.3	500	3.475 [3.236, 3.713]	8.386 [7.714, 9.043]	2.971 [2.746, 3.196]	0 [(0
filtered_sft_pilot_s0	missing_0.3	500	3.528 [3.281, 3.776]	8.394 [7.710, 9.076]	3.014 [2.778, 3.240]	0 [(0
filtered_sft_pilot_s1	missing_0.3	500	3.584 [3.339, 3.825]	8.520 [7.853, 9.163]	3.028 [2.803, 3.248]	0 [(0
filtered_sft_pilot_s2	missing_0.3	500	3.595 [3.343, 3.838]	8.715 [8.030, 9.436]	3.052 [2.821, 3.274]	0 [(0
m0_pilot_s0	missing_0.3	500	4.309 [4.041, 4.591]	10.347 [9.582, 11.123]	3.628 [3.377, 3.882]	0 [(0
m0_pilot_s1	missing_0.3	500	4.348 [4.042, 4.640]	10.162 [9.364, 10.958]	3.717 [3.440, 3.991]	0 [(0
m0_pilot_s2	missing_0.3	500	4.110 [3.829, 4.389]	9.837 [9.117, 10.592]	3.496 [3.238, 3.758]	0 [(0
m1_pilot_s0	missing_0.3	500	3.594 [3.330, 3.857]	8.441 [7.763, 9.125]	3.047 [2.798, 3.285]	0 [(0
m1_pilot_s1	missing_0.3	500	3.502 [3.255, 3.745]	8.491 [7.840, 9.158]	2.999 [2.766, 3.234]	0 [(0
m1_pilot_s2	missing_0.3	500	3.523 [3.278, 3.759]	8.365 [7.687, 9.017]	2.976 [2.747, 3.200]	0 [(0
mind_pilot_s0	missing_0.3	500	4.434 [4.119, 4.735]	10.728 [9.914, 11.532]	3.805 [3.517, 4.078]	0 [(0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
mind_pilot_s1	missing_0.3	500	4.788 [4.488, 5.092]	11.472 [10.617, 12.287]	3.976 [3.710, 4.253]	0 [0
mind_pilot_s2	missing_0.3	500	4.532 [4.247, 4.833]	10.679 [9.913, 11.453]	3.749 [3.486, 4.019]	0 [0
oracle_log	missing_0.3	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	missing_0.3	500	4.491 [4.215, 4.749]	11.010 [10.233, 11.778]	3.999 [3.722, 4.245]	0 [0
rl_pilot_s1	missing_0.3	500	4.419 [4.143, 4.690]	10.884 [10.155, 11.612]	3.843 [3.586, 4.093]	0 [0
rl_pilot_s2	missing_0.3	500	4.223 [3.943, 4.496]	10.327 [9.540, 11.119]	3.662 [3.393, 3.909]	0 [0
select_m1_s0	missing_0.3	500	3.448 [3.182, 3.709]	8.348 [7.584, 9.120]	2.953 [2.709, 3.194]	0 [0
select_m1_s1	missing_0.3	500	3.426 [3.175, 3.664]	8.159 [7.527, 8.821]	2.974 [2.732, 3.202]	0 [0
select_m1_s2	missing_0.3	500	3.551 [3.291, 3.803]	8.548 [7.828, 9.285]	3.075 [2.827, 3.315]	0 [0
cp	missing_0.5	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	missing_0.5	500	10.762 [10.179, 11.342]	27.588 [26.084, 29.067]	10.758 [10.173, 11.333]	0 [0
extra_sft_pilot_s0	missing_0.5	500	3.637 [3.383, 3.908]	8.746 [8.045, 9.471]	3.115 [2.877, 3.371]	0 [0
extra_sft_pilot_s1	missing_0.5	500	3.612 [3.368, 3.853]	8.625 [7.984, 9.307]	3.084 [2.846, 3.315]	0 [0
extra_sft_pilot_s2	missing_0.5	500	3.792 [3.526, 4.047]	9.302 [8.565, 10.032]	3.267 [3.025, 3.518]	0 [0
filtered_sft_pilot_s0	missing_0.5	500	3.754 [3.506, 3.996]	8.643 [7.985, 9.312]	3.210 [2.987, 3.433]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s	h
filtered_sft_pilot_s1	missing_0.5	500	3.696 [3.455, 3.958]	8.882 [8.199, 9.592]	3.179 [2.942, 3.421]	0 [0
filtered_sft_pilot_s2	missing_0.5	500	3.784 [3.513, 4.053]	9.080 [8.344, 9.817]	3.246 [2.992, 3.499]	0 [0
m0_pilot_s0	missing_0.5	500	4.776 [4.461, 5.097]	11.535 [10.622, 12.433]	4.090 [3.789, 4.396]	0 [0
m0_pilot_s1	missing_0.5	500	4.937 [4.615, 5.253]	11.375 [10.561, 12.219]	4.175 [3.878, 4.473]	0 [0
m0_pilot_s2	missing_0.5	500	4.630 [4.336, 4.910]	10.898 [10.135, 11.717]	3.926 [3.648, 4.193]	0 [0
m1_pilot_s0	missing_0.5	500	3.658 [3.395, 3.924]	8.805 [8.080, 9.518]	3.132 [2.887, 3.376]	0 [0
m1_pilot_s1	missing_0.5	500	3.612 [3.363, 3.865]	8.566 [7.904, 9.232]	3.044 [2.822, 3.275]	0 [0
m1_pilot_s2	missing_0.5	500	3.762 [3.486, 4.017]	9.002 [8.221, 9.752]	3.219 [2.954, 3.468]	0 [0
mind_pilot_s0	missing_0.5	500	5.148 [4.826, 5.461]	12.264 [11.403, 13.098]	4.302 [4.007, 4.586]	0 [0
mind_pilot_s1	missing_0.5	500	5.464 [5.114, 5.815]	13.065 [12.148, 14.035]	4.541 [4.230, 4.854]	0 [0
mind_pilot_s2	missing_0.5	500	4.921 [4.617, 5.234]	11.471 [10.713, 12.326]	4.086 [3.808, 4.376]	0 [0
oracle_log	missing_0.5	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	missing_0.5	500	4.975 [4.671, 5.292]	12.220 [11.365, 13.055]	4.470 [4.176, 4.771]	0 [0
rl_pilot_s1	missing_0.5	500	4.766 [4.475, 5.053]	11.766 [10.964, 12.540]	4.201 [3.922, 4.475]	0 [0
rl_pilot_s2	missing_0.5	500	4.379 [4.087, 4.660]	10.681 [9.926, 11.445]	3.847 [3.575, 4.110]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
select_m1_s0	missing_0.5	500	3.566 [3.293, 3.832]	8.597 [7.853, 9.355]	3.066 [2.809, 3.320]	0 [0
select_m1_s1	missing_0.5	500	3.510 [3.261, 3.756]	8.253 [7.596, 8.949]	3.001 [2.773, 3.232]	0 [0
select_m1_s2	missing_0.5	500	3.570 [3.304, 3.838]	8.535 [7.833, 9.231]	3.082 [2.830, 3.337]	0 [0
cp	missing_0.7	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	missing_0.7	500	11.427 [10.781, 12.093]	28.673 [27.113, 30.278]	11.427 [10.775, 12.090]	0 [0
extra_sft_pilot_s0	missing_0.7	500	4.244 [3.942, 4.537]	9.661 [8.965, 10.341]	3.604 [3.324, 3.885]	0 [0
extra_sft_pilot_s1	missing_0.7	500	4.151 [3.865, 4.455]	9.334 [8.640, 10.056]	3.577 [3.299, 3.866]	0 [0
extra_sft_pilot_s2	missing_0.7	500	4.163 [3.873, 4.477]	9.571 [8.833, 10.330]	3.570 [3.303, 3.857]	0 [0
filtered_sft_pilot_s0	missing_0.7	500	4.500 [4.189, 4.819]	10.070 [9.313, 10.863]	3.808 [3.532, 4.093]	0 [0
filtered_sft_pilot_s1	missing_0.7	500	4.445 [4.122, 4.792]	10.203 [9.418, 11.007]	3.830 [3.526, 4.150]	0 [0
filtered_sft_pilot_s2	missing_0.7	500	4.413 [4.074, 4.777]	10.063 [9.194, 10.875]	3.838 [3.505, 4.181]	0 [0
m0_pilot_s0	missing_0.7	500	5.819 [5.424, 6.233]	13.362 [12.418, 14.315]	4.951 [4.592, 5.330]	0 [0
m0_pilot_s1	missing_0.7	500	6.211 [5.800, 6.658]	14.030 [12.995, 15.126]	5.260 [4.905, 5.657]	0 [0
m0_pilot_s2	missing_0.7	500	5.835 [5.475, 6.207]	13.334 [12.454, 14.311]	4.956 [4.626, 5.303]	0 [0
m1_pilot_s0	missing_0.7	500	4.208 [3.911, 4.508]	9.571 [8.898, 10.268]	3.568 [3.292, 3.856]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
m1_pilot_s1	missing_0.7	500	4.219 [3.907, 4.561]	9.557 [8.790, 10.357]	3.635 [3.347, 3.946]	0 [0
m1_pilot_s2	missing_0.7	500	4.194 [3.912, 4.483]	9.599 [8.896, 10.356]	3.610 [3.341, 3.883]	0 [0
mind_pilot_s0	missing_0.7	500	6.139 [5.746, 6.537]	14.106 [13.208, 15.063]	5.232 [4.877, 5.599]	0 [0
mind_pilot_s1	missing_0.7	500	7.069 [6.590, 7.576]	16.427 [15.306, 17.596]	5.979 [5.549, 6.464]	0 [0
mind_pilot_s2	missing_0.7	500	6.309 [5.916, 6.730]	14.253 [13.265, 15.291]	5.461 [5.087, 5.858]	0 [0
oracle_log	missing_0.7	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	missing_0.7	500	5.527 [5.148, 5.875]	13.140 [12.280, 14.003]	5.022 [4.657, 5.354]	0 [0
rl_pilot_s1	missing_0.7	500	5.653 [5.232, 6.083]	12.951 [12.034, 13.906]	5.075 [4.660, 5.499]	0 [0
rl_pilot_s2	missing_0.7	500	5.051 [4.689, 5.418]	11.716 [10.820, 12.628]	4.444 [4.106, 4.800]	0 [0
select_m1_s0	missing_0.7	500	3.989 [3.708, 4.296]	8.946 [8.246, 9.657]	3.456 [3.176, 3.745]	0 [0
select_m1_s1	missing_0.7	500	4.132 [3.825, 4.456]	9.385 [8.634, 10.169]	3.571 [3.291, 3.876]	0 [0
select_m1_s2	missing_0.7	500	3.941 [3.654, 4.242]	9.047 [8.302, 9.800]	3.409 [3.136, 3.688]	0 [0
cp	noise_0.1	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	noise_0.1	500	10.941 [10.362, 11.522]	27.861 [26.335, 29.377]	10.935 [10.352, 11.515]	0 [0
extra_sft_pilot_s0	noise_0.1	500	4.118 [3.844, 4.396]	9.431 [8.671, 10.129]	3.502 [3.256, 3.758]	0 [0

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s	h
extra_sft_pilot_s1	noise_0.1	500	4.007 [3.760, 4.266]	9.185 [8.494, 9.868]	3.400 [3.167, 3.644]	0 [0
extra_sft_pilot_s2	noise_0.1	500	4.123 [3.851, 4.393]	9.524 [8.782, 10.248]	3.516 [3.258, 3.771]	0 [0
filtered_sft_pilot_s0	noise_0.1	500	4.084 [3.811, 4.345]	9.388 [8.641, 10.065]	3.471 [3.224, 3.717]	0 [0
filtered_sft_pilot_s1	noise_0.1	500	4.115 [3.850, 4.381]	9.584 [8.895, 10.297]	3.519 [3.268, 3.770]	0 [0
filtered_sft_pilot_s2	noise_0.1	500	4.118 [3.836, 4.402]	9.468 [8.734, 10.208]	3.509 [3.243, 3.772]	0 [0
m0_pilot_s0	noise_0.1	500	6.226 [5.849, 6.606]	13.746 [12.764, 14.732]	5.251 [4.926, 5.585]	0 [0
m0_pilot_s1	noise_0.1	500	6.066 [5.726, 6.403]	13.583 [12.724, 14.469]	5.276 [4.957, 5.594]	0 [0
m0_pilot_s2	noise_0.1	500	5.710 [5.387, 6.042]	12.832 [12.013, 13.694]	4.903 [4.599, 5.204]	0 [0
m1_pilot_s0	noise_0.1	500	4.205 [3.936, 4.487]	9.662 [8.926, 10.362]	3.467 [3.222, 3.719]	0 [0
m1_pilot_s1	noise_0.1	500	3.943 [3.689, 4.198]	9.150 [8.501, 9.843]	3.361 [3.119, 3.605]	0 [0
m1_pilot_s2	noise_0.1	500	4.033 [3.766, 4.299]	9.280 [8.595, 10.011]	3.459 [3.214, 3.699]	0 [0
mind_pilot_s0	noise_0.1	500	6.600 [6.200, 7.010]	14.539 [13.573, 15.601]	5.683 [5.310, 6.058]	0 [0
mind_pilot_s1	noise_0.1	500	6.227 [5.840, 6.603]	13.702 [12.758, 14.678]	5.300 [4.951, 5.642]	0 [0
mind_pilot_s2	noise_0.1	500	5.535 [5.186, 5.860]	12.570 [11.678, 13.483]	4.740 [4.423, 5.033]	0 [0
oracle_log	noise_0.1	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s	c 1 h
rl_pilot_s0	noise_0.1	500	5.621 [5.301, 5.932]	13.065 [12.232, 13.895]	4.943 [4.631, 5.247]	0 [(0
rl_pilot_s1	noise_0.1	500	5.602 [5.280, 5.914]	13.277 [12.464, 14.057]	4.989 [4.675, 5.291]	0 [(0
rl_pilot_s2	noise_0.1	500	5.050 [4.754, 5.345]	11.819 [11.038, 12.625]	4.391 [4.107, 4.685]	0 [(0
select_m1_s0	noise_0.1	500	3.851 [3.588, 4.108]	8.795 [8.115, 9.473]	3.297 [3.056, 3.542]	0 [(0
select_m1_s1	noise_0.1	500	3.892 [3.624, 4.157]	8.999 [8.274, 9.741]	3.352 [3.094, 3.601]	0 [(0
select_m1_s2	noise_0.1	500	4.084 [3.771, 4.404]	9.603 [8.776, 10.450]	3.482 [3.193, 3.777]	0 [(0
cp	noise_0.3	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [(0
cv	noise_0.3	500	11.631 [11.098, 12.187]	28.848 [27.384, 30.278]	11.628 [11.094, 12.186]	0 [(0
extra_sft_pilot_s0	noise_0.3	500	4.949 [4.674, 5.230]	10.786 [10.014, 11.545]	4.125 [3.876, 4.385]	0 [(0
extra_sft_pilot_s1	noise_0.3	500	4.751 [4.499, 5.000]	10.432 [9.733, 11.118]	3.966 [3.736, 4.200]	0 [(0
extra_sft_pilot_s2	noise_0.3	500	4.862 [4.578, 5.146]	10.665 [9.896, 11.420]	4.036 [3.776, 4.291]	0 [(0
filtered_sft_pilot_s0	noise_0.3	500	5.105 [4.831, 5.379]	11.073 [10.324, 11.804]	4.322 [4.066, 4.580]	0 [(0
filtered_sft_pilot_s1	noise_0.3	500	5.100 [4.821, 5.377]	11.142 [10.427, 11.914]	4.303 [4.049, 4.561]	0 [(0
filtered_sft_pilot_s2	noise_0.3	500	5.006 [4.732, 5.276]	11.105 [10.325, 11.861]	4.128 [3.871, 4.380]	0 [(0
m0_pilot_s0	noise_0.3	500	12.417 [11.855, 12.942]	26.141 [24.902, 27.420]	10.425 [9.911, 10.894]	0 [(0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
m0_pilot_s1	noise_0.3	500	11.094 [10.567, 11.611]	23.840 [22.570, 25.145]	9.662 [9.173, 10.146]	0 [0
m0_pilot_s2	noise_0.3	500	11.428 [10.915, 11.953]	25.190 [24.029, 26.478]	10.107 [9.611, 10.603]	0 [0
m1_pilot_s0	noise_0.3	500	4.940 [4.678, 5.205]	10.918 [10.182, 11.646]	4.115 [3.863, 4.374]	0 [0
m1_pilot_s1	noise_0.3	500	4.857 [4.604, 5.112]	10.455 [9.789, 11.107]	4.007 [3.769, 4.257]	0 [0
m1_pilot_s2	noise_0.3	500	4.831 [4.560, 5.105]	10.721 [9.997, 11.459]	3.982 [3.745, 4.231]	0 [0
mind_pilot_s0	noise_0.3	500	12.800 [12.262, 13.310]	27.253 [26.041, 28.581]	10.941 [10.440, 11.440]	0 [0
mind_pilot_s1	noise_0.3	500	12.466 [11.906, 13.021]	27.252 [25.852, 28.592]	10.796 [10.274, 11.314]	0 [0
mind_pilot_s2	noise_0.3	500	11.034 [10.530, 11.566]	23.635 [22.321, 24.990]	9.700 [9.205, 10.213]	0 [0
oracle_log	noise_0.3	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	noise_0.3	500	7.223 [6.900, 7.560]	16.086 [15.206, 16.998]	6.425 [6.104, 6.755]	0 [0
rl_pilot_s1	noise_0.3	500	7.005 [6.663, 7.357]	15.631 [14.726, 16.539]	6.275 [5.930, 6.608]	0 [0
rl_pilot_s2	noise_0.3	500	6.299 [5.969, 6.629]	14.259 [13.402, 15.122]	5.590 [5.269, 5.904]	0 [0
select_m1_s0	noise_0.3	500	4.675 [4.397, 4.960]	10.369 [9.614, 11.117]	3.953 [3.701, 4.211]	0 [0
select_m1_s1	noise_0.3	500	4.612 [4.331, 4.890]	10.047 [9.317, 10.759]	3.830 [3.569, 4.093]	0 [0
select_m1_s2	noise_0.3	500	4.558 [4.283, 4.831]	10.074 [9.342, 10.818]	3.831 [3.579, 4.083]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s	c 1 h
cp	noise_0.5	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [(0
cv	noise_0.5	500	12.586 [12.052, 13.128]	30.255 [28.800, 31.669]	12.579 [12.045, 13.122]	0 [(0
extra_sft_pilot_s0	noise_0.5	500	6.123 [5.824, 6.409]	12.973 [12.227, 13.732]	5.078 [4.803, 5.342]	0 [(0
extra_sft_pilot_s1	noise_0.5	500	5.847 [5.571, 6.141]	12.508 [11.822, 13.227]	4.892 [4.621, 5.159]	0 [(0
extra_sft_pilot_s2	noise_0.5	500	5.636 [5.379, 5.900]	11.975 [11.271, 12.641]	4.619 [4.382, 4.865]	0 [(0
filtered_sft_pilot_s0	noise_0.5	500	6.278 [5.972, 6.576]	13.124 [12.364, 13.906]	5.229 [4.932, 5.515]	0 [(0
filtered_sft_pilot_s1	noise_0.5	500	6.251 [5.962, 6.546]	13.210 [12.461, 13.964]	5.204 [4.939, 5.487]	0 [(0
filtered_sft_pilot_s2	noise_0.5	500	5.770 [5.481, 6.074]	12.238 [11.521, 12.953]	4.819 [4.558, 5.095]	0 [(0
m0_pilot_s0	noise_0.5	500	15.216 [14.603, 15.801]	32.422 [30.932, 33.967]	12.905 [12.317, 13.458]	0 [(0
m0_pilot_s1	noise_0.5	500	13.618 [13.051, 14.229]	29.756 [28.343, 31.287]	11.825 [11.295, 12.395]	0 [(0
m0_pilot_s2	noise_0.5	500	14.817 [14.195, 15.438]	32.472 [31.078, 33.959]	12.928 [12.322, 13.541]	0 [(0
m1_pilot_s0	noise_0.5	500	6.117 [5.815, 6.419]	12.745 [11.928, 13.498]	5.074 [4.810, 5.356]	0 [(0
m1_pilot_s1	noise_0.5	500	5.961 [5.672, 6.250]	12.593 [11.846, 13.362]	4.948 [4.680, 5.218]	0 [(0
m1_pilot_s2	noise_0.5	500	5.697 [5.422, 5.965]	11.973 [11.245, 12.730]	4.692 [4.436, 4.943]	0 [(0
mind_pilot_s0	noise_0.5	500	15.551 [14.996, 16.110]	33.091 [31.724, 34.483]	12.881 [12.380, 13.374]	0 [(0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
mind_pilot_s1	noise_0.5	500	15.329 [14.671, 15.989]	33.480 [31.925, 35.055]	13.213 [12.613, 13.817]	0 [0
mind_pilot_s2	noise_0.5	500	14.537 [13.963, 15.165]	31.200 [29.783, 32.688]	12.949 [12.408, 13.520]	0 [0
oracle_log	noise_0.5	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	noise_0.5	500	8.649 [8.262, 9.050]	18.649 [17.711, 19.600]	7.766 [7.387, 8.151]	0 [0
rl_pilot_s1	noise_0.5	500	8.257 [7.874, 8.621]	18.006 [17.072, 18.982]	7.321 [6.955, 7.681]	0 [0
rl_pilot_s2	noise_0.5	500	7.425 [7.081, 7.760]	16.275 [15.393, 17.162]	6.506 [6.165, 6.824]	0 [0
select_m1_s0	noise_0.5	500	5.605 [5.320, 5.914]	12.045 [11.244, 12.806]	4.655 [4.388, 4.943]	0 [0
select_m1_s1	noise_0.5	500	5.417 [5.132, 5.694]	11.605 [10.887, 12.307]	4.495 [4.235, 4.748]	0 [0
select_m1_s2	noise_0.5	500	5.294 [5.012, 5.563]	11.303 [10.579, 12.026]	4.385 [4.134, 4.634]	0 [0
cp	noise_1	500	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.727 [24.994, 28.589]	0 [0
cv	noise_1	500	16.053 [15.473, 16.659]	35.912 [34.431, 37.395]	16.043 [15.459, 16.656]	0 [0
extra_sft_pilot_s0	noise_1	500	9.516 [9.147, 9.893]	19.294 [18.378, 20.178]	8.081 [7.714, 8.431]	0 [0
extra_sft_pilot_s1	noise_1	500	9.350 [8.998, 9.726]	18.761 [17.878, 19.662]	7.790 [7.451, 8.135]	0 [0
extra_sft_pilot_s2	noise_1	500	9.039 [8.698, 9.408]	18.305 [17.455, 19.127]	7.665 [7.352, 8.000]	0 [0
filtered_sft_pilot_s0	noise_1	500	9.563 [9.192, 9.945]	19.408 [18.513, 20.271]	8.100 [7.763, 8.450]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s	h
filtered_sft_pilot_s1	noise_1	500	9.434 [9.045, 9.844]	18.728 [17.851, 19.631]	7.990 [7.627, 8.365]	0 [0
filtered_sft_pilot_s2	noise_1	500	9.088 [8.723, 9.461]	18.417 [17.567, 19.266]	7.786 [7.452, 8.135]	0 [0
m0_pilot_s0	noise_1	500	17.987 [17.401, 18.594]	39.057 [37.527, 40.571]	15.042 [14.503, 15.614]	0 [0
m0_pilot_s1	noise_1	500	16.976 [16.287, 17.658]	37.214 [35.472, 38.902]	15.249 [14.591, 15.887]	0 [0
m0_pilot_s2	noise_1	500	20.223 [19.501, 20.935]	44.563 [42.744, 46.383]	17.208 [16.545, 17.857]	0 [0
m1_pilot_s0	noise_1	500	9.510 [9.157, 9.871]	19.267 [18.371, 20.093]	8.003 [7.657, 8.349]	0 [0
m1_pilot_s1	noise_1	500	9.284 [8.933, 9.662]	18.639 [17.698, 19.521]	7.641 [7.317, 7.984]	0 [0
m1_pilot_s2	noise_1	500	9.252 [8.908, 9.645]	18.723 [17.863, 19.631]	7.785 [7.446, 8.140]	0 [0
mind_pilot_s0	noise_1	500	19.390 [18.775, 20.027]	41.488 [39.837, 43.151]	15.785 [15.203, 16.406]	0 [0
mind_pilot_s1	noise_1	500	18.175 [17.523, 18.869]	39.656 [37.971, 41.336]	15.686 [15.066, 16.328]	0 [0
mind_pilot_s2	noise_1	500	18.827 [18.161, 19.481]	40.571 [38.942, 42.322]	16.423 [15.772, 17.036]	0 [0
oracle_log	noise_1	500	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	n [n
rl_pilot_s0	noise_1	500	11.837 [11.368, 12.310]	24.176 [23.124, 25.247]	10.578 [10.129, 11.021]	0 [0
rl_pilot_s1	noise_1	500	11.122 [10.693, 11.576]	22.769 [21.726, 23.834]	9.781 [9.367, 10.212]	0 [0
rl_pilot_s2	noise_1	500	10.860 [10.422, 11.294]	22.758 [21.757, 23.804]	9.632 [9.195, 10.046]	0 [0

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s	h
select_m1_s0	noise_1	500	8.572 [8.220, 8.941]	17.445 [16.558, 18.315]	7.212 [6.887, 7.544]	0 [0
select_m1_s1	noise_1	500	8.447 [8.077, 8.835]	17.133 [16.252, 18.038]	7.054 [6.695, 7.427]	0 [0
select_m1_s2	noise_1	500	8.332 [7.983, 8.716]	16.778 [15.949, 17.636]	7.109 [6.785, 7.454]	0 [0

Paired differences vs `m0_pilot_s0` (method − reference; group-level bootstrap 95% CI; groups = heuristic source-segment components, D-025):

method	condition	Δ joint minADE@68s	rel. Δ	Δ overlap rate (pp)	n
cp	clean	+22.966 [+21.252, +24.762]	+608.2% [+562.9%, +655.8%]	-14.56 [-16.43, -12.74]	500
cp	compound_0	+16.690 [+14.942, +18.595]	+166.0% [+148.6%, +185.0%]	-16.37 [-18.69, -14.03]	500
cp	missing_0.1	+22.801 [+21.119, +24.616]	+578.6% [+535.9%, +624.7%]	-15.03 [-17.00, -13.16]	500
cp	missing_0.3	+22.433 [+20.755, +24.223]	+520.6% [+481.6%, +562.1%]	-18.14 [-20.47, -16.13]	500
cp	missing_0.5	+21.966 [+20.294, +23.759]	+459.9% [+424.9%, +497.5%]	-17.57 [-19.85, -15.46]	500
cp	missing_0.7	+20.922 [+19.265, +22.724]	+359.5% [+331.0%, +390.5%]	-19.77 [-22.20, -17.61]	500
cp	noise_0.1	+20.515 [+18.807, +22.335]	+329.5% [+302.0%, +358.7%]	-16.77 [-18.90, -14.50]	500
cp	noise_0.3	+14.324 [+12.581, +16.308]	+115.4% [+101.3%, +131.3%]	-13.13 [-14.99, -11.36]	500
cp	noise_0.5	+11.526 [+9.697, +13.513]	+75.8% [+63.7%, +88.8%]	-12.32 [-14.12, -10.63]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
cp	noise_1	+8.755 [+6.867, +10.780]	+48.7% [+38.2%, +59.9%]	-11.96 [-13.84, -10.27]	500
cv	clean	+7.046 [+6.443, +7.673]	+186.6% [+170.6%, +203.2%]	+2.07 [-1.30, +5.41]	500
cv	compound_0	+1.815 [+1.092, +2.558]	+18.1% [+10.9%, +25.4%]	+1.47 [-1.70, +4.75]	500
cv	missing_0.1	+6.873 [+6.285, +7.475]	+174.4% [+159.5%, +189.7%]	+1.60 [-1.77, +5.06]	500
cv	missing_0.3	+6.459 [+5.858, +7.072]	+149.9% [+135.9%, +164.1%]	-1.90 [-5.25, +1.50]	500
cv	missing_0.5	+5.986 [+5.375, +6.584]	+125.3% [+112.5%, +137.9%]	-0.73 [-3.98, +2.43]	500
cv	missing_0.7	+5.607 [+4.986, +6.213]	+96.4% [+85.7%, +106.8%]	-3.54 [-6.83, -0.30]	500
cv	noise_0.1	+4.714 [+4.064, +5.388]	+75.7% [+65.3%, +86.5%]	+0.07 [-3.39, +3.65]	500
cv	noise_0.3	-0.786 [-1.519, +0.053]	-6.3% [-12.2%, +0.4%]	+4.51 [+1.03, +7.91]	500
cv	noise_0.5	-2.629 [-3.480, -1.774]	-17.3% [-22.9%, -11.7%]	+3.51 [+0.53, +6.81]	500
cv	noise_1	-1.934 [-2.724, -1.139]	-10.8% [-15.1%, -6.3%]	+0.47 [-2.32, +3.03]	500
extra_sft_pilot_s0	clean	-0.322 [-0.506, -0.125]	-8.5% [-13.4%, -3.3%]	+1.27 [-0.74, +3.43]	500
extra_sft_pilot_s0	compound_0	-4.705 [-5.176, -4.242]	-46.8% [-51.5%, -42.2%]	+0.60 [-1.91, +3.23]	500
extra_sft_pilot_s0	missing_0.1	-0.501 [-0.700, -0.307]	-12.7% [-17.8%, -7.8%]	-0.53 [-2.47, +1.47]	500
extra_sft_pilot_s0	missing_0.3	-0.765 [-0.988, -0.558]	-17.8% [-22.9%, -12.9%]	-1.17 [-3.41, +0.93]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
extra_sft_pilot_s0	missing_0.5	-1.139 [-1.408, -0.876]	-23.8% [-29.5%, -18.3%]	-1.27 [-3.51, +0.90]	500
extra_sft_pilot_s0	missing_0.7	-1.576 [-1.910, -1.288]	-27.1% [-32.8%, -22.1%]	-0.23 [-2.74, +2.27]	500
extra_sft_pilot_s0	noise_0.1	-2.108 [-2.438, -1.785]	-33.9% [-39.2%, -28.7%]	+1.50 [-0.97, +4.08]	500
extra_sft_pilot_s0	noise_0.3	-7.469 [-8.019, -6.897]	-60.1% [-64.6%, -55.5%]	+4.71 [+2.34, +7.03]	500
extra_sft_pilot_s0	noise_0.5	-9.092 [-9.710, -8.439]	-59.8% [-63.8%, -55.5%]	+3.61 [+1.30, +6.08]	500
extra_sft_pilot_s0	noise_1	-8.471 [-9.101, -7.884]	-47.1% [-50.6%, -43.8%]	+2.57 [+0.60, +4.52]	500
extra_sft_pilot_s1	clean	-0.306 [-0.519, -0.102]	-8.1% [-13.8%, -2.7%]	-1.00 [-3.04, +0.94]	500
extra_sft_pilot_s1	compound_0	-4.860 [-5.360, -4.375]	-48.3% [-53.3%, -43.5%]	+0.43 [-2.23, +3.17]	500
extra_sft_pilot_s1	missing_0.1	-0.429 [-0.645, -0.216]	-10.9% [-16.4%, -5.5%]	-0.67 [-2.83, +1.53]	500
extra_sft_pilot_s1	missing_0.3	-0.797 [-1.046, -0.567]	-18.5% [-24.3%, -13.1%]	-3.81 [-6.31, -1.37]	500
extra_sft_pilot_s1	missing_0.5	-1.164 [-1.439, -0.906]	-24.4% [-30.1%, -19.0%]	-2.30 [-4.80, +0.10]	500
extra_sft_pilot_s1	missing_0.7	-1.669 [-2.004, -1.349]	-28.7% [-34.4%, -23.2%]	-3.01 [-5.63, -0.34]	500
extra_sft_pilot_s1	noise_0.1	-2.220 [-2.580, -1.894]	-35.7% [-41.4%, -30.4%]	+0.03 [-2.38, +2.58]	500
extra_sft_pilot_s1	noise_0.3	-7.666 [-8.219, -7.088]	-61.7% [-66.2%, -57.1%]	+4.28 [+1.57, +6.78]	500
extra_sft_pilot_s1	noise_0.5	-9.369 [-10.000, -8.729]	-61.6% [-65.7%, -57.4%]	+3.47 [+1.10, +5.75]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
extra_sft_pilot_s1	noise_1	-8.637 [-9.301, -8.033]	-48.0% [-51.7%, -44.7%]	+3.41 [+1.41, +5.40]	500
extra_sft_pilot_s2	clean	-0.283 [-0.462, -0.105]	-7.5% [-12.2%, -2.8%]	-0.20 [-2.23, +1.81]	500
extra_sft_pilot_s2	compound_0	-4.731 [-5.208, -4.277]	-47.1% [-51.8%, -42.5%]	+0.84 [-1.74, +3.55]	500
extra_sft_pilot_s2	missing_0.1	-0.328 [-0.523, -0.131]	-8.3% [-13.3%, -3.3%]	-0.43 [-2.66, +1.67]	500
extra_sft_pilot_s2	missing_0.3	-0.834 [-1.072, -0.610]	-19.4% [-24.9%, -14.2%]	-3.51 [-5.88, -1.20]	500
extra_sft_pilot_s2	missing_0.5	-0.984 [-1.252, -0.727]	-20.6% [-26.2%, -15.2%]	-4.24 [-6.45, -1.97]	500
extra_sft_pilot_s2	missing_0.7	-1.657 [-2.001, -1.320]	-28.5% [-34.4%, -22.7%]	-3.04 [-5.72, -0.60]	500
extra_sft_pilot_s2	noise_0.1	-2.104 [-2.441, -1.772]	-33.8% [-39.2%, -28.5%]	-0.30 [-2.80, +2.40]	500
extra_sft_pilot_s2	noise_0.3	-7.555 [-8.124, -6.988]	-60.8% [-65.4%, -56.3%]	+5.91 [+3.27, +8.48]	500
extra_sft_pilot_s2	noise_0.5	-9.580 [-10.221, -8.944]	-63.0% [-67.2%, -58.8%]	+3.97 [+1.54, +6.51]	500
extra_sft_pilot_s2	noise_1	-8.948 [-9.566, -8.348]	-49.7% [-53.2%, -46.4%]	+2.94 [+0.96, +4.92]	500
filtered_sft_pilot_s0	clean	-0.167 [-0.353, +0.017]	-4.4% [-9.4%, +0.4%]	+0.67 [-1.14, +2.74]	500
filtered_sft_pilot_s0	compound_0	-4.637 [-5.106, -4.185]	-46.1% [-50.8%, -41.6%]	+0.23 [-2.40, +2.98]	500
filtered_sft_pilot_s0	missing_0.1	-0.417 [-0.607, -0.227]	-10.6% [-15.4%, -5.8%]	+0.33 [-1.47, +2.27]	500
filtered_sft_pilot_s0	missing_0.3	-0.782 [-1.004, -0.565]	-18.1% [-23.3%, -13.1%]	-2.44 [-4.64, -0.36]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
filtered_sft_pilot_s0	missing_0.5	-1.022 [-1.285, -0.768]	-21.4% [-26.9%, -16.1%]	-2.74 [-5.01, -0.57]	500
filtered_sft_pilot_s0	missing_0.7	-1.319 [-1.636, -1.010]	-22.7% [-28.1%, -17.4%]	-3.01 [-5.46, -0.64]	500
filtered_sft_pilot_s0	noise_0.1	-2.142 [-2.480, -1.811]	-34.4% [-39.8%, -29.1%]	+0.47 [-2.04, +3.23]	500
filtered_sft_pilot_s0	noise_0.3	-7.313 [-7.837, -6.751]	-58.9% [-63.1%, -54.4%]	+3.34 [+0.83, +5.71]	500
filtered_sft_pilot_s0	noise_0.5	-8.938 [-9.550, -8.286]	-58.7% [-62.8%, -54.5%]	+3.24 [+0.90, +5.58]	500
filtered_sft_pilot_s0	noise_1	-8.424 [-9.018, -7.848]	-46.8% [-50.1%, -43.6%]	+2.47 [+0.44, +4.48]	500
filtered_sft_pilot_s1	clean	-0.217 [-0.418, -0.011]	-5.7% [-11.1%, -0.3%]	-0.27 [-2.31, +1.77]	500
filtered_sft_pilot_s1	compound_0	-4.429 [-4.922, -3.961]	-44.1% [-49.0%, -39.4%]	-2.07 [-4.73, +0.50]	500
filtered_sft_pilot_s1	missing_0.1	-0.351 [-0.554, -0.142]	-8.9% [-14.1%, -3.6%]	-2.17 [-4.29, -0.10]	500
filtered_sft_pilot_s1	missing_0.3	-0.725 [-0.959, -0.495]	-16.8% [-22.3%, -11.5%]	-4.07 [-6.55, -1.74]	500
filtered_sft_pilot_s1	missing_0.5	-1.080 [-1.358, -0.824]	-22.6% [-28.4%, -17.3%]	-3.47 [-5.94, -1.04]	500
filtered_sft_pilot_s1	missing_0.7	-1.375 [-1.705, -1.047]	-23.6% [-29.3%, -18.0%]	-3.41 [-6.16, -0.80]	500
filtered_sft_pilot_s1	noise_0.1	-2.112 [-2.450, -1.783]	-33.9% [-39.3%, -28.6%]	-1.44 [-3.97, +1.17]	500
filtered_sft_pilot_s1	noise_0.3	-7.318 [-7.859, -6.759]	-58.9% [-63.3%, -54.4%]	+3.64 [+1.16, +6.15]	500
filtered_sft_pilot_s1	noise_0.5	-8.965 [-9.589, -8.328]	-58.9% [-63.0%, -54.7%]	+1.87 [-0.53, +4.12]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
filtered_sft_pilot_s1	noise_1	-8.553 [-9.169, -7.944]	-47.6% [-51.0%, -44.2%]	+4.44 [+2.16, +6.64]	500
filtered_sft_pilot_s2	clean	-0.288 [-0.483, -0.097]	-7.6% [-12.8%, -2.6%]	-0.84 [-2.99, +1.23]	500
filtered_sft_pilot_s2	compound_0	-4.613 [-5.097, -4.154]	-45.9% [-50.7%, -41.3%]	+0.57 [-2.04, +3.37]	500
filtered_sft_pilot_s2	missing_0.1	-0.396 [-0.597, -0.192]	-10.1% [-15.2%, -4.9%]	-0.63 [-2.83, +1.44]	500
filtered_sft_pilot_s2	missing_0.3	-0.714 [-0.936, -0.503]	-16.6% [-21.7%, -11.7%]	-3.57 [-6.05, -1.14]	500
filtered_sft_pilot_s2	missing_0.5	-0.992 [-1.249, -0.731]	-20.8% [-26.2%, -15.3%]	-4.21 [-6.67, -1.81]	500
filtered_sft_pilot_s2	missing_0.7	-1.406 [-1.740, -1.076]	-24.2% [-29.9%, -18.5%]	-3.74 [-6.15, -1.31]	500
filtered_sft_pilot_s2	noise_0.1	-2.109 [-2.453, -1.766]	-33.9% [-39.4%, -28.4%]	-1.17 [-3.59, +1.36]	500
filtered_sft_pilot_s2	noise_0.3	-7.411 [-7.959, -6.831]	-59.7% [-64.1%, -55.0%]	+3.41 [+0.80, +5.78]	500
filtered_sft_pilot_s2	noise_0.5	-9.446 [-10.040, -8.826]	-62.1% [-66.0%, -58.0%]	+1.84 [-0.37, +4.08]	500
filtered_sft_pilot_s2	noise_1	-8.899 [-9.489, -8.333]	-49.5% [-52.8%, -46.3%]	+2.10 [+0.07, +4.07]	500
m0_pilot_s1	clean	-0.110 [-0.323, +0.103]	-2.9% [-8.5%, +2.7%]	+0.63 [-1.31, +2.61]	500
m0_pilot_s1	compound_0	+0.352 [-0.168, +0.900]	+3.5% [-1.7%, +9.0%]	-0.53 [-2.84, +1.79]	500
m0_pilot_s1	missing_0.1	-0.156 [-0.356, +0.055]	-4.0% [-9.0%, +1.4%]	+1.00 [-1.10, +3.05]	500
m0_pilot_s1	missing_0.3	+0.039 [-0.221, +0.297]	+0.9% [-5.1%, +6.9%]	-0.50 [-2.87, +1.81]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m0_pilot_s1	missing_0.5	+0.161 [-0.142, +0.453]	+3.4% [-3.0%, +9.5%]	+2.77 [+0.40, +5.21]	500
m0_pilot_s1	missing_0.7	+0.392 [+0.029, +0.763]	+6.7% [+0.5%, +13.1%]	+1.10 [-1.47, +3.68]	500
m0_pilot_s1	noise_0.1	-0.161 [-0.481, +0.186]	-2.6% [-7.7%, +3.0%]	-0.10 [-2.54, +2.38]	500
m0_pilot_s1	noise_0.3	-1.323 [-1.850, -0.776]	-10.7% [-14.9%, -6.2%]	+0.70 [-1.66, +2.92]	500
m0_pilot_s1	noise_0.5	-1.598 [-2.210, -0.952]	-10.5% [-14.5%, -6.3%]	-2.04 [-4.07, +0.07]	500
m0_pilot_s1	noise_1	-1.011 [-1.759, -0.286]	-5.6% [-9.8%, -1.6%]	+0.10 [-1.67, +1.94]	500
m0_pilot_s2	clean	-0.010 [-0.217, +0.187]	-0.3% [-5.8%, +4.9%]	+0.80 [-1.20, +2.83]	500
m0_pilot_s2	compound_0	+0.220 [-0.276, +0.704]	+2.2% [-2.7%, +7.0%]	-1.64 [-3.88, +0.74]	500
m0_pilot_s2	missing_0.1	+0.009 [-0.176, +0.191]	+0.2% [-4.5%, +4.9%]	+2.47 [+0.33, +4.45]	500
m0_pilot_s2	missing_0.3	-0.200 [-0.448, +0.040]	-4.6% [-10.4%, +0.9%]	+0.70 [-1.78, +3.05]	500
m0_pilot_s2	missing_0.5	-0.146 [-0.435, +0.132]	-3.1% [-9.1%, +2.8%]	+4.18 [+1.74, +6.75]	500
m0_pilot_s2	missing_0.7	+0.015 [-0.315, +0.357]	+0.3% [-5.4%, +6.1%]	+4.01 [+1.17, +6.59]	500
m0_pilot_s2	noise_0.1	-0.516 [-0.869, -0.155]	-8.3% [-14.0%, -2.5%]	-1.20 [-3.81, +1.33]	500
m0_pilot_s2	noise_0.3	-0.989 [-1.556, -0.401]	-8.0% [-12.5%, -3.2%]	+1.37 [-1.07, +3.70]	500
m0_pilot_s2	noise_0.5	-0.398 [-1.080, +0.345]	-2.6% [-7.1%, +2.3%]	+0.27 [-1.94, +2.50]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m0_pilot_s2	noise_1	+2.236 [+1.458, +3.017]	+12.4% [+8.1%, +16.8%]	+1.00 [-0.80, +2.77]	500
m1_pilot_s0	clean	-0.214 [-0.391, -0.036]	-5.7% [-10.3%, -0.9%]	+1.27 [-0.53, +3.21]	500
m1_pilot_s0	compound_0	-4.718 [-5.187, -4.238]	-46.9% [-51.6%, -42.2%]	+0.40 [-2.11, +3.02]	500
m1_pilot_s0	missing_0.1	-0.479 [-0.657, -0.295]	-12.2% [-16.7%, -7.5%]	+0.10 [-1.87, +2.10]	500
m1_pilot_s0	missing_0.3	-0.716 [-0.934, -0.512]	-16.6% [-21.7%, -11.9%]	-2.47 [-4.71, -0.30]	500
m1_pilot_s0	missing_0.5	-1.118 [-1.377, -0.862]	-23.4% [-28.8%, -18.1%]	-2.87 [-5.22, -0.57]	500
m1_pilot_s0	missing_0.7	-1.612 [-1.931, -1.318]	-27.7% [-33.2%, -22.7%]	-1.17 [-3.64, +1.24]	500
m1_pilot_s0	noise_0.1	-2.021 [-2.365, -1.688]	-32.5% [-38.0%, -27.1%]	+1.30 [-1.11, +3.83]	500
m1_pilot_s0	noise_0.3	-7.478 [-8.014, -6.927]	-60.2% [-64.5%, -55.8%]	+4.18 [+1.64, +6.59]	500
m1_pilot_s0	noise_0.5	-9.099 [-9.698, -8.479]	-59.8% [-63.7%, -55.7%]	+3.57 [+1.27, +5.99]	500
m1_pilot_s0	noise_1	-8.477 [-9.066, -7.874]	-47.1% [-50.4%, -43.8%]	+3.37 [+1.21, +5.43]	500
m1_pilot_s1	clean	-0.295 [-0.498, -0.106]	-7.8% [-13.2%, -2.8%]	-0.33 [-2.35, +1.71]	500
m1_pilot_s1	compound_0	-4.788 [-5.277, -4.299]	-47.6% [-52.5%, -42.8%]	+0.23 [-2.33, +3.03]	500
m1_pilot_s1	missing_0.1	-0.408 [-0.612, -0.194]	-10.3% [-15.5%, -4.9%]	-0.30 [-2.46, +1.81]	500
m1_pilot_s1	missing_0.3	-0.807 [-1.037, -0.574]	-18.7% [-24.1%, -13.3%]	-1.10 [-3.61, +1.27]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m1_pilot_s1	missing_0.5	-1.164 [-1.437, -0.897]	-24.4% [-30.1%, -18.8%]	-0.94 [-3.40, +1.54]	500
m1_pilot_s1	missing_0.7	-1.601 [-1.950, -1.275]	-27.5% [-33.5%, -21.9%]	-1.14 [-3.67, +1.34]	500
m1_pilot_s1	noise_0.1	-2.283 [-2.635, -1.959]	-36.7% [-42.3%, -31.5%]	+1.27 [-1.37, +3.99]	500
m1_pilot_s1	noise_0.3	-7.561 [-8.113, -6.986]	-60.9% [-65.3%, -56.3%]	+4.74 [+2.06, +7.34]	500
m1_pilot_s1	noise_0.5	-9.254 [-9.880, -8.623]	-60.8% [-64.9%, -56.7%]	+3.67 [+1.34, +6.10]	500
m1_pilot_s1	noise_1	-8.703 [-9.324, -8.099]	-48.4% [-51.8%, -45.0%]	+1.80 [-0.03, +3.79]	500
m1_pilot_s2	clean	-0.281 [-0.472, -0.103]	-7.4% [-12.5%, -2.7%]	+0.17 [-2.07, +2.41]	500
m1_pilot_s2	compound_0	-4.788 [-5.262, -4.331]	-47.6% [-52.4%, -43.1%]	+0.37 [-2.27, +3.11]	500
m1_pilot_s2	missing_0.1	-0.477 [-0.660, -0.298]	-12.1% [-16.8%, -7.6%]	-0.20 [-2.34, +1.87]	500
m1_pilot_s2	missing_0.3	-0.786 [-0.990, -0.587]	-18.2% [-23.0%, -13.6%]	-2.47 [-4.92, -0.17]	500
m1_pilot_s2	missing_0.5	-1.013 [-1.279, -0.766]	-21.2% [-26.8%, -16.0%]	-2.34 [-4.77, +0.13]	500
m1_pilot_s2	missing_0.7	-1.625 [-1.954, -1.292]	-27.9% [-33.6%, -22.2%]	-2.67 [-5.16, -0.20]	500
m1_pilot_s2	noise_0.1	-2.194 [-2.527, -1.863]	-35.2% [-40.6%, -29.9%]	+0.20 [-2.24, +2.93]	500
m1_pilot_s2	noise_0.3	-7.586 [-8.129, -7.007]	-61.1% [-65.5%, -56.4%]	+4.18 [+1.64, +6.51]	500
m1_pilot_s2	noise_0.5	-9.518 [-10.155, -8.876]	-62.6% [-66.7%, -58.3%]	+3.57 [+1.17, +5.91]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m1_pilot_s2	noise_1	-8.735 [-9.337, -8.131]	-48.6% [-51.9%, -45.2%]	+2.51 [+0.60, +4.38]	500
mind_pilot_s0	clean	+0.095 [-0.104, +0.289]	+2.5% [-2.7%, +7.7%]	+3.14 [+1.07, +5.32]	500
mind_pilot_s0	compound_0	+0.303 [-0.150, +0.736]	+3.0% [-1.5%, +7.3%]	-1.14 [-3.16, +0.90]	500
mind_pilot_s0	missing_0.1	+0.098 [-0.100, +0.294]	+2.5% [-2.5%, +7.5%]	+3.84 [+1.54, +5.98]	500
mind_pilot_s0	missing_0.3	+0.124 [-0.121, +0.368]	+2.9% [-2.8%, +8.5%]	+1.47 [-1.00, +3.77]	500
mind_pilot_s0	missing_0.5	+0.372 [+0.093, +0.653]	+7.8% [+1.9%, +13.7%]	+3.24 [+1.00, +5.62]	500
mind_pilot_s0	missing_0.7	+0.320 [-0.009, +0.624]	+5.5% [-0.1%, +10.7%]	+1.04 [-1.36, +3.55]	500
mind_pilot_s0	noise_0.1	+0.374 [+0.036, +0.696]	+6.0% [+0.6%, +11.2%]	+1.34 [-0.90, +3.64]	500
mind_pilot_s0	noise_0.3	+0.382 [-0.094, +0.901]	+3.1% [-0.8%, +7.3%]	+0.20 [-1.70, +2.18]	500
mind_pilot_s0	noise_0.5	+0.336 [-0.291, +0.944]	+2.2% [-1.9%, +6.2%]	-0.50 [-2.31, +1.37]	500
mind_pilot_s0	noise_1	+1.403 [+0.749, +2.120]	+7.8% [+4.2%, +11.8%]	+0.23 [-1.60, +1.98]	500
mind_pilot_s1	clean	+0.245 [+0.022, +0.463]	+6.5% [+0.6%, +12.3%]	+3.81 [+1.63, +5.98]	500
mind_pilot_s1	compound_0	+1.271 [+0.739, +1.799]	+12.6% [+7.4%, +17.9%]	-2.34 [-4.61, -0.10]	500
mind_pilot_s1	missing_0.1	+0.135 [-0.073, +0.352]	+3.4% [-1.8%, +8.9%]	+2.84 [+0.53, +5.02]	500
mind_pilot_s1	missing_0.3	+0.479 [+0.220, +0.740]	+11.1% [+5.1%, +17.2%]	-0.10 [-2.61, +2.38]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
mind_pilot_s1	missing_0.5	+0.688 [+0.388, +1.013]	+14.4% [+8.1%, +21.2%]	+2.67 [+0.23, +5.20]	500
mind_pilot_s1	missing_0.7	+1.250 [+0.876, +1.643]	+21.5% [+15.1%, +28.2%]	-1.00 [-3.61, +1.34]	500
mind_pilot_s1	noise_0.1	+0.001 [-0.359, +0.369]	+0.0% [-5.8%, +5.9%]	+0.80 [-1.74, +3.28]	500
mind_pilot_s1	noise_0.3	+0.048 [-0.484, +0.629]	+0.4% [-3.9%, +5.1%]	-1.70 [-3.94, +0.44]	500
mind_pilot_s1	noise_0.5	+0.114 [-0.571, +0.832]	+0.7% [-3.8%, +5.5%]	-2.04 [-3.87, -0.13]	500
mind_pilot_s1	noise_1	+0.188 [-0.483, +0.907]	+1.0% [-2.7%, +5.0%]	-1.30 [-3.03, +0.47]	500
mind_pilot_s2	clean	+0.021 [-0.188, +0.222]	+0.5% [-5.0%, +5.9%]	+5.71 [+3.48, +7.93]	500
mind_pilot_s2	compound_0	-0.243 [-0.757, +0.269]	-2.4% [-7.5%, +2.7%]	+0.90 [-1.84, +3.70]	500
mind_pilot_s2	missing_0.1	+0.232 [+0.013, +0.454]	+5.9% [+0.3%, +11.5%]	+5.91 [+3.76, +8.25]	500
mind_pilot_s2	missing_0.3	+0.223 [-0.008, +0.461]	+5.2% [-0.2%, +10.7%]	+5.44 [+2.95, +8.03]	500
mind_pilot_s2	missing_0.5	+0.145 [-0.156, +0.439]	+3.0% [-3.3%, +9.2%]	+6.11 [+3.59, +8.55]	500
mind_pilot_s2	missing_0.7	+0.490 [+0.140, +0.842]	+8.4% [+2.4%, +14.5%]	+4.11 [+1.44, +6.73]	500
mind_pilot_s2	noise_0.1	-0.691 [-1.075, -0.323]	-11.1% [-17.3%, -5.2%]	+2.54 [-0.00, +5.19]	500
mind_pilot_s2	noise_0.3	-1.384 [-1.955, -0.766]	-11.1% [-15.7%, -6.2%]	+2.34 [+0.00, +4.80]	500
mind_pilot_s2	noise_0.5	-0.678 [-1.371, +0.032]	-4.5% [-9.0%, +0.2%]	-0.33 [-2.47, +1.81]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
mind_pilot_s2	noise_1	+0.840 [+0.083, +1.572]	+4.7% [+0.5%, +8.7%]	+0.73 [-1.26, +2.78]	500
oracle_log	clean	-3.776 [-4.037, -3.522]	-100.0% [-106.9%, -93.3%]	+nan [+nan, +nan]	500
oracle_log	compound_0	-10.052 [-10.533, -9.579]	-100.0% [-104.8%, -95.3%]	+nan [+nan, +nan]	500
oracle_log	missing_0.1	-3.941 [-4.195, -3.691]	-100.0% [-106.4%, -93.7%]	+nan [+nan, +nan]	500
oracle_log	missing_0.3	-4.309 [-4.591, -4.041]	-100.0% [-106.5%, -93.8%]	+nan [+nan, +nan]	500
oracle_log	missing_0.5	-4.776 [-5.097, -4.461]	-100.0% [-106.7%, -93.4%]	+nan [+nan, +nan]	500
oracle_log	missing_0.7	-5.819 [-6.233, -5.424]	-100.0% [-107.1%, -93.2%]	+nan [+nan, +nan]	500
oracle_log	noise_0.1	-6.226 [-6.606, -5.849]	-100.0% [-106.1%, -93.9%]	+nan [+nan, +nan]	500
oracle_log	noise_0.3	-12.417 [-12.942, -11.855]	-100.0% [-104.2%, -95.5%]	+nan [+nan, +nan]	500
oracle_log	noise_0.5	-15.216 [-15.801, -14.603]	-100.0% [-103.8%, -96.0%]	+nan [+nan, +nan]	500
oracle_log	noise_1	-17.987 [-18.594, -17.401]	-100.0% [-103.4%, -96.7%]	+nan [+nan, +nan]	500
rl_pilot_s0	clean	+0.729 [+0.520, +0.944]	+19.3% [+13.8%, +25.0%]	-9.29 [-11.01, -7.48]	500
rl_pilot_s0	compound_0	-2.336 [-2.829, -1.875]	-23.2% [-28.1%, -18.7%]	-9.08 [-11.34, -6.69]	500
rl_pilot_s0	missing_0.1	+0.544 [+0.317, +0.780]	+13.8% [+8.0%, +19.8%]	-10.62 [-12.52, -8.75]	500
rl_pilot_s0	missing_0.3	+0.182 [-0.082, +0.445]	+4.2% [-1.9%, +10.3%]	-13.33 [-15.66, -11.20]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
rl_pilot_s0	missing_0.5	+0.199 [-0.113, +0.499]	+4.2% [-2.4%, +10.4%]	-13.66 [-16.06, -11.38]	500
rl_pilot_s0	missing_0.7	-0.293 [-0.641, +0.067]	-5.0% [-11.0%, +1.2%]	-14.93 [-17.43, -12.61]	500
rl_pilot_s0	noise_0.1	-0.605 [-0.948, -0.251]	-9.7% [-15.2%, -4.0%]	-11.26 [-13.39, -9.00]	500
rl_pilot_s0	noise_0.3	-5.194 [-5.689, -4.660]	-41.8% [-45.8%, -37.5%]	-5.81 [-7.92, -3.71]	500
rl_pilot_s0	noise_0.5	-6.567 [-7.187, -5.924]	-43.2% [-47.2%, -38.9%]	-4.04 [-6.00, -2.10]	500
rl_pilot_s0	noise_1	-6.150 [-6.771, -5.540]	-34.2% [-37.6%, -30.8%]	-2.67 [-4.57, -0.80]	500
rl_pilot_s1	clean	+0.634 [+0.412, +0.853]	+16.8% [+10.9%, +22.6%]	-10.29 [-12.22, -8.35]	500
rl_pilot_s1	compound_0	-2.558 [-3.051, -2.068]	-25.4% [-30.4%, -20.6%]	-9.22 [-11.58, -6.99]	500
rl_pilot_s1	missing_0.1	+0.487 [+0.257, +0.728]	+12.4% [+6.5%, +18.5%]	-10.72 [-12.86, -8.66]	500
rl_pilot_s1	missing_0.3	+0.110 [-0.168, +0.384]	+2.6% [-3.9%, +8.9%]	-13.93 [-16.37, -11.78]	500
rl_pilot_s1	missing_0.5	-0.010 [-0.307, +0.282]	-0.2% [-6.4%, +5.9%]	-13.19 [-15.57, -10.95]	500
rl_pilot_s1	missing_0.7	-0.167 [-0.536, +0.187]	-2.9% [-9.2%, +3.2%]	-13.66 [-16.13, -11.30]	500
rl_pilot_s1	noise_0.1	-0.624 [-0.986, -0.287]	-10.0% [-15.8%, -4.6%]	-11.76 [-14.09, -9.42]	500
rl_pilot_s1	noise_0.3	-5.413 [-5.958, -4.828]	-43.6% [-48.0%, -38.9%]	-6.11 [-8.42, -3.90]	500
rl_pilot_s1	noise_0.5	-6.958 [-7.602, -6.316]	-45.7% [-50.0%, -41.5%]	-4.44 [-6.37, -2.40]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
rl_pilot_s1	noise_1	-6.865 [-7.483, -6.244]	-38.2% [-41.6%, -34.7%]	-2.44 [-4.25, -0.74]	500
rl_pilot_s2	clean	+0.364 [+0.139, +0.588]	+9.6% [+3.7%, +15.6%]	-8.32 [-10.22, -6.36]	500
rl_pilot_s2	compound_0	-3.286 [-3.778, -2.795]	-32.7% [-37.6%, -27.8%]	-8.92 [-11.29, -6.54]	500
rl_pilot_s2	missing_0.1	+0.256 [+0.045, +0.476]	+6.5% [+1.1%, +12.1%]	-9.35 [-11.38, -7.29]	500
rl_pilot_s2	missing_0.3	-0.086 [-0.340, +0.171]	-2.0% [-7.9%, +4.0%]	-12.16 [-14.50, -10.00]	500
rl_pilot_s2	missing_0.5	-0.397 [-0.669, -0.135]	-8.3% [-14.0%, -2.8%]	-11.76 [-14.06, -9.49]	500
rl_pilot_s2	missing_0.7	-0.769 [-1.128, -0.431]	-13.2% [-19.4%, -7.4%]	-13.09 [-15.57, -10.79]	500
rl_pilot_s2	noise_0.1	-1.177 [-1.512, -0.842]	-18.9% [-24.3%, -13.5%]	-10.62 [-13.01, -8.15]	500
rl_pilot_s2	noise_0.3	-6.119 [-6.678, -5.519]	-49.3% [-53.8%, -44.4%]	-4.78 [-7.06, -2.64]	500
rl_pilot_s2	noise_0.5	-7.790 [-8.401, -7.149]	-51.2% [-55.2%, -47.0%]	-3.71 [-5.87, -1.53]	500
rl_pilot_s2	noise_1	-7.127 [-7.774, -6.480]	-39.6% [-43.2%, -36.0%]	-1.40 [-3.43, +0.64]	500
select_m1_s0	clean	-0.445 [-0.629, -0.259]	-11.8% [-16.7%, -6.9%]	-13.49 [-15.36, -11.72]	500
select_m1_s0	compound_0	-5.051 [-5.539, -4.584]	-50.2% [-55.1%, -45.6%]	-15.23 [-17.47, -13.03]	500
select_m1_s0	missing_0.1	-0.556 [-0.760, -0.341]	-14.1% [-19.3%, -8.7%]	-13.93 [-15.90, -12.11]	500
select_m1_s0	missing_0.3	-0.861 [-1.085, -0.648]	-20.0% [-25.2%, -15.0%]	-16.97 [-19.28, -14.96]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
select_m1_s0	missing_0.5	-1.209 [-1.469, -0.947]	-25.3% [-30.8%, -19.8%]	-16.60 [-18.80, -14.60]	500
select_m1_s0	missing_0.7	-1.830 [-2.162, -1.522]	-31.5% [-37.1%, -26.2%]	-18.97 [-21.44, -16.70]	500
select_m1_s0	noise_0.1	-2.376 [-2.726, -2.041]	-38.2% [-43.8%, -32.8%]	-16.33 [-18.50, -14.11]	500
select_m1_s0	noise_0.3	-7.743 [-8.297, -7.198]	-62.4% [-66.8%, -58.0%]	-12.73 [-14.60, -10.98]	500
select_m1_s0	noise_0.5	-9.611 [-10.226, -8.963]	-63.2% [-67.2%, -58.9%]	-11.66 [-13.45, -9.86]	500
select_m1_s0	noise_1	-9.415 [-10.014, -8.808]	-52.3% [-55.7%, -49.0%]	-10.92 [-12.59, -9.28]	500
select_m1_s1	clean	-0.503 [-0.713, -0.304]	-13.3% [-18.9%, -8.0%]	-13.99 [-15.77, -12.16]	500
select_m1_s1	compound_0	-4.939 [-5.444, -4.459]	-49.1% [-54.2%, -44.4%]	-15.06 [-17.40, -12.84]	500
select_m1_s1	missing_0.1	-0.651 [-0.862, -0.435]	-16.5% [-21.9%, -11.0%]	-14.43 [-16.37, -12.58]	500
select_m1_s1	missing_0.3	-0.883 [-1.124, -0.652]	-20.5% [-26.1%, -15.1%]	-17.03 [-19.24, -15.07]	500
select_m1_s1	missing_0.5	-1.265 [-1.529, -1.011]	-26.5% [-32.0%, -21.2%]	-16.30 [-18.44, -14.26]	500
select_m1_s1	missing_0.7	-1.687 [-2.018, -1.361]	-29.0% [-34.7%, -23.4%]	-18.20 [-20.55, -16.07]	500
select_m1_s1	noise_0.1	-2.335 [-2.687, -1.997]	-37.5% [-43.2%, -32.1%]	-16.30 [-18.47, -14.07]	500
select_m1_s1	noise_0.3	-7.806 [-8.355, -7.216]	-62.9% [-67.3%, -58.1%]	-12.59 [-14.59, -10.82]	500
select_m1_s1	noise_0.5	-9.798 [-10.397, -9.162]	-64.4% [-68.3%, -60.2%]	-11.92 [-13.72, -10.25]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
select_m1_s1	noise_1	-9.540 [-10.173, -8.922]	-53.0% [-56.6%, -49.6%]	-11.22 [-12.93, -9.57]	500
select_m1_s2	clean	-0.442 [-0.632, -0.249]	-11.7% [-16.7%, -6.6%]	-12.73 [-14.67, -10.80]	500
select_m1_s2	compound_0	-4.946 [-5.453, -4.490]	-49.2% [-54.3%, -44.7%]	-14.76 [-17.07, -12.54]	500
select_m1_s2	missing_0.1	-0.544 [-0.732, -0.355]	-13.8% [-18.6%, -9.0%]	-13.69 [-15.74, -11.72]	500
select_m1_s2	missing_0.3	-0.758 [-0.997, -0.521]	-17.6% [-23.1%, -12.1%]	-17.10 [-19.44, -14.99]	500
select_m1_s2	missing_0.5	-1.206 [-1.460, -0.964]	-25.3% [-30.6%, -20.2%]	-16.40 [-18.64, -14.29]	500
select_m1_s2	missing_0.7	-1.878 [-2.202, -1.560]	-32.3% [-37.8%, -26.8%]	-18.50 [-20.85, -16.33]	500
select_m1_s2	noise_0.1	-2.142 [-2.500, -1.814]	-34.4% [-40.1%, -29.1%]	-15.97 [-18.10, -13.73]	500
select_m1_s2	noise_0.3	-7.860 [-8.411, -7.276]	-63.3% [-67.7%, -58.6%]	-12.73 [-14.63, -10.97]	500
select_m1_s2	noise_0.5	-9.922 [-10.555, -9.296]	-65.2% [-69.4%, -61.1%]	-11.69 [-13.44, -9.99]	500
select_m1_s2	noise_1	-9.655 [-10.275, -9.033]	-53.7% [-57.1%, -50.2%]	-10.65 [-12.32, -9.02]	500
cp	CORE_SUITE	+18.928 [+17.208, +20.764]	+242.2% [+220.2%, +265.7%]	-15.49 [-16.74, -14.36]	500
cv	CORE_SUITE	+3.436 [+2.829, +4.064]	+44.0% [+36.2%, +52.0%]	+1.17 [-1.11, +3.57]	500
extra_sft_pilot_s0	CORE_SUITE	-3.512 [-3.720, -3.295]	-44.9% [-47.6%, -42.2%]	+1.14 [-0.07, +2.40]	500
extra_sft_pilot_s1	CORE_SUITE	-3.608 [-3.833, -3.382]	-46.2% [-49.0%, -43.3%]	+0.17 [-1.20, +1.53]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
extra_sft_pilot_s2	CORE_SUITE	-3.564 [-3.780, -3.347]	-45.6% [-48.4%, -42.8%]	+0.23 [-1.12, +1.58]	500
filtered_sft_pilot_s0	CORE_SUITE	-3.435 [-3.632, -3.235]	-44.0% [-46.5%, -41.4%]	+0.37 [-0.90, +1.61]	500
filtered_sft_pilot_s1	CORE_SUITE	-3.425 [-3.639, -3.207]	-43.8% [-46.6%, -41.0%]	-0.94 [-2.34, +0.37]	500
filtered_sft_pilot_s2	CORE_SUITE	-3.511 [-3.716, -3.295]	-44.9% [-47.5%, -42.2%]	-0.72 [-2.11, +0.69]	500
m0_pilot_s1	CORE_SUITE	-0.506 [-0.704, -0.293]	-6.5% [-9.0%, -3.8%]	+0.31 [-0.81, +1.38]	500
m0_pilot_s2	CORE_SUITE	-0.373 [-0.581, -0.163]	-4.8% [-7.4%, -2.1%]	+1.30 [+0.17, +2.45]	500
m1_pilot_s0	CORE_SUITE	-3.485 [-3.698, -3.270]	-44.6% [-47.3%, -41.9%]	+0.63 [-0.66, +1.94]	500
m1_pilot_s1	CORE_SUITE	-3.580 [-3.799, -3.358]	-45.8% [-48.6%, -43.0%]	+1.22 [-0.19, +2.62]	500
m1_pilot_s2	CORE_SUITE	-3.596 [-3.810, -3.381]	-46.0% [-48.8%, -43.3%]	+0.49 [-0.85, +1.86]	500
mind_pilot_s0	CORE_SUITE	+0.281 [+0.107, +0.465]	+3.6% [+1.4%, +5.9%]	+1.60 [+0.55, +2.63]	500
mind_pilot_s1	CORE_SUITE	+0.244 [+0.042, +0.457]	+3.1% [+0.5%, +5.8%]	+0.41 [-0.70, +1.46]	500
mind_pilot_s2	CORE_SUITE	-0.359 [-0.570, -0.136]	-4.6% [-7.3%, -1.7%]	+3.67 [+2.46, +4.80]	500
oracle_log	CORE_SUITE	-7.814 [-8.069, -7.550]	-100.0% [-103.3%, -96.6%]	+nan [+nan, +nan]	500
rl_pilot_s0	CORE_SUITE	-1.907 [-2.133, -1.676]	-24.4% [-27.3%, -21.4%]	-9.79 [-10.99, -8.61]	500
rl_pilot_s1	CORE_SUITE	-2.068 [-2.289, -1.831]	-26.5% [-29.3%, -23.4%]	-10.03 [-11.33, -8.81]	500

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
rl_pilot_s2	CORE_SUITE	-2.552 [-2.774, -2.314]	-32.7% [-35.5%, -29.6%]	-8.73 [-10.03, -7.43]	500
select_m1_s0	CORE_SUITE	-3.726 [-3.945, -3.499]	-47.7% [-50.5%, -44.8%]	-14.70 [-15.93, -13.62]	500
select_m1_s1	CORE_SUITE	-3.790 [-4.014, -3.560]	-48.5% [-51.4%, -45.6%]	-14.76 [-15.96, -13.68]	500
select_m1_s2	CORE_SUITE	-3.739 [-3.961, -3.520]	-47.8% [-50.7%, -45.1%]	-14.60 [-15.82, -13.45]	500

Denominators: `overlap rate (6, full horizon)` = fraction of (scene, sample) pairs with any box overlap over the full 80-step generated horizon, scenes with initial overlap excluded, NaN for logged futures; `overlap jointly-valid` = same test restricted to sub-steps where both targets have valid logged labels (defined for generated and logged trajectories on identical intervals; the oracle_log row is the reference rate); `offroad frac` = vehicle targets only, full horizon (logged: valid steps only); cp/cv are deterministic single trajectories; learned models use 6 joint samples (32 for the audit columns).

3a. Per-seed development comparison

Development comparison (500 dev scenes; bootstrap unit = heuristic source-segment group, 95% CI)

Sampling: learned models 6 samples (joint metrics) + 32-sample audit; cp/cv deterministic (1 sample, listed separately).

Per-seed results (joint minADE@6 8 s, m; overlap and offroad rates as fractions)

method	seed	clean jminADE	clean jminFDE	core-suite jminADE	held-out jminADE	clean overlap
cp	det.	26.742 [25.024, 28.612]	51.409 [48.093, 54.779]	26.742 [25.024, 28.612]	26.742 [25.024, 28.612]	0.0000 [0.0000, 0.0000]
cv	det.	10.822 [10.236, 11.404]	27.732 [26.190, 29.256]	11.250 [10.708, 11.804]	13.116 [12.576, 13.676]	0.1663 [0.1345, 0.2000]
oracle_log	det.	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	—

method	seed	clean jminADE	clean jminFDE	core-suite jminADE	held-out jminADE	clean overlap
m0_pilot_s0	0	3.776 [3.522, 4.037]	9.195 [8.468, 9.916]	7.814 [7.550, 8.069]	11.286 [10.949, 11.620]	0.1456 [0.1274, 0.1643]
m0_pilot_s1	1	3.666 [3.417, 3.915]	8.672 [8.003, 9.338]	7.308 [7.047, 7.572]	11.197 [10.827, 11.578]	0.1520 [0.1310, 0.1730]
m0_pilot_s2	2	3.765 [3.491, 4.027]	9.125 [8.428, 9.842]	7.441 [7.175, 7.698]	12.110 [11.738, 12.464]	0.1536 [0.1333, 0.1757]
mind_pilot_s0	0	3.871 [3.598, 4.137]	9.522 [8.764, 10.272]	8.095 [7.836, 8.349]	11.962 [11.655, 12.267]	0.1770 [0.1557, 0.1995]
mind_pilot_s1	1	4.020 [3.755, 4.277]	9.724 [9.001, 10.440]	8.058 [7.778, 8.329]	12.189 [11.811, 12.556]	0.1837 [0.1617, 0.2063]
mind_pilot_s2	2	3.796 [3.544, 4.053]	9.109 [8.437, 9.796]	7.455 [7.178, 7.733]	11.648 [11.278, 11.999]	0.2027 [0.1804, 0.2256]
m1_pilot_s0	0	3.561 [3.310, 3.811]	8.418 [7.739, 9.089]	4.329 [4.093, 4.565]	6.351 [6.121, 6.598]	0.1583 [0.1379, 0.1813]
m1_pilot_s1	1	3.481 [3.234, 3.724]	8.424 [7.734, 9.092]	4.235 [4.012, 4.459]	6.256 [6.011, 6.497]	0.1423 [0.1218, 0.1630]
m1_pilot_s2	2	3.495 [3.246, 3.728]	8.360 [7.680, 9.019]	4.218 [3.988, 4.447]	6.237 [6.012, 6.489]	0.1473 [0.1263, 0.1700]
select_m1_s0	0	3.331 [3.081, 3.586]	7.818 [7.131, 8.509]	4.088 [3.855, 4.321]	5.854 [5.623, 6.087]	0.0107 [0.0034, 0.0194]
select_m1_s1	1	3.273 [3.040, 3.509]	7.899 [7.269, 8.549]	4.024 [3.797, 4.246]	5.897 [5.648, 6.159]	0.0057 [0.0013, 0.0114]
select_m1_s2	2	3.334 [3.102, 3.564]	8.073 [7.412, 8.728]	4.076 [3.831, 4.309]	5.793 [5.556, 6.043]	0.0184 [0.0084, 0.0301]
extra_sft_pilot_s0	0	3.454 [3.200, 3.706]	8.442 [7.750, 9.133]	4.302 [4.062, 4.532]	6.369 [6.122, 6.608]	0.1583 [0.1379, 0.1810]
extra_sft_pilot_s1	1	3.470 [3.241, 3.691]	8.401 [7.767, 9.049]	4.207 [3.981, 4.433]	6.231 [5.994, 6.476]	0.1356 [0.1155, 0.1570]
extra_sft_pilot_s2	2	3.493 [3.238, 3.732]	8.375 [7.672, 9.053]	4.250 [4.021, 4.482]	6.174 [5.941, 6.416]	0.1436 [0.1233, 0.1653]

method	seed	clean jminADE	clean jminFDE	core-suite jminADE	held-out jminADE	clean overlap
filtered_sft_pilot_s0	0	3.609 [3.334, 3.877]	8.667 [7.953, 9.391]	4.379 [4.150, 4.611]	6.493 [6.248, 6.752]	0.1523 [0.1323, 0.1750]
filtered_sft_pilot_s1	1	3.559 [3.321, 3.803]	8.677 [8.000, 9.368]	4.389 [4.161, 4.616]	6.501 [6.245, 6.771]	0.1430 [0.1214, 0.1653]
filtered_sft_pilot_s2	2	3.488 [3.234, 3.728]	8.411 [7.690, 9.084]	4.303 [4.070, 4.542]	6.313 [6.056, 6.580]	0.1373 [0.1172, 0.1579]
rl_pilot_s0	0	4.504 [4.225, 4.755]	11.114 [10.349, 11.874]	5.907 [5.630, 6.179]	8.360 [8.033, 8.684]	0.0528 [0.0406, 0.0647]
rl_pilot_s1	1	4.409 [4.147, 4.663]	10.856 [10.135, 11.545]	5.746 [5.477, 6.007]	8.089 [7.762, 8.404]	0.0428 [0.0315, 0.0552]
rl_pilot_s2	2	4.139 [3.864, 4.404]	10.119 [9.368, 10.894]	5.262 [4.997, 5.513]	7.559 [7.258, 7.858]	0.0625 [0.0487, 0.0773]

Q1 Joint vs independent decoder (m0 − mind; negative = joint better)

seed	clean jminADE Δ	clean jminFDE Δ	clean overlap Δ (pp)	clean overlap32 Δ (pp)
0	-0.095 [-0.289, +0.104] (-2.5%)	-0.327 [-0.900, +0.276] (-3.4%)	-3.14 [-5.32, -1.07]	-2.82 [-4.36, -1.16]
1	-0.354 [-0.548, -0.165] (-8.8%)	-1.052 [-1.599, -0.498] (-10.8%)	-3.17 [-5.33, -0.97]	-2.46 [-4.00, -0.87]
2	-0.031 [-0.232, +0.159] (-0.8%)	+0.015 [-0.614, +0.603] (+0.2%)	-4.91 [-7.08, -2.73]	-3.69 [-5.30, -1.98]
sign agreement	consistent	MIXED	consistent	consistent

Q2 Corruption-trained vs clean-trained under degraded inputs (m1 − m0; negative = m1 better)

seed	core-suite jminADE Δ	held-out jminADE Δ	core-suite jminFDE Δ	core-suite overlap Δ (pp)
0	-3.485 [-3.698, -3.270] (-44.6%)	-4.935 [-5.239, -4.643] (-43.7%)	-7.454 [-8.028, -6.901] (-43.1%)	+0.63 [-0.66, +1.94]

seed	core-suite jminADE Δ	held-out jminADE Δ	core-suite jminFDE Δ	core-suite overlap Δ (pp)
1	-3.073 [-3.278, -2.870] (-42.1%)	-4.942 [-5.290, -4.591] (-44.1%)	-6.655 [-7.208, -6.104] (-40.8%)	+0.92 [-0.47, +2.26]
2	-3.222 [-3.433, -3.007] (-43.3%)	-5.873 [-6.193, -5.535] (-48.5%)	-7.182 [-7.740, -6.647] (-42.7%)	-0.81 [-2.08, +0.54]
sign agreement	consistent	consistent	consistent	MIXED

Q3 Clean accuracy and physical consistency (m1 − m0 on clean)

seed	clean jminADE Δ	clean overlap Δ (pp)	clean offroad Δ (pp)	clean implausible Δ (pp)
0	-0.214 [-0.391, -0.036] (-5.7%)	+1.27 [-0.53, +3.21]	-0.14 [-0.77, +0.47]	-0.05 [-0.13, +0.00]
1	-0.185 [-0.354, -0.016] (-5.1%)	-0.97 [-3.02, +1.07]	-1.15 [-1.83, -0.52]	+0.02 [-0.03, +0.08]
2	-0.270 [-0.471, -0.067] (-7.2%)	-0.63 [-2.61, +1.48]	-2.03 [-2.66, -1.43]	+0.05 [+0.00, +0.13]
sign agreement	consistent	MIXED	consistent	MIXED

Q5a Bounded RL vs its M1 initialisation (rl − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+0.943 [+0.781, +1.114] (+26.5%)	+1.578 [+1.401, +1.750] (+36.5%)	-10.55 [-12.45, -8.85]	-10.42 [-11.99, -9.03]	-10.38 [-11.94, -8.99]	-1.18 [-1.6 -0.70
1	+0.928 [+0.760, +1.098] (+26.7%)	+1.512 [+1.335, +1.680] (+35.7%)	-9.95 [-11.91, -8.17]	-11.90 [-13.57, -10.34]	-11.82 [-13.53, -10.29]	-1.33 [-1.7 -0.90
2	+0.644 [+0.463, +0.831] (+18.4%)	+1.044 [+0.893, +1.197] (+24.7%)	-8.48 [-10.34, -6.68]	-9.01 [-10.40, -7.62]	-8.94 [-10.34, -7.56]	-1.16 [-1.5 -0.74

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q5b Extra-SFT control vs M1 (extra_sft − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	clea offro Δ (p
0	-0.107 [-0.244, +0.030] (-3.0%)	-0.027 [-0.087, +0.040] (-0.6%)	+0.00 [-1.57, +1.57]	+1.05 [+0.34, +1.78]	+1.06 [+0.36, +1.78]	+0.09 [-0.21 +0.40
1	-0.011 [-0.140, +0.130] (-0.3%)	-0.028 [-0.090, +0.035] (-0.7%)	-0.67 [-2.04, +0.77]	-1.03 [-1.76, -0.32]	-1.01 [-1.74, -0.29]	-0.21 [-0.56 +0.12
2	-0.003 [-0.126, +0.120] (-0.1%)	+0.032 [-0.034, +0.103] (+0.8%)	-0.37 [-1.77, +1.07]	-0.06 [-0.83, +0.67]	-0.01 [-0.77, +0.71]	+0.22 [-0.11 +0.55
sign agreement	consistent	MIXED	MIXED	MIXED	MIXED	MIXI

Q5c Reward-filtered-SFT control vs M1 (filtered_sft − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	clea offro Δ (p
0	+0.047 [-0.080, +0.173] (+1.3%)	+0.050 [-0.020, +0.115] (+1.1%)	-0.60 [-2.13, +1.00]	+0.33 [-0.42, +1.11]	+0.29 [-0.44, +1.05]	+0.16 [-0.15 +0.46

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	clea offro Δ (p
1	+0.078 [-0.037, +0.192] (+2.2%)	+0.154 [+0.087, +0.224] (+3.6%)	+0.07 [-1.44, +1.50]	-1.88 [-2.68, -1.08]	-1.92 [-2.72, -1.12]	-0.10 [-0.4 +0.23
2	-0.007 [-0.126, +0.111] (-0.2%)	+0.084 [+0.007, +0.161] (+2.0%)	-1.00 [-2.61, +0.57]	-1.33 [-2.14, -0.58]	-1.29 [-2.10, -0.53]	+0.11 [-0.21 +0.44
sign agreement	MIXED	consistent	MIXED	MIXED	MIXED	MIXI

Q5d Reward-based candidate selection vs M1 (select − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	-0.230 [-0.348, -0.108] (-6.5%)	-0.241 [-0.319, -0.164] (-5.6%)	-14.76 [-16.93, -12.86]	+0.19 [-0.43, +0.79]	+0.15 [-0.47, +0.74]	-3.26 [-3.76 -2.78
1	-0.208 [-0.322, -0.090] (-6.0%)	-0.210 [-0.283, -0.141] (-5.0%)	-13.66 [-15.67, -11.68]	+0.06 [-0.61, +0.72]	+0.04 [-0.64, +0.68]	-3.07 [-3.56 -2.57
2	-0.162 [-0.284, -0.039] (-4.6%)	-0.143 [-0.224, -0.063] (-3.4%)	-12.89 [-14.96, -11.00]	-0.59 [-1.23, +0.04]	-0.56 [-1.20, +0.07]	-3.05 [-3.54 -2.58
sign agreement	consistent	consistent	consistent	MIXED	MIXED	consi

Q5e RL vs extra-SFT control (rl − extra_sft)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+1.050 [+0.864, +1.234] (+30.4%)	+1.606 [+1.431, +1.781] (+37.3%)	-10.55 [-12.63, -8.73]	-11.47 [-13.13, -9.96]	-11.45 [-13.12, -9.92]	-1.28 [-1.79 -0.80
1	+0.940 [+0.769, +1.113] (+27.1%)	+1.540 [+1.365, +1.710] (+36.6%)	-9.29 [-11.19, -7.56]	-10.87 [-12.48, -9.41]	-10.80 [-12.39, -9.35]	-1.12 [-1.52 -0.70
2	+0.647 [+0.465, +0.838] (+18.5%)	+1.012 [+0.863, +1.157] (+23.8%)	-8.12 [-9.87, -6.31]	-8.94 [-10.24, -7.58]	-8.93 [-10.23, -7.57]	-1.38 [-1.84 -0.94
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q5f RL vs reward-filtered-SFT control (rl − filtered_sft)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+0.896 [+0.715, +1.069] (+24.8%)	+1.529 [+1.369, +1.695] (+34.9%)	-9.95 [-12.00, -8.15]	-10.75 [-12.31, -9.38]	-10.67 [-12.21, -9.31]	-1.35 [-1.81 -0.89
1	+0.851 [+0.688, +1.016] (+23.9%)	+1.357 [+1.206, +1.506] (+30.9%)	-10.02 [-12.01, -8.32]	-10.02 [-11.48, -8.66]	-9.90 [-11.34, -8.53]	-1.23 [-1.66 -0.80
2	+0.651 [+0.483, +0.832] (+18.7%)	+0.959 [+0.820, +1.099] (+22.3%)	-7.48 [-9.17, -5.88]	-7.67 [-8.90, -6.43]	-7.65 [-8.87, -6.42]	-1.27 [-1.78 -0.80
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q5g RL vs reward-based candidate selection (rl − select_m1; select draws 32 candidates per scene)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+1.173 [+0.998, +1.339] (+35.2%)	+1.819 [+1.653, +1.987] (+44.5%)	+4.21 [+3.01, +5.41]	-10.61 [-12.18, -9.20]	-10.53 [-12.10, -9.13]	+2.08 [+1.6 +2.57
1	+1.137 [+0.952, +1.325] (+34.7%)	+1.722 [+1.558, +1.885] (+42.8%)	+3.71 [+2.61, +4.88]	-11.96 [-13.59, -10.42]	-11.85 [-13.51, -10.26]	+1.73 [+1.3 +2.15
2	+0.806 [+0.627, +0.995] (+24.2%)	+1.187 [+1.047, +1.332] (+29.1%)	+4.41 [+3.21, +5.81]	-8.41 [-9.75, -7.08]	-8.38 [-9.72, -7.07]	+1.88 [+1.4 +2.34
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q6a Excessive-stopping diagnostic (descriptive; clean dev; 500 scenes; identical scenes/sampling; eligible = agent with valid logged final step and logged final displacement > 5 m; stopped = predicted final displacement < 1 m; counted per (agent, joint sample); 6 samples/scene (select_m1: best 6 of 32 by reward); torch seed 0; descriptive only, no threshold)

method	eligible (agent, sample) pairs	stopped-while-logged-moved	frequency	seeds
m1_pilot	17082	73	0.43 %	0,1,2
select_m1	17082	75	0.44 %	0,1,2
extra_sft_pilot	17082	74	0.43 %	0,1,2
filtered_sft_pilot	17082	99	0.58 %	0,1,2
rl_pilot	17082	596	3.49 %	0,1,2
m0_pilot	17082	89	0.52 %	0,1,2

Q6b Post-training behaviour summary (clean dev; scene-level means over seeds; 6-sample metrics unless 32 noted)

method	speed_mean (m/s)	GT speed_mean	final_disp_mean (m)	GT final_disp	final / G
m1_pilot	6.61	6.66	51.08	51.56	0.991
select_m1	6.52	6.66	50.50	51.56	0.979
extra_sft_pilot	6.57	6.66	50.78	51.56	0.985
filtered_sft_pilot	6.56	6.66	50.66	51.56	0.982
rl_pilot	5.78	6.66	44.97	51.56	0.872

Sampling cost and wall-clock (clean condition)

Wall-clock seconds per scene are DESCRIPTIVE measurements logged by evaluate.py on a GPU shared with unrelated jobs; they do not establish a reliable relative speed ranking. The deterministic cost driver is the number of joint samples drawn per scene (and, for select_m1, the reward scoring of all 32 candidates).

method	samples drawn per scene	s / scene (descriptive)
cp	1	0.0000
cv	1	0.0000
extra_sft_pilot_s0	6	0.0073
extra_sft_pilot_s1	6	0.0072
extra_sft_pilot_s2	6	0.0074
filtered_sft_pilot_s0	6	0.0074
filtered_sft_pilot_s1	6	0.0074
filtered_sft_pilot_s2	6	0.0075
m0_pilot_s0	6	0.0038
m0_pilot_s1	6	0.0032
m0_pilot_s2	6	0.0032
m1_pilot_s0	6	0.0032
m1_pilot_s1	6	0.0032
m1_pilot_s2	6	0.0032
mind_pilot_s0	6	0.0031
mind_pilot_s1	6	0.0031
mind_pilot_s2	6	0.0032
oracle_log	1	0.0000
rl_pilot_s0	6	0.0078
rl_pilot_s1	6	0.0075
rl_pilot_s2	6	0.0075
select_m1_s0	32 + reward scoring of 32	0.0521

method	samples drawn per scene	s / scene (descriptive)
select_m1_s1	32 + reward scoring of 32	0.0512
select_m1_s2	32 + reward scoring of 32	0.0512

Per-condition joint minADE@6 (mean over seeds; per-seed values in summary.json)

condition	cv	m0_pilot	mind_pilot	m1_pilot	select_m1	rl_pilot
clean	10.822	3.736 (±0.049, n=3)	3.896 (±0.093, n=3)	3.513 (±0.035, n=3)	3.313 (±0.028, n=3)	4.351 (±0.155, n=3)
missing_0.1	10.814	3.892 (±0.076, n=3)	4.096 (±0.057, n=3)	3.486 (±0.033, n=3)	3.357 (±0.048, n=3)	4.370 (±0.125, n=3)
missing_0.3	10.768	4.256 (±0.105, n=3)	4.584 (±0.149, n=3)	3.540 (±0.039, n=3)	3.475 (±0.054, n=3)	4.378 (±0.113, n=3)
missing_0.5	10.762	4.781 (±0.125, n=3)	5.178 (±0.223, n=3)	3.677 (±0.063, n=3)	3.549 (±0.027, n=3)	4.707 (±0.247, n=3)
noise_0.1	10.941	6.001 (±0.216, n=3)	6.121 (±0.441, n=3)	4.060 (±0.109, n=3)	3.942 (±0.102, n=3)	5.424 (±0.265, n=3)
noise_0.3	11.631	11.647 (±0.562, n=3)	12.100 (±0.766, n=3)	4.876 (±0.046, n=3)	4.615 (±0.048, n=3)	6.842 (±0.395, n=3)
noise_0.5	12.586	14.550 (±0.679, n=3)	15.139 (±0.435, n=3)	5.925 (±0.173, n=3)	5.439 (±0.128, n=3)	8.110 (±0.510, n=3)
missing_0.7	11.427	5.955 (±0.181, n=3)	6.506 (±0.404, n=3)	4.207 (±0.010, n=3)	4.021 (±0.081, n=3)	5.410 (±0.259, n=3)
noise_1	16.053	18.395 (±1.357, n=3)	18.797 (±0.496, n=3)	9.349 (±0.115, n=3)	8.450 (±0.098, n=3)	11.273 (±0.413, n=3)
compound_0	11.868	10.243 (±0.145, n=3)	10.496 (±0.626, n=3)	5.288 (±0.033, n=3)	5.074 (±0.051, n=3)	7.326 (±0.406, n=3)

Preregistered thresholds (m1 vs m0, per seed)

seed	core-suite rel. Δ (need ≤ −5%)	clean rel. Δ (need ≤ +3%)	core-suite overlap Δ pp (need ≤ +0.5)	verdict
0	-44.6% [-47.3, -41.9]	-5.7% [-10.3, -0.9]	+0.63 [-0.66, +1.94]	FAIL
1	-42.1% [-44.8, -39.3]	-5.1% [-9.7, -0.4]	+0.92 [-0.47, +2.26]	FAIL

seed	core-suite rel. Δ (need ≤ −5%)	clean rel. Δ (need ≤ +3%)	core-suite overlap Δ pp (need ≤ +0.5)	verdict
2	-43.3% [-46.1, -40.4]	-7.2% [-12.5, -1.8]	-0.81 [-2.08, +0.54]	PASS

5. Locked final evaluation (official validation shards; one pass per frozen method)

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s
cp	clean	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	clean	1000	11.114 [10.681, 11.562]	28.456 [27.264, 29.601]	11.092 [10.657, 11.539]
extra_sft_pilot_s0	clean	1000	3.123 [2.976, 3.282]	7.667 [7.258, 8.145]	2.638 [2.502, 2.781]
extra_sft_pilot_s1	clean	1000	3.078 [2.925, 3.234]	7.505 [7.073, 7.960]	2.604 [2.463, 2.749]
extra_sft_pilot_s2	clean	1000	3.156 [3.000, 3.322]	7.784 [7.333, 8.258]	2.672 [2.528, 2.835]
filtered_sft_pilot_s0	clean	1000	3.178 [3.036, 3.345]	7.724 [7.302, 8.196]	2.677 [2.545, 2.825]
filtered_sft_pilot_s1	clean	1000	3.180 [3.032, 3.331]	7.705 [7.283, 8.143]	2.693 [2.559, 2.835]
filtered_sft_pilot_s2	clean	1000	3.218 [3.059, 3.396]	7.986 [7.524, 8.463]	2.727 [2.574, 2.888]
m0_pilot_s0	clean	1000	3.516 [3.357, 3.687]	8.600 [8.150, 9.111]	2.959 [2.810, 3.114]
m0_pilot_s1	clean	1000	3.425 [3.262, 3.594]	8.470 [8.029, 8.956]	2.854 [2.704, 3.008]
m0_pilot_s2	clean	1000	3.541 [3.372, 3.721]	8.755 [8.261, 9.258]	2.999 [2.839, 3.170]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s
m1_pilot_s0	clean	1000	3.200 [3.055, 3.364]	7.875 [7.433, 8.351]	2.720 [2.583, 2.878]
m1_pilot_s1	clean	1000	3.143 [2.991, 3.303]	7.694 [7.273, 8.155]	2.649 [2.513, 2.789]
m1_pilot_s2	clean	1000	3.251 [3.088, 3.424]	7.975 [7.517, 8.471]	2.758 [2.610, 2.921]
mind_pilot_s0	clean	1000	3.689 [3.527, 3.866]	9.175 [8.706, 9.695]	3.061 [2.910, 3.222]
mind_pilot_s1	clean	1000	3.575 [3.417, 3.752]	8.844 [8.370, 9.350]	2.962 [2.811, 3.121]
mind_pilot_s2	clean	1000	3.691 [3.525, 3.870]	8.997 [8.512, 9.494]	3.039 [2.891, 3.197]
oracle_log	clean	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	clean	1000	4.043 [3.871, 4.232]	10.165 [9.642, 10.706]	3.563 [3.397, 3.745]
rl_pilot_s1	clean	1000	4.059 [3.875, 4.246]	10.300 [9.778, 10.848]	3.579 [3.403, 3.761]
rl_pilot_s2	clean	1000	3.795 [3.627, 3.977]	9.738 [9.242, 10.237]	3.327 [3.165, 3.502]
select_m1_s0	clean	1000	3.041 [2.894, 3.204]	7.379 [6.941, 7.838]	2.602 [2.460, 2.756]
select_m1_s1	clean	1000	2.994 [2.851, 3.146]	7.307 [6.886, 7.734]	2.563 [2.428, 2.711]
select_m1_s2	clean	1000	3.045 [2.892, 3.218]	7.564 [7.099, 8.041]	2.620 [2.479, 2.780]
cp	compound_0	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	compound_0	1000	12.051 [11.625, 12.476]	29.755 [28.615, 30.892]	12.027 [11.602, 12.449]

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s
extra_sft_pilot_s0	compound_0	1000	4.986 [4.801, 5.176]	10.682 [10.189, 11.207]	4.187 [4.009, 4.360]
extra_sft_pilot_s1	compound_0	1000	4.912 [4.726, 5.103]	10.671 [10.203, 11.188]	4.096 [3.930, 4.268]
extra_sft_pilot_s2	compound_0	1000	4.947 [4.762, 5.148]	10.770 [10.274, 11.275]	4.154 [3.981, 4.351]
filtered_sft_pilot_s0	compound_0	1000	5.159 [4.970, 5.349]	10.979 [10.507, 11.468]	4.343 [4.173, 4.513]
filtered_sft_pilot_s1	compound_0	1000	5.048 [4.862, 5.241]	10.969 [10.486, 11.475]	4.299 [4.132, 4.467]
filtered_sft_pilot_s2	compound_0	1000	5.023 [4.831, 5.224]	11.040 [10.523, 11.569]	4.238 [4.058, 4.431]
m0_pilot_s0	compound_0	1000	10.108 [9.757, 10.456]	21.449 [20.625, 22.307]	8.667 [8.347, 9.002]
m0_pilot_s1	compound_0	1000	10.579 [10.205, 10.936]	22.885 [21.993, 23.807]	9.229 [8.871, 9.569]
m0_pilot_s2	compound_0	1000	9.973 [9.628, 10.312]	21.708 [20.838, 22.589]	8.752 [8.435, 9.062]
m1_pilot_s0	compound_0	1000	5.005 [4.825, 5.186]	10.788 [10.300, 11.279]	4.216 [4.050, 4.390]
m1_pilot_s1	compound_0	1000	4.910 [4.719, 5.090]	10.659 [10.176, 11.165]	4.147 [3.970, 4.316]
m1_pilot_s2	compound_0	1000	4.911 [4.722, 5.104]	10.742 [10.243, 11.233]	4.168 [3.993, 4.350]
mind_pilot_s0	compound_0	1000	10.746 [10.386, 11.090]	22.606 [21.818, 23.525]	9.258 [8.922, 9.580]
mind_pilot_s1	compound_0	1000	11.376 [10.993, 11.754]	25.004 [24.089, 25.947]	9.784 [9.436, 10.137]
mind_pilot_s2	compound_0	1000	9.725 [9.384, 10.042]	20.731 [19.930, 21.541]	8.616 [8.295, 8.914]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s
oracle_log	compound_0	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	compound_0	1000	7.206 [6.959, 7.454]	15.999 [15.334, 16.644]	6.439 [6.197, 6.681]
rl_pilot_s1	compound_0	1000	7.112 [6.861, 7.361]	15.992 [15.324, 16.650]	6.336 [6.085, 6.578]
rl_pilot_s2	compound_0	1000	6.343 [6.117, 6.580]	14.474 [13.861, 15.069]	5.588 [5.368, 5.818]
select_m1_s0	compound_0	1000	4.815 [4.631, 5.008]	10.333 [9.845, 10.836]	4.067 [3.896, 4.241]
select_m1_s1	compound_0	1000	4.655 [4.480, 4.837]	10.061 [9.566, 10.571]	3.965 [3.797, 4.134]
select_m1_s2	compound_0	1000	4.694 [4.509, 4.893]	10.323 [9.825, 10.858]	3.983 [3.806, 4.175]
cp	missing_0.1	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	missing_0.1	1000	11.099 [10.668, 11.545]	28.427 [27.236, 29.574]	11.077 [10.645, 11.525]
extra_sft_pilot_s0	missing_0.1	1000	3.155 [3.011, 3.313]	7.692 [7.296, 8.132]	2.634 [2.498, 2.783]
extra_sft_pilot_s1	missing_0.1	1000	3.134 [2.983, 3.288]	7.717 [7.292, 8.179]	2.665 [2.530, 2.809]
extra_sft_pilot_s2	missing_0.1	1000	3.146 [2.988, 3.319]	7.848 [7.385, 8.320]	2.660 [2.510, 2.820]
filtered_sft_pilot_s0	missing_0.1	1000	3.177 [3.021, 3.343]	7.732 [7.307, 8.222]	2.680 [2.527, 2.839]
filtered_sft_pilot_s1	missing_0.1	1000	3.183 [3.030, 3.341]	7.819 [7.388, 8.290]	2.681 [2.543, 2.826]
filtered_sft_pilot_s2	missing_0.1	1000	3.269 [3.099, 3.451]	8.185 [7.701, 8.692]	2.775 [2.622, 2.942]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s
m0_pilot_s0	missing_0.1	1000	3.556 [3.397, 3.730]	8.669 [8.194, 9.154]	2.952 [2.804, 3.108]
m0_pilot_s1	missing_0.1	1000	3.489 [3.334, 3.656]	8.535 [8.086, 9.007]	2.942 [2.804, 3.094]
m0_pilot_s2	missing_0.1	1000	3.600 [3.441, 3.778]	8.692 [8.227, 9.191]	2.996 [2.850, 3.154]
m1_pilot_s0	missing_0.1	1000	3.199 [3.048, 3.369]	7.826 [7.373, 8.307]	2.698 [2.563, 2.855]
m1_pilot_s1	missing_0.1	1000	3.145 [2.995, 3.292]	7.709 [7.295, 8.167]	2.675 [2.532, 2.814]
m1_pilot_s2	missing_0.1	1000	3.155 [2.998, 3.327]	7.893 [7.418, 8.359]	2.708 [2.561, 2.868]
mind_pilot_s0	missing_0.1	1000	3.664 [3.501, 3.841]	9.056 [8.593, 9.575]	3.075 [2.922, 3.235]
mind_pilot_s1	missing_0.1	1000	3.695 [3.526, 3.877]	9.122 [8.641, 9.607]	3.045 [2.897, 3.206]
mind_pilot_s2	missing_0.1	1000	3.761 [3.593, 3.943]	9.162 [8.663, 9.688]	3.113 [2.959, 3.277]
oracle_log	missing_0.1	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	missing_0.1	1000	4.108 [3.931, 4.290]	10.350 [9.831, 10.908]	3.604 [3.435, 3.783]
rl_pilot_s1	missing_0.1	1000	4.100 [3.925, 4.290]	10.397 [9.850, 10.952]	3.622 [3.449, 3.812]
rl_pilot_s2	missing_0.1	1000	3.818 [3.644, 4.001]	9.724 [9.223, 10.239]	3.304 [3.139, 3.469]
select_m1_s0	missing_0.1	1000	3.038 [2.894, 3.197]	7.498 [7.059, 7.969]	2.565 [2.434, 2.712]
select_m1_s1	missing_0.1	1000	3.010 [2.861, 3.173]	7.374 [6.935, 7.837]	2.575 [2.437, 2.731]

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s
select_m1_s2	missing_0.1	1000	3.062 [2.911, 3.227]	7.734 [7.269, 8.189]	2.630 [2.484, 2.779]
cp	missing_0.3	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	missing_0.3	1000	11.060 [10.626, 11.505]	28.365 [27.181, 29.506]	11.038 [10.604, 11.481]
extra_sft_pilot_s0	missing_0.3	1000	3.132 [2.990, 3.300]	7.640 [7.240, 8.099]	2.615 [2.489, 2.758]
extra_sft_pilot_s1	missing_0.3	1000	3.142 [2.988, 3.304]	7.685 [7.239, 8.143]	2.668 [2.528, 2.820]
extra_sft_pilot_s2	missing_0.3	1000	3.152 [3.003, 3.313]	7.879 [7.421, 8.362]	2.675 [2.535, 2.822]
filtered_sft_pilot_s0	missing_0.3	1000	3.267 [3.111, 3.436]	7.973 [7.522, 8.468]	2.741 [2.600, 2.895]
filtered_sft_pilot_s1	missing_0.3	1000	3.166 [3.021, 3.326]	7.721 [7.285, 8.171]	2.672 [2.538, 2.812]
filtered_sft_pilot_s2	missing_0.3	1000	3.253 [3.096, 3.412]	7.974 [7.516, 8.461]	2.771 [2.629, 2.927]
m0_pilot_s0	missing_0.3	1000	3.978 [3.797, 4.169]	9.627 [9.107, 10.186]	3.317 [3.157, 3.494]
m0_pilot_s1	missing_0.3	1000	3.906 [3.741, 4.082]	9.296 [8.812, 9.794]	3.307 [3.154, 3.466]
m0_pilot_s2	missing_0.3	1000	3.911 [3.727, 4.098]	9.500 [9.012, 10.048]	3.299 [3.132, 3.472]
m1_pilot_s0	missing_0.3	1000	3.203 [3.059, 3.368]	7.816 [7.373, 8.277]	2.686 [2.557, 2.834]
m1_pilot_s1	missing_0.3	1000	3.145 [3.000, 3.310]	7.659 [7.242, 8.104]	2.648 [2.512, 2.792]
m1_pilot_s2	missing_0.3	1000	3.195 [3.048, 3.353]	7.892 [7.436, 8.357]	2.708 [2.573, 2.848]

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s
mind_pilot_s0	missing_0.3	1000	4.184 [4.003, 4.377]	10.303 [9.771, 10.830]	3.535 [3.375, 3.709]
mind_pilot_s1	missing_0.3	1000	4.166 [3.986, 4.360]	10.083 [9.577, 10.594]	3.462 [3.295, 3.643]
mind_pilot_s2	missing_0.3	1000	4.196 [4.016, 4.394]	10.100 [9.588, 10.613]	3.511 [3.350, 3.692]
oracle_log	missing_0.3	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	missing_0.3	1000	4.212 [4.031, 4.401]	10.520 [9.978, 11.073]	3.713 [3.531, 3.896]
rl_pilot_s1	missing_0.3	1000	4.139 [3.959, 4.333]	10.399 [9.859, 10.935]	3.682 [3.509, 3.874]
rl_pilot_s2	missing_0.3	1000	3.822 [3.655, 3.996]	9.535 [9.039, 10.062]	3.308 [3.151, 3.479]
select_m1_s0	missing_0.3	1000	3.045 [2.905, 3.213]	7.381 [6.944, 7.828]	2.599 [2.471, 2.750]
select_m1_s1	missing_0.3	1000	3.046 [2.894, 3.203]	7.396 [6.976, 7.838]	2.589 [2.449, 2.738]
select_m1_s2	missing_0.3	1000	3.122 [2.960, 3.294]	7.741 [7.275, 8.241]	2.707 [2.556, 2.869]
cp	missing_0.5	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	missing_0.5	1000	11.045 [10.606, 11.491]	28.322 [27.147, 29.477]	11.023 [10.591, 11.467]
extra_sft_pilot_s0	missing_0.5	1000	3.321 [3.165, 3.490]	8.030 [7.587, 8.500]	2.759 [2.617, 2.911]
extra_sft_pilot_s1	missing_0.5	1000	3.306 [3.143, 3.479]	8.019 [7.566, 8.507]	2.772 [2.625, 2.927]
extra_sft_pilot_s2	missing_0.5	1000	3.387 [3.226, 3.560]	8.238 [7.751, 8.738]	2.866 [2.714, 3.031]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s
filtered_sft_pilot_s0	missing_0.5	1000	3.326 [3.170, 3.496]	7.981 [7.527, 8.466]	2.828 [2.681, 2.985]
filtered_sft_pilot_s1	missing_0.5	1000	3.353 [3.191, 3.523]	8.053 [7.610, 8.531]	2.865 [2.713, 3.034]
filtered_sft_pilot_s2	missing_0.5	1000	3.429 [3.261, 3.609]	8.328 [7.843, 8.811]	2.920 [2.767, 3.086]
m0_pilot_s0	missing_0.5	1000	4.371 [4.186, 4.576]	10.395 [9.862, 10.929]	3.648 [3.476, 3.836]
m0_pilot_s1	missing_0.5	1000	4.586 [4.398, 4.786]	10.710 [10.189, 11.248]	3.895 [3.721, 4.083]
m0_pilot_s2	missing_0.5	1000	4.291 [4.107, 4.491]	10.126 [9.601, 10.669]	3.612 [3.441, 3.802]
m1_pilot_s0	missing_0.5	1000	3.299 [3.148, 3.462]	8.001 [7.560, 8.462]	2.767 [2.626, 2.918]
m1_pilot_s1	missing_0.5	1000	3.346 [3.188, 3.529]	8.086 [7.637, 8.584]	2.797 [2.656, 2.957]
m1_pilot_s2	missing_0.5	1000	3.315 [3.157, 3.496]	8.136 [7.644, 8.619]	2.820 [2.676, 2.990]
mind_pilot_s0	missing_0.5	1000	4.633 [4.438, 4.835]	11.147 [10.607, 11.719]	3.886 [3.706, 4.072]
mind_pilot_s1	missing_0.5	1000	4.931 [4.715, 5.163]	11.818 [11.212, 12.464]	4.112 [3.914, 4.323]
mind_pilot_s2	missing_0.5	1000	4.835 [4.624, 5.067]	11.366 [10.791, 11.988]	4.071 [3.873, 4.283]
oracle_log	missing_0.5	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	missing_0.5	1000	4.441 [4.255, 4.642]	11.087 [10.552, 11.645]	3.931 [3.753, 4.123]
rl_pilot_s1	missing_0.5	1000	4.409 [4.202, 4.628]	10.900 [10.334, 11.488]	3.907 [3.711, 4.116]

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s
rl_pilot_s2	missing_0.5	1000	4.030 [3.855, 4.219]	10.078 [9.566, 10.606]	3.528 [3.363, 3.712]
select_m1_s0	missing_0.5	1000	3.205 [3.057, 3.375]	7.844 [7.384, 8.315]	2.715 [2.572, 2.875]
select_m1_s1	missing_0.5	1000	3.139 [2.978, 3.312]	7.519 [7.082, 8.000]	2.641 [2.492, 2.791]
select_m1_s2	missing_0.5	1000	3.219 [3.054, 3.396]	7.868 [7.374, 8.375]	2.773 [2.621, 2.936]
cp	missing_0.7	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	missing_0.7	1000	11.439 [10.974, 11.943]	28.892 [27.733, 30.111]	11.418 [10.952, 11.921]
extra_sft_pilot_s0	missing_0.7	1000	3.724 [3.546, 3.914]	8.754 [8.297, 9.238]	3.143 [2.976, 3.319]
extra_sft_pilot_s1	missing_0.7	1000	3.748 [3.573, 3.935]	8.831 [8.369, 9.318]	3.181 [3.013, 3.357]
extra_sft_pilot_s2	missing_0.7	1000	3.822 [3.623, 4.020]	9.156 [8.656, 9.724]	3.261 [3.077, 3.447]
filtered_sft_pilot_s0	missing_0.7	1000	3.896 [3.699, 4.104]	9.097 [8.602, 9.642]	3.348 [3.165, 3.544]
filtered_sft_pilot_s1	missing_0.7	1000	3.915 [3.716, 4.120]	9.100 [8.595, 9.637]	3.303 [3.118, 3.494]
filtered_sft_pilot_s2	missing_0.7	1000	4.002 [3.793, 4.212]	9.415 [8.910, 9.972]	3.400 [3.217, 3.579]
m0_pilot_s0	missing_0.7	1000	5.492 [5.210, 5.794]	12.685 [12.019, 13.383]	4.583 [4.320, 4.865]
m0_pilot_s1	missing_0.7	1000	6.126 [5.817, 6.444]	14.154 [13.464, 14.912]	5.235 [4.951, 5.539]
m0_pilot_s2	missing_0.7	1000	5.543 [5.284, 5.821]	12.719 [12.092, 13.394]	4.674 [4.430, 4.926]

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s
m1_pilot_s0	missing_0.7	1000	3.732 [3.550, 3.927]	8.801 [8.331, 9.301]	3.158 [2.999, 3.337]
m1_pilot_s1	missing_0.7	1000	3.703 [3.518, 3.902]	8.784 [8.294, 9.319]	3.144 [2.972, 3.332]
m1_pilot_s2	missing_0.7	1000	3.787 [3.588, 3.989]	8.855 [8.352, 9.383]	3.220 [3.037, 3.401]
mind_pilot_s0	missing_0.7	1000	5.939 [5.655, 6.245]	13.831 [13.180, 14.542]	5.007 [4.744, 5.295]
mind_pilot_s1	missing_0.7	1000	6.664 [6.371, 6.984]	15.590 [14.854, 16.312]	5.519 [5.252, 5.787]
mind_pilot_s2	missing_0.7	1000	5.892 [5.618, 6.153]	13.588 [12.923, 14.271]	4.953 [4.704, 5.197]
oracle_log	missing_0.7	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	missing_0.7	1000	5.226 [4.963, 5.478]	12.712 [12.083, 13.357]	4.680 [4.428, 4.924]
rl_pilot_s1	missing_0.7	1000	5.305 [5.010, 5.585]	12.650 [11.989, 13.351]	4.755 [4.471, 5.036]
rl_pilot_s2	missing_0.7	1000	4.642 [4.393, 4.887]	11.205 [10.601, 11.888]	4.101 [3.875, 4.328]
select_m1_s0	missing_0.7	1000	3.678 [3.490, 3.872]	8.625 [8.132, 9.169]	3.151 [2.975, 3.339]
select_m1_s1	missing_0.7	1000	3.603 [3.402, 3.811]	8.471 [7.978, 8.991]	3.067 [2.886, 3.261]
select_m1_s2	missing_0.7	1000	3.793 [3.590, 4.007]	8.866 [8.364, 9.450]	3.290 [3.097, 3.486]
cp	noise_0.1	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	noise_0.1	1000	11.204 [10.773, 11.642]	28.549 [27.381, 29.701]	11.183 [10.753, 11.620]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s
extra_sft_pilot_s0	noise_0.1	1000	3.599 [3.447, 3.757]	8.283 [7.858, 8.729]	2.985 [2.850, 3.134]
extra_sft_pilot_s1	noise_0.1	1000	3.665 [3.495, 3.838]	8.528 [8.083, 9.024]	3.072 [2.918, 3.220]
extra_sft_pilot_s2	noise_0.1	1000	3.841 [3.666, 4.036]	9.043 [8.526, 9.563]	3.236 [3.074, 3.423]
filtered_sft_pilot_s0	noise_0.1	1000	3.688 [3.528, 3.848]	8.597 [8.169, 9.065]	3.086 [2.938, 3.233]
filtered_sft_pilot_s1	noise_0.1	1000	3.809 [3.631, 3.986]	8.850 [8.393, 9.351]	3.214 [3.051, 3.375]
filtered_sft_pilot_s2	noise_0.1	1000	3.866 [3.693, 4.059]	9.045 [8.547, 9.547]	3.284 [3.118, 3.468]
m0_pilot_s0	noise_0.1	1000	5.818 [5.568, 6.070]	12.890 [12.265, 13.566]	4.995 [4.758, 5.235]
m0_pilot_s1	noise_0.1	1000	5.861 [5.615, 6.110]	13.224 [12.574, 13.916]	5.082 [4.854, 5.316]
m0_pilot_s2	noise_0.1	1000	5.711 [5.467, 5.955]	13.008 [12.386, 13.674]	4.955 [4.726, 5.183]
m1_pilot_s0	noise_0.1	1000	3.760 [3.607, 3.931]	8.679 [8.232, 9.172]	3.112 [2.975, 3.272]
m1_pilot_s1	noise_0.1	1000	3.774 [3.600, 3.954]	8.819 [8.355, 9.333]	3.165 [3.004, 3.329]
m1_pilot_s2	noise_0.1	1000	3.804 [3.631, 3.995]	8.981 [8.493, 9.482]	3.203 [3.045, 3.380]
mind_pilot_s0	noise_0.1	1000	6.204 [5.941, 6.477]	13.755 [13.121, 14.457]	5.397 [5.150, 5.651]
mind_pilot_s1	noise_0.1	1000	6.271 [6.018, 6.536]	14.276 [13.650, 14.947]	5.324 [5.086, 5.564]
mind_pilot_s2	noise_0.1	1000	5.479 [5.250, 5.702]	12.316 [11.732, 12.944]	4.621 [4.409, 4.830]

method	condition	n	joint minADE@6 8s	joint minFDE@6 8s	marginal minADE 8s
oracle_log	noise_0.1	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	noise_0.1	1000	5.204 [5.006, 5.415]	12.252 [11.688, 12.836]	4.610 [4.421, 4.815]
rl_pilot_s1	noise_0.1	1000	5.359 [5.137, 5.584]	12.787 [12.185, 13.423]	4.783 [4.568, 5.010]
rl_pilot_s2	noise_0.1	1000	4.730 [4.543, 4.945]	11.384 [10.844, 11.929]	4.122 [3.936, 4.318]
select_m1_s0	noise_0.1	1000	3.559 [3.393, 3.734]	8.193 [7.742, 8.674]	3.028 [2.876, 3.188]
select_m1_s1	noise_0.1	1000	3.537 [3.374, 3.708]	8.133 [7.705, 8.615]	3.022 [2.872, 3.186]
select_m1_s2	noise_0.1	1000	3.646 [3.474, 3.839]	8.569 [8.079, 9.067]	3.109 [2.949, 3.288]
cp	noise_0.3	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	noise_0.3	1000	11.886 [11.463, 12.311]	29.571 [28.445, 30.691]	11.869 [11.443, 12.282]
extra_sft_pilot_s0	noise_0.3	1000	4.641 [4.473, 4.833]	10.207 [9.726, 10.717]	3.826 [3.672, 4.004]
extra_sft_pilot_s1	noise_0.3	1000	4.469 [4.293, 4.668]	9.947 [9.472, 10.452]	3.712 [3.553, 3.891]
extra_sft_pilot_s2	noise_0.3	1000	4.421 [4.250, 4.613]	9.814 [9.357, 10.322]	3.644 [3.490, 3.815]
filtered_sft_pilot_s0	noise_0.3	1000	4.604 [4.433, 4.793]	10.193 [9.714, 10.697]	3.825 [3.667, 4.004]
filtered_sft_pilot_s1	noise_0.3	1000	4.685 [4.514, 4.864]	10.404 [9.926, 10.913]	3.919 [3.764, 4.084]
filtered_sft_pilot_s2	noise_0.3	1000	4.579 [4.397, 4.778]	10.309 [9.830, 10.847]	3.807 [3.645, 3.994]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s
m0_pilot_s0	noise_0.3	1000	12.523 [12.120, 12.913]	26.615 [25.713, 27.629]	10.632 [10.264, 10.988]
m0_pilot_s1	noise_0.3	1000	11.495 [11.120, 11.878]	25.197 [24.250, 26.155]	10.063 [9.701, 10.422]
m0_pilot_s2	noise_0.3	1000	10.995 [10.634, 11.388]	24.630 [23.736, 25.574]	9.661 [9.318, 10.010]
m1_pilot_s0	noise_0.3	1000	4.522 [4.364, 4.692]	9.844 [9.417, 10.300]	3.757 [3.617, 3.913]
m1_pilot_s1	noise_0.3	1000	4.440 [4.282, 4.613]	9.742 [9.301, 10.201]	3.659 [3.514, 3.814]
m1_pilot_s2	noise_0.3	1000	4.510 [4.329, 4.701]	10.119 [9.628, 10.630]	3.756 [3.594, 3.933]
mind_pilot_s0	noise_0.3	1000	12.604 [12.213, 12.989]	26.453 [25.510, 27.422]	10.825 [10.450, 11.193]
mind_pilot_s1	noise_0.3	1000	12.612 [12.206, 13.022]	27.604 [26.635, 28.630]	10.890 [10.513, 11.274]
mind_pilot_s2	noise_0.3	1000	10.602 [10.260, 10.945]	22.855 [22.036, 23.792]	9.323 [8.996, 9.646]
oracle_log	noise_0.3	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	noise_0.3	1000	6.772 [6.544, 7.002]	15.145 [14.516, 15.793]	5.985 [5.761, 6.215]
rl_pilot_s1	noise_0.3	1000	6.726 [6.478, 6.972]	15.397 [14.734, 16.063]	5.995 [5.756, 6.241]
rl_pilot_s2	noise_0.3	1000	5.973 [5.764, 6.189]	13.750 [13.173, 14.354]	5.235 [5.037, 5.444]
select_m1_s0	noise_0.3	1000	4.241 [4.077, 4.414]	9.362 [8.908, 9.843]	3.530 [3.381, 3.689]
select_m1_s1	noise_0.3	1000	4.184 [4.012, 4.356]	9.394 [8.928, 9.880]	3.483 [3.326, 3.645]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s
select_m1_s2	noise_0.3	1000	4.141 [3.976, 4.318]	9.280 [8.814, 9.762]	3.480 [3.326, 3.645]
cp	noise_0.5	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	noise_0.5	1000	12.769 [12.361, 13.172]	30.762 [29.616, 31.893]	12.747 [12.337, 13.146]
extra_sft_pilot_s0	noise_0.5	1000	5.775 [5.609, 5.950]	12.132 [11.661, 12.618]	4.739 [4.585, 4.902]
extra_sft_pilot_s1	noise_0.5	1000	5.540 [5.353, 5.727]	11.615 [11.124, 12.155]	4.531 [4.370, 4.707]
extra_sft_pilot_s2	noise_0.5	1000	5.289 [5.114, 5.464]	11.404 [10.943, 11.890]	4.364 [4.216, 4.522]
filtered_sft_pilot_s0	noise_0.5	1000	5.752 [5.589, 5.932]	12.021 [11.560, 12.475]	4.777 [4.625, 4.936]
filtered_sft_pilot_s1	noise_0.5	1000	5.792 [5.597, 5.980]	12.200 [11.717, 12.722]	4.849 [4.674, 5.017]
filtered_sft_pilot_s2	noise_0.5	1000	5.530 [5.341, 5.715]	11.785 [11.290, 12.304]	4.600 [4.433, 4.772]
m0_pilot_s0	noise_0.5	1000	15.068 [14.648, 15.465]	32.524 [31.500, 33.563]	12.675 [12.296, 13.054]
m0_pilot_s1	noise_0.5	1000	13.748 [13.354, 14.148]	30.197 [29.128, 31.294]	11.992 [11.624, 12.371]
m0_pilot_s2	noise_0.5	1000	14.572 [14.154, 14.974]	32.429 [31.337, 33.508]	12.578 [12.198, 12.947]
m1_pilot_s0	noise_0.5	1000	5.719 [5.543, 5.901]	11.975 [11.500, 12.493]	4.713 [4.553, 4.877]
m1_pilot_s1	noise_0.5	1000	5.492 [5.311, 5.671]	11.499 [11.054, 11.961]	4.445 [4.291, 4.598]
m1_pilot_s2	noise_0.5	1000	5.483 [5.309, 5.660]	11.637 [11.176, 12.124]	4.510 [4.353, 4.678]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s
mind_pilot_s0	noise_0.5	1000	15.917 [15.506, 16.347]	33.984 [33.016, 35.012]	13.375 [12.982, 13.772]
mind_pilot_s1	noise_0.5	1000	15.059 [14.635, 15.488]	33.131 [32.073, 34.250]	12.924 [12.532, 13.314]
mind_pilot_s2	noise_0.5	1000	14.430 [14.015, 14.854]	31.330 [30.350, 32.432]	12.650 [12.254, 13.045]
oracle_log	noise_0.5	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	noise_0.5	1000	8.006 [7.762, 8.257]	17.317 [16.627, 17.992]	7.074 [6.831, 7.317]
rl_pilot_s1	noise_0.5	1000	7.893 [7.652, 8.137]	17.464 [16.816, 18.143]	7.035 [6.798, 7.271]
rl_pilot_s2	noise_0.5	1000	7.159 [6.932, 7.395]	15.994 [15.386, 16.603]	6.236 [6.016, 6.461]
select_m1_s0	noise_0.5	1000	5.248 [5.073, 5.427]	10.982 [10.511, 11.486]	4.311 [4.153, 4.473]
select_m1_s1	noise_0.5	1000	5.242 [5.057, 5.442]	11.058 [10.567, 11.597]	4.356 [4.187, 4.538]
select_m1_s2	noise_0.5	1000	5.018 [4.840, 5.194]	10.647 [10.168, 11.125]	4.123 [3.966, 4.281]
cp	noise_1	1000	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.915 [24.746, 27.150]
cv	noise_1	1000	15.875 [15.422, 16.288]	35.628 [34.571, 36.690]	15.857 [15.403, 16.272]
extra_sft_pilot_s0	noise_1	1000	9.147 [8.883, 9.421]	18.345 [17.763, 18.957]	7.563 [7.313, 7.816]
extra_sft_pilot_s1	noise_1	1000	9.191 [8.924, 9.478]	18.289 [17.683, 18.911]	7.678 [7.427, 7.947]
extra_sft_pilot_s2	noise_1	1000	9.040 [8.774, 9.306]	18.479 [17.872, 19.095]	7.609 [7.364, 7.854]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE8s
filtered_sft_pilot_s0	noise_1	1000	9.406 [9.120, 9.682]	18.839 [18.227, 19.486]	7.946 [7.671, 8.210]
filtered_sft_pilot_s1	noise_1	1000	9.485 [9.185, 9.756]	19.016 [18.393, 19.667]	8.026 [7.751, 8.282]
filtered_sft_pilot_s2	noise_1	1000	9.301 [9.020, 9.575]	18.807 [18.148, 19.447]	7.931 [7.657, 8.194]
m0_pilot_s0	noise_1	1000	18.556 [18.122, 19.007]	40.108 [39.097, 41.230]	15.297 [14.881, 15.692]
m0_pilot_s1	noise_1	1000	16.699 [16.232, 17.156]	36.794 [35.643, 38.019]	14.930 [14.492, 15.363]
m0_pilot_s2	noise_1	1000	19.518 [19.013, 20.026]	43.152 [41.933, 44.496]	16.624 [16.149, 17.101]
m1_pilot_s0	noise_1	1000	9.219 [8.937, 9.490]	18.418 [17.770, 19.066]	7.653 [7.392, 7.916]
m1_pilot_s1	noise_1	1000	9.110 [8.846, 9.395]	18.141 [17.551, 18.763]	7.636 [7.386, 7.890]
m1_pilot_s2	noise_1	1000	9.047 [8.783, 9.311]	18.374 [17.773, 18.969]	7.614 [7.359, 7.872]
mind_pilot_s0	noise_1	1000	19.273 [18.829, 19.708]	41.320 [40.196, 42.399]	15.833 [15.424, 16.218]
mind_pilot_s1	noise_1	1000	18.293 [17.841, 18.750]	39.742 [38.545, 41.014]	15.844 [15.409, 16.266]
mind_pilot_s2	noise_1	1000	18.986 [18.490, 19.465]	40.479 [39.347, 41.696]	16.516 [16.053, 16.963]
oracle_log	noise_1	1000	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
rl_pilot_s0	noise_1	1000	11.560 [11.221, 11.893]	23.837 [23.036, 24.648]	10.300 [9.970, 10.630]
rl_pilot_s1	noise_1	1000	11.249 [10.910, 11.582]	23.366 [22.584, 24.135]	9.989 [9.668, 10.310]

method	condition	n	joint minADE@68s	joint minFDE@68s	marginal minADE 8s
rl_pilot_s2	noise_1	1000	10.867 [10.539, 11.183]	22.933 [22.169, 23.708]	9.638 [9.325, 9.934]
select_m1_s0	noise_1	1000	8.314 [8.049, 8.587]	16.530 [15.932, 17.109]	6.857 [6.615, 7.107]
select_m1_s1	noise_1	1000	8.406 [8.136, 8.682]	16.929 [16.311, 17.539]	7.003 [6.748, 7.258]
select_m1_s2	noise_1	1000	8.426 [8.144, 8.702]	17.302 [16.683, 17.899]	7.054 [6.799, 7.311]

Paired differences vs `m0_pilot_s0` (method − reference; group-level bootstrap 95% CI; groups = heuristic source-segment components, D-025):

method	condition	Δ joint minADE@68s	rel. Δ	Δ overlap rate (pp)	n
cp	clean	+22.421 [+21.264, +23.663]	+637.7% [+604.8%, +673.0%]	-14.96 [-16.38, -13.55]	1000
cp	compound_0	+15.829 [+14.667, +17.113]	+156.6% [+145.1%, +169.3%]	-15.20 [-16.60, -13.76]	1000
cp	missing_0.1	+22.381 [+21.233, +23.612]	+629.4% [+597.2%, +664.1%]	-16.06 [-17.55, -14.66]	1000
cp	missing_0.3	+21.959 [+20.806, +23.169]	+552.0% [+523.0%, +582.4%]	-18.74 [-20.43, -17.12]	1000
cp	missing_0.5	+21.566 [+20.406, +22.779]	+493.4% [+466.9%, +521.2%]	-20.88 [-22.54, -19.17]	1000
cp	missing_0.7	+20.444 [+19.318, +21.658]	+372.2% [+351.7%, +394.3%]	-21.78 [-23.49, -20.11]	1000
cp	noise_0.1	+20.119 [+18.972, +21.370]	+345.8% [+326.1%, +367.3%]	-17.02 [-18.70, -15.48]	1000
cp	noise_0.3	+13.414 [+12.179, +14.676]	+107.1% [+97.3%, +117.2%]	-12.87 [-14.38, -11.53]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
cp	noise_0.5	+10.869 [+9.661, +12.153]	+72.1% [+64.1%, +80.7%]	-12.96 [-14.22, -11.67]	1000
cp	noise_1	+7.380 [+6.115, +8.695]	+39.8% [+33.0%, +46.9%]	-12.77 [-14.16, -11.50]	1000
cv	clean	+7.598 [+7.149, +8.030]	+216.1% [+203.3%, +228.4%]	+2.29 [-0.02, +4.69]	1000
cv	compound_0	+1.943 [+1.370, +2.477]	+19.2% [+13.6%, +24.5%]	+0.15 [-2.16, +2.54]	1000
cv	missing_0.1	+7.543 [+7.095, +7.979]	+212.1% [+199.5%, +224.4%]	+1.19 [-1.27, +3.77]	1000
cv	missing_0.3	+7.082 [+6.634, +7.510]	+178.0% [+166.8%, +188.8%]	-0.99 [-3.48, +1.52]	1000
cv	missing_0.5	+6.674 [+6.236, +7.105]	+152.7% [+142.7%, +162.6%]	-3.83 [-6.24, -1.29]	1000
cv	missing_0.7	+5.947 [+5.502, +6.393]	+108.3% [+100.2%, +116.4%]	-4.53 [-6.65, -2.39]	1000
cv	noise_0.1	+5.386 [+4.892, +5.882]	+92.6% [+84.1%, +101.1%]	+0.43 [-2.25, +2.95]	1000
cv	noise_0.3	-0.637 [-1.253, -0.047]	-5.1% [-10.0%, -0.4%]	+4.58 [+2.11, +7.09]	1000
cv	noise_0.5	-2.299 [-2.910, -1.664]	-15.3% [-19.3%, -11.0%]	+4.20 [+1.94, +6.56]	1000
cv	noise_1	-2.681 [-3.259, -2.067]	-14.5% [-17.6%, -11.1%]	+1.67 [-0.50, +3.83]	1000
extra_sft_pilot_s0	clean	-0.392 [-0.527, -0.264]	-11.2% [-15.0%, -7.5%]	+0.45 [-1.00, +1.94]	1000
extra_sft_pilot_s0	compound_0	-5.121 [-5.463, -4.787]	-50.7% [-54.0%, -47.4%]	+2.39 [+0.59, +4.13]	1000
extra_sft_pilot_s0	missing_0.1	-0.401 [-0.535, -0.275]	-11.3% [-15.1%, -7.7%]	-0.72 [-2.14, +0.62]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
extra_sft_pilot_s0	missing_0.3	-0.846 [-0.998, -0.701]	-21.3% [-25.1%, -17.6%]	-3.59 [-5.28, -1.98]	1000
extra_sft_pilot_s0	missing_0.5	-1.050 [-1.211, -0.891]	-24.0% [-27.7%, -20.4%]	-5.78 [-7.41, -4.13]	1000
extra_sft_pilot_s0	missing_0.7	-1.768 [-2.009, -1.534]	-32.2% [-36.6%, -27.9%]	-4.41 [-6.31, -2.61]	1000
extra_sft_pilot_s0	noise_0.1	-2.219 [-2.440, -1.990]	-38.1% [-41.9%, -34.2%]	+1.25 [-0.65, +2.96]	1000
extra_sft_pilot_s0	noise_0.3	-7.882 [-8.273, -7.490]	-62.9% [-66.1%, -59.8%]	+5.23 [+3.42, +7.00]	1000
extra_sft_pilot_s0	noise_0.5	-9.293 [-9.705, -8.877]	-61.7% [-64.4%, -58.9%]	+5.40 [+3.73, +7.02]	1000
extra_sft_pilot_s0	noise_1	-9.409 [-9.831, -8.974]	-50.7% [-53.0%, -48.4%]	+2.21 [+0.80, +3.63]	1000
extra_sft_pilot_s1	clean	-0.438 [-0.573, -0.313]	-12.5% [-16.3%, -8.9%]	+0.80 [-0.65, +2.18]	1000
extra_sft_pilot_s1	compound_0	-5.195 [-5.542, -4.849]	-51.4% [-54.8%, -48.0%]	+3.08 [+1.19, +4.81]	1000
extra_sft_pilot_s1	missing_0.1	-0.422 [-0.568, -0.287]	-11.9% [-16.0%, -8.1%]	-0.38 [-1.87, +1.10]	1000
extra_sft_pilot_s1	missing_0.3	-0.836 [-0.993, -0.687]	-21.0% [-25.0%, -17.3%]	-2.77 [-4.41, -1.12]	1000
extra_sft_pilot_s1	missing_0.5	-1.065 [-1.229, -0.905]	-24.4% [-28.1%, -20.7%]	-4.30 [-6.17, -2.34]	1000
extra_sft_pilot_s1	missing_0.7	-1.745 [-1.989, -1.503]	-31.8% [-36.2%, -27.4%]	-4.46 [-6.36, -2.66]	1000
extra_sft_pilot_s1	noise_0.1	-2.153 [-2.384, -1.922]	-37.0% [-41.0%, -33.0%]	+1.69 [-0.30, +3.60]	1000
extra_sft_pilot_s1	noise_0.3	-8.054 [-8.435, -7.664]	-64.3% [-67.4%, -61.2%]	+6.72 [+4.86, +8.53]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
extra_sft_pilot_s1	noise_0.5	-9.528 [-9.952, -9.119]	-63.2% [-66.0%, -60.5%]	+4.86 [+3.25, +6.45]	1000
extra_sft_pilot_s1	noise_1	-9.365 [-9.812, -8.906]	-50.5% [-52.9%, -48.0%]	+2.07 [+0.67, +3.56]	1000
extra_sft_pilot_s2	clean	-0.359 [-0.506, -0.213]	-10.2% [-14.4%, -6.1%]	+0.65 [-0.77, +2.06]	1000
extra_sft_pilot_s2	compound_0	-5.161 [-5.513, -4.821]	-51.1% [-54.5%, -47.7%]	+3.26 [+1.40, +5.04]	1000
extra_sft_pilot_s2	missing_0.1	-0.410 [-0.554, -0.274]	-11.5% [-15.6%, -7.7%]	-0.15 [-1.67, +1.32]	1000
extra_sft_pilot_s2	missing_0.3	-0.826 [-0.992, -0.655]	-20.8% [-24.9%, -16.5%]	-2.64 [-4.41, -0.90]	1000
extra_sft_pilot_s2	missing_0.5	-0.984 [-1.162, -0.810]	-22.5% [-26.6%, -18.5%]	-4.43 [-6.21, -2.66]	1000
extra_sft_pilot_s2	missing_0.7	-1.670 [-1.904, -1.428]	-30.4% [-34.7%, -26.0%]	-4.38 [-6.13, -2.58]	1000
extra_sft_pilot_s2	noise_0.1	-1.976 [-2.210, -1.736]	-34.0% [-38.0%, -29.8%]	+1.54 [-0.37, +3.49]	1000
extra_sft_pilot_s2	noise_0.3	-8.102 [-8.490, -7.709]	-64.7% [-67.8%, -61.6%]	+6.74 [+4.74, +8.58]	1000
extra_sft_pilot_s2	noise_0.5	-9.779 [-10.187, -9.362]	-64.9% [-67.6%, -62.1%]	+6.34 [+4.63, +8.06]	1000
extra_sft_pilot_s2	noise_1	-9.516 [-9.973, -9.082]	-51.3% [-53.7%, -48.9%]	+2.24 [+0.77, +3.75]	1000
filtered_sft_pilot_s0	clean	-0.338 [-0.469, -0.214]	-9.6% [-13.3%, -6.1%]	-0.99 [-2.41, +0.40]	1000
filtered_sft_pilot_s0	compound_0	-4.948 [-5.282, -4.599]	-49.0% [-52.3%, -45.5%]	+1.54 [-0.33, +3.31]	1000
filtered_sft_pilot_s0	missing_0.1	-0.379 [-0.525, -0.246]	-10.7% [-14.8%, -6.9%]	-1.77 [-3.23, -0.35]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
filtered_sft_pilot_s0	missing_0.3	-0.711 [-0.868, -0.561]	-17.9% [-21.8%, -14.1%]	-4.90 [-6.57, -3.38]	1000
filtered_sft_pilot_s0	missing_0.5	-1.044 [-1.214, -0.881]	-23.9% [-27.8%, -20.2%]	-6.82 [-8.58, -5.03]	1000
filtered_sft_pilot_s0	missing_0.7	-1.596 [-1.816, -1.376]	-29.1% [-33.1%, -25.0%]	-7.02 [-8.80, -5.27]	1000
filtered_sft_pilot_s0	noise_0.1	-2.129 [-2.341, -1.913]	-36.6% [-40.2%, -32.9%]	-0.48 [-2.34, +1.25]	1000
filtered_sft_pilot_s0	noise_0.3	-7.919 [-8.311, -7.520]	-63.2% [-66.4%, -60.1%]	+5.43 [+3.53, +7.20]	1000
filtered_sft_pilot_s0	noise_0.5	-9.315 [-9.720, -8.898]	-61.8% [-64.5%, -59.1%]	+4.81 [+3.10, +6.44]	1000
filtered_sft_pilot_s0	noise_1	-9.150 [-9.588, -8.700]	-49.3% [-51.7%, -46.9%]	+2.22 [+0.67, +3.74]	1000
filtered_sft_pilot_s1	clean	-0.336 [-0.473, -0.207]	-9.5% [-13.4%, -5.9%]	-0.30 [-1.79, +1.14]	1000
filtered_sft_pilot_s1	compound_0	-5.059 [-5.391, -4.721]	-50.1% [-53.3%, -46.7%]	+2.94 [+1.09, +4.77]	1000
filtered_sft_pilot_s1	missing_0.1	-0.373 [-0.514, -0.243]	-10.5% [-14.5%, -6.8%]	-2.02 [-3.56, -0.57]	1000
filtered_sft_pilot_s1	missing_0.3	-0.812 [-0.972, -0.668]	-20.4% [-24.4%, -16.8%]	-4.65 [-6.40, -2.96]	1000
filtered_sft_pilot_s1	missing_0.5	-1.018 [-1.183, -0.859]	-23.3% [-27.1%, -19.6%]	-5.82 [-7.77, -3.94]	1000
filtered_sft_pilot_s1	missing_0.7	-1.577 [-1.828, -1.352]	-28.7% [-33.3%, -24.6%]	-6.15 [-8.00, -4.26]	1000
filtered_sft_pilot_s1	noise_0.1	-2.008 [-2.245, -1.772]	-34.5% [-38.6%, -30.5%]	+0.59 [-1.45, +2.48]	1000
filtered_sft_pilot_s1	noise_0.3	-7.838 [-8.214, -7.466]	-62.6% [-65.6%, -59.6%]	+5.35 [+3.37, +7.17]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
filtered_sft_pilot_s1	noise_0.5	-9.276 [-9.714, -8.866]	-61.6% [-64.5%, -58.8%]	+4.50 [+2.84, +6.14]	1000
filtered_sft_pilot_s1	noise_1	-9.071 [-9.555, -8.613]	-48.9% [-51.5%, -46.4%]	+2.59 [+1.08, +4.05]	1000
filtered_sft_pilot_s2	clean	-0.297 [-0.441, -0.153]	-8.5% [-12.5%, -4.4%]	+0.50 [-1.03, +1.94]	1000
filtered_sft_pilot_s2	compound_0	-5.084 [-5.428, -4.741]	-50.3% [-53.7%, -46.9%]	+2.07 [+0.33, +3.89]	1000
filtered_sft_pilot_s2	missing_0.1	-0.287 [-0.428, -0.146]	-8.1% [-12.0%, -4.1%]	-0.67 [-2.24, +0.85]	1000
filtered_sft_pilot_s2	missing_0.3	-0.725 [-0.883, -0.572]	-18.2% [-22.2%, -14.4%]	-3.16 [-4.93, -1.49]	1000
filtered_sft_pilot_s2	missing_0.5	-0.941 [-1.112, -0.770]	-21.5% [-25.4%, -17.6%]	-6.03 [-7.81, -4.28]	1000
filtered_sft_pilot_s2	missing_0.7	-1.490 [-1.728, -1.251]	-27.1% [-31.5%, -22.8%]	-5.97 [-7.62, -4.20]	1000
filtered_sft_pilot_s2	noise_0.1	-1.952 [-2.188, -1.712]	-33.6% [-37.6%, -29.4%]	+0.22 [-1.71, +2.05]	1000
filtered_sft_pilot_s2	noise_0.3	-7.944 [-8.334, -7.555]	-63.4% [-66.5%, -60.3%]	+4.78 [+2.92, +6.54]	1000
filtered_sft_pilot_s2	noise_0.5	-9.538 [-9.944, -9.119]	-63.3% [-66.0%, -60.5%]	+4.85 [+3.18, +6.45]	1000
filtered_sft_pilot_s2	noise_1	-9.255 [-9.696, -8.805]	-49.9% [-52.3%, -47.4%]	+2.32 [+0.92, +3.82]	1000
m0_pilot_s1	clean	-0.091 [-0.226, +0.042]	-2.6% [-6.4%, +1.2%]	+1.42 [-0.10, +2.81]	1000
m0_pilot_s1	compound_0	+0.471 [+0.125, +0.816]	+4.7% [+1.2%, +8.1%]	-2.66 [-4.15, -1.27]	1000
m0_pilot_s1	missing_0.1	-0.067 [-0.200, +0.071]	-1.9% [-5.6%, +2.0%]	+0.50 [-1.10, +2.04]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m0_pilot_s1	missing_0.3	-0.072 [-0.250, +0.099]	-1.8% [-6.3%, +2.5%]	+0.32 [-1.37, +2.04]	1000
m0_pilot_s1	missing_0.5	+0.215 [+0.021, +0.416]	+4.9% [+0.5%, +9.5%]	-0.45 [-2.29, +1.32]	1000
m0_pilot_s1	missing_0.7	+0.634 [+0.379, +0.895]	+11.5% [+6.9%, +16.3%]	+0.74 [-1.04, +2.57]	1000
m0_pilot_s1	noise_0.1	+0.044 [-0.198, +0.290]	+0.8% [-3.4%, +5.0%]	+0.57 [-1.30, +2.28]	1000
m0_pilot_s1	noise_0.3	-1.028 [-1.413, -0.625]	-8.2% [-11.3%, -5.0%]	+0.37 [-1.20, +1.81]	1000
m0_pilot_s1	noise_0.5	-1.320 [-1.782, -0.864]	-8.8% [-11.8%, -5.7%]	-0.74 [-2.11, +0.70]	1000
m0_pilot_s1	noise_1	-1.857 [-2.370, -1.337]	-10.0% [-12.8%, -7.2%]	-0.82 [-2.23, +0.52]	1000
m0_pilot_s2	clean	+0.025 [-0.119, +0.171]	+0.7% [-3.4%, +4.9%]	+1.59 [+0.10, +3.08]	1000
m0_pilot_s2	compound_0	-0.134 [-0.503, +0.249]	-1.3% [-5.0%, +2.5%]	-0.05 [-1.79, +1.61]	1000
m0_pilot_s2	missing_0.1	+0.044 [-0.108, +0.178]	+1.2% [-3.0%, +5.0%]	+2.01 [+0.42, +3.50]	1000
m0_pilot_s2	missing_0.3	-0.067 [-0.240, +0.100]	-1.7% [-6.0%, +2.5%]	+0.08 [-1.64, +1.79]	1000
m0_pilot_s2	missing_0.5	-0.079 [-0.258, +0.108]	-1.8% [-5.9%, +2.5%]	+0.69 [-1.20, +2.47]	1000
m0_pilot_s2	missing_0.7	+0.051 [-0.199, +0.298]	+0.9% [-3.6%, +5.4%]	+3.34 [+1.35, +5.22]	1000
m0_pilot_s2	noise_0.1	-0.107 [-0.346, +0.133]	-1.8% [-6.0%, +2.3%]	+0.67 [-1.27, +2.50]	1000
m0_pilot_s2	noise_0.3	-1.528 [-1.948, -1.100]	-12.2% [-15.6%, -8.8%]	+2.11 [+0.45, +3.83]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m0_pilot_s2	noise_0.5	-0.496 [-0.977, -0.025]	-3.3% [-6.5%, -0.2%]	+0.30 [-1.15, +1.70]	1000
m0_pilot_s2	noise_1	+0.962 [+0.401, +1.507]	+5.2% [+2.2%, +8.1%]	-0.32 [-1.64, +1.04]	1000
m1_pilot_s0	clean	-0.315 [-0.457, -0.185]	-9.0% [-13.0%, -5.3%]	-0.23 [-1.64, +1.21]	1000
m1_pilot_s0	compound_0	-5.103 [-5.446, -4.759]	-50.5% [-53.9%, -47.1%]	+1.92 [+0.17, +3.71]	1000
m1_pilot_s0	missing_0.1	-0.357 [-0.480, -0.233]	-10.0% [-13.5%, -6.5%]	-1.05 [-2.50, +0.37]	1000
m1_pilot_s0	missing_0.3	-0.775 [-0.930, -0.628]	-19.5% [-23.4%, -15.8%]	-4.30 [-5.91, -2.76]	1000
m1_pilot_s0	missing_0.5	-1.072 [-1.233, -0.920]	-24.5% [-28.2%, -21.0%]	-6.64 [-8.39, -4.96]	1000
m1_pilot_s0	missing_0.7	-1.760 [-2.007, -1.531]	-32.1% [-36.5%, -27.9%]	-4.93 [-6.78, -3.19]	1000
m1_pilot_s0	noise_0.1	-2.058 [-2.277, -1.837]	-35.4% [-39.1%, -31.6%]	+0.48 [-1.44, +2.27]	1000
m1_pilot_s0	noise_0.3	-8.001 [-8.385, -7.611]	-63.9% [-67.0%, -60.8%]	+5.42 [+3.47, +7.25]	1000
m1_pilot_s0	noise_0.5	-9.349 [-9.756, -8.937]	-62.0% [-64.7%, -59.3%]	+4.50 [+2.83, +6.14]	1000
m1_pilot_s0	noise_1	-9.337 [-9.780, -8.878]	-50.3% [-52.7%, -47.8%]	+2.59 [+1.11, +4.07]	1000
m1_pilot_s1	clean	-0.373 [-0.508, -0.240]	-10.6% [-14.4%, -6.8%]	+1.92 [+0.42, +3.38]	1000
m1_pilot_s1	compound_0	-5.198 [-5.524, -4.873]	-51.4% [-54.7%, -48.2%]	+4.06 [+2.14, +5.90]	1000
m1_pilot_s1	missing_0.1	-0.411 [-0.558, -0.281]	-11.6% [-15.7%, -7.9%]	+0.65 [-0.85, +2.11]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m1_pilot_s1	missing_0.3	-0.833 [-0.988, -0.684]	-20.9% [-24.8%, -17.2%]	-2.17 [-3.80, -0.64]	1000
m1_pilot_s1	missing_0.5	-1.024 [-1.183, -0.864]	-23.4% [-27.1%, -19.8%]	-3.28 [-5.18, -1.41]	1000
m1_pilot_s1	missing_0.7	-1.789 [-2.035, -1.551]	-32.6% [-37.0%, -28.2%]	-4.36 [-6.15, -2.66]	1000
m1_pilot_s1	noise_0.1	-2.043 [-2.284, -1.799]	-35.1% [-39.3%, -30.9%]	+3.88 [+1.87, +5.77]	1000
m1_pilot_s1	noise_0.3	-8.083 [-8.450, -7.698]	-64.5% [-67.5%, -61.5%]	+6.55 [+4.73, +8.36]	1000
m1_pilot_s1	noise_0.5	-9.576 [-9.996, -9.162]	-63.6% [-66.3%, -60.8%]	+5.27 [+3.61, +6.83]	1000
m1_pilot_s1	noise_1	-9.446 [-9.878, -8.993]	-50.9% [-53.2%, -48.5%]	+2.96 [+1.44, +4.44]	1000
m1_pilot_s2	clean	-0.265 [-0.414, -0.117]	-7.5% [-11.8%, -3.3%]	+0.94 [-0.52, +2.35]	1000
m1_pilot_s2	compound_0	-5.197 [-5.534, -4.854]	-51.4% [-54.8%, -48.0%]	+3.04 [+1.11, +4.85]	1000
m1_pilot_s2	missing_0.1	-0.401 [-0.539, -0.266]	-11.3% [-15.2%, -7.5%]	-0.23 [-1.75, +1.22]	1000
m1_pilot_s2	missing_0.3	-0.783 [-0.935, -0.623]	-19.7% [-23.5%, -15.7%]	-2.59 [-4.33, -0.87]	1000
m1_pilot_s2	missing_0.5	-1.055 [-1.228, -0.880]	-24.1% [-28.1%, -20.1%]	-5.23 [-6.96, -3.51]	1000
m1_pilot_s2	missing_0.7	-1.705 [-1.945, -1.469]	-31.1% [-35.4%, -26.8%]	-4.25 [-6.07, -2.45]	1000
m1_pilot_s2	noise_0.1	-2.014 [-2.257, -1.764]	-34.6% [-38.8%, -30.3%]	+1.05 [-0.79, +2.86]	1000
m1_pilot_s2	noise_0.3	-8.013 [-8.402, -7.609]	-64.0% [-67.1%, -60.8%]	+7.36 [+5.52, +9.23]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
m1_pilot_s2	noise_0.5	-9.585 [-9.990, -9.165]	-63.6% [-66.3%, -60.8%]	+5.87 [+4.13, +7.55]	1000
m1_pilot_s2	noise_1	-9.509 [-9.962, -9.074]	-51.2% [-53.7%, -48.9%]	+2.49 [+1.05, +3.97]	1000
mind_pilot_s0	clean	+0.174 [+0.045, +0.301]	+4.9% [+1.3%, +8.6%]	+3.56 [+2.04, +5.00]	1000
mind_pilot_s0	compound_0	+0.639 [+0.319, +0.952]	+6.3% [+3.2%, +9.4%]	-0.82 [-2.29, +0.60]	1000
mind_pilot_s0	missing_0.1	+0.108 [-0.026, +0.242]	+3.0% [-0.7%, +6.8%]	+3.23 [+1.67, +4.76]	1000
mind_pilot_s0	missing_0.3	+0.206 [+0.052, +0.355]	+5.2% [+1.3%, +8.9%]	+1.17 [-0.62, +2.88]	1000
mind_pilot_s0	missing_0.5	+0.262 [+0.087, +0.434]	+6.0% [+2.0%, +9.9%]	+1.30 [-0.30, +2.91]	1000
mind_pilot_s0	missing_0.7	+0.447 [+0.233, +0.654]	+8.1% [+4.2%, +11.9%]	+1.65 [+0.07, +3.21]	1000
mind_pilot_s0	noise_0.1	+0.387 [+0.182, +0.592]	+6.6% [+3.1%, +10.2%]	+1.15 [-0.47, +2.81]	1000
mind_pilot_s0	noise_0.3	+0.081 [-0.288, +0.448]	+0.6% [-2.3%, +3.6%]	+0.55 [-0.87, +2.04]	1000
mind_pilot_s0	noise_0.5	+0.849 [+0.411, +1.299]	+5.6% [+2.7%, +8.6%]	-0.77 [-2.07, +0.49]	1000
mind_pilot_s0	noise_1	+0.717 [+0.228, +1.198]	+3.9% [+1.2%, +6.5%]	+0.13 [-1.10, +1.39]	1000
mind_pilot_s1	clean	+0.060 [-0.074, +0.203]	+1.7% [-2.1%, +5.8%]	+3.31 [+1.79, +4.90]	1000
mind_pilot_s1	compound_0	+1.268 [+0.876, +1.638]	+12.5% [+8.7%, +16.2%]	-2.64 [-4.14, -1.17]	1000
mind_pilot_s1	missing_0.1	+0.139 [-0.011, +0.294]	+3.9% [-0.3%, +8.3%]	+3.04 [+1.37, +4.67]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
mind_pilot_s1	missing_0.3	+0.188 [+0.003, +0.378]	+4.7% [+0.1%, +9.5%]	+2.16 [+0.53, +3.87]	1000
mind_pilot_s1	missing_0.5	+0.560 [+0.360, +0.769]	+12.8% [+8.2%, +17.6%]	+0.85 [-1.04, +2.69]	1000
mind_pilot_s1	missing_0.7	+1.172 [+0.908, +1.461]	+21.3% [+16.5%, +26.6%]	-0.40 [-2.10, +1.37]	1000
mind_pilot_s1	noise_0.1	+0.453 [+0.203, +0.704]	+7.8% [+3.5%, +12.1%]	+2.07 [+0.32, +3.82]	1000
mind_pilot_s1	noise_0.3	+0.089 [-0.341, +0.502]	+0.7% [-2.7%, +4.0%]	-0.85 [-2.34, +0.57]	1000
mind_pilot_s1	noise_0.5	-0.009 [-0.487, +0.436]	-0.1% [-3.2%, +2.9%]	-1.50 [-2.83, -0.20]	1000
mind_pilot_s1	noise_1	-0.263 [-0.792, +0.251]	-1.4% [-4.3%, +1.4%]	-1.82 [-3.20, -0.52]	1000
mind_pilot_s2	clean	+0.175 [+0.022, +0.329]	+5.0% [+0.6%, +9.4%]	+5.48 [+3.85, +7.06]	1000
mind_pilot_s2	compound_0	-0.383 [-0.752, -0.022]	-3.8% [-7.4%, -0.2%]	-0.89 [-2.60, +0.85]	1000
mind_pilot_s2	missing_0.1	+0.205 [+0.050, +0.353]	+5.8% [+1.4%, +9.9%]	+4.65 [+3.06, +6.26]	1000
mind_pilot_s2	missing_0.3	+0.218 [+0.033, +0.406]	+5.5% [+0.8%, +10.2%]	+3.54 [+1.67, +5.40]	1000
mind_pilot_s2	missing_0.5	+0.464 [+0.272, +0.661]	+10.6% [+6.2%, +15.1%]	+3.06 [+1.17, +4.96]	1000
mind_pilot_s2	missing_0.7	+0.400 [+0.165, +0.630]	+7.3% [+3.0%, +11.5%]	+2.98 [+1.05, +4.79]	1000
mind_pilot_s2	noise_0.1	-0.339 [-0.568, -0.093]	-5.8% [-9.8%, -1.6%]	+2.98 [+1.07, +4.86]	1000
mind_pilot_s2	noise_0.3	-1.921 [-2.297, -1.521]	-15.3% [-18.3%, -12.1%]	+2.46 [+0.75, +4.14]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
mind_pilot_s2	noise_0.5	-0.638 [-1.135, -0.115]	-4.2% [-7.5%, -0.8%]	+0.27 [-1.36, +1.79]	1000
mind_pilot_s2	noise_1	+0.430 [-0.126, +0.995]	+2.3% [-0.7%, +5.4%]	+0.33 [-1.15, +1.72]	1000
oracle_log	clean	-3.516 [-3.687, -3.357]	-100.0% [-104.9%, -95.5%]	+nan [+nan, +nan]	1000
oracle_log	compound_0	-10.108 [-10.456, -9.757]	-100.0% [-103.5%, -96.5%]	+nan [+nan, +nan]	1000
oracle_log	missing_0.1	-3.556 [-3.730, -3.397]	-100.0% [-104.9%, -95.6%]	+nan [+nan, +nan]	1000
oracle_log	missing_0.3	-3.978 [-4.169, -3.797]	-100.0% [-104.8%, -95.4%]	+nan [+nan, +nan]	1000
oracle_log	missing_0.5	-4.371 [-4.576, -4.186]	-100.0% [-104.7%, -95.8%]	+nan [+nan, +nan]	1000
oracle_log	missing_0.7	-5.492 [-5.794, -5.210]	-100.0% [-105.5%, -94.9%]	+nan [+nan, +nan]	1000
oracle_log	noise_0.1	-5.818 [-6.070, -5.568]	-100.0% [-104.3%, -95.7%]	+nan [+nan, +nan]	1000
oracle_log	noise_0.3	-12.523 [-12.913, -12.120]	-100.0% [-103.1%, -96.8%]	+nan [+nan, +nan]	1000
oracle_log	noise_0.5	-15.068 [-15.465, -14.648]	-100.0% [-102.6%, -97.2%]	+nan [+nan, +nan]	1000
oracle_log	noise_1	-18.556 [-19.007, -18.122]	-100.0% [-102.4%, -97.7%]	+nan [+nan, +nan]	1000
rl_pilot_s0	clean	+0.528 [+0.373, +0.678]	+15.0% [+10.6%, +19.3%]	-11.08 [-12.42, -9.79]	1000
rl_pilot_s0	compound_0	-2.902 [-3.253, -2.558]	-28.7% [-32.2%, -25.3%]	-8.11 [-9.63, -6.61]	1000
rl_pilot_s0	missing_0.1	+0.552 [+0.404, +0.697]	+15.5% [+11.4%, +19.6%]	-12.05 [-13.48, -10.74]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
rl_pilot_s0	missing_0.3	+0.234 [+0.063, +0.389]	+5.9% [+1.6%, +9.8%]	-14.74 [-16.42, -13.25]	1000
rl_pilot_s0	missing_0.5	+0.070 [-0.114, +0.240]	+1.6% [-2.6%, +5.5%]	-16.35 [-18.19, -14.61]	1000
rl_pilot_s0	missing_0.7	-0.266 [-0.508, -0.026]	-4.8% [-9.2%, -0.5%]	-16.27 [-18.10, -14.51]	1000
rl_pilot_s0	noise_0.1	-0.613 [-0.851, -0.368]	-10.5% [-14.6%, -6.3%]	-11.90 [-13.57, -10.29]	1000
rl_pilot_s0	noise_0.3	-5.751 [-6.142, -5.362]	-45.9% [-49.0%, -42.8%]	-5.90 [-7.46, -4.39]	1000
rl_pilot_s0	noise_0.5	-7.062 [-7.490, -6.626]	-46.9% [-49.7%, -44.0%]	-4.88 [-6.27, -3.61]	1000
rl_pilot_s0	noise_1	-6.996 [-7.467, -6.529]	-37.7% [-40.2%, -35.2%]	-3.63 [-4.99, -2.28]	1000
rl_pilot_s1	clean	+0.543 [+0.388, +0.695]	+15.4% [+11.0%, +19.8%]	-10.41 [-11.79, -9.08]	1000
rl_pilot_s1	compound_0	-2.995 [-3.332, -2.654]	-29.6% [-33.0%, -26.3%]	-7.04 [-8.71, -5.45]	1000
rl_pilot_s1	missing_0.1	+0.544 [+0.392, +0.690]	+15.3% [+11.0%, +19.4%]	-10.73 [-12.23, -9.33]	1000
rl_pilot_s1	missing_0.3	+0.161 [-0.005, +0.316]	+4.0% [-0.1%, +7.9%]	-13.86 [-15.60, -12.28]	1000
rl_pilot_s1	missing_0.5	+0.038 [-0.143, +0.222]	+0.9% [-3.3%, +5.1%]	-15.58 [-17.42, -13.74]	1000
rl_pilot_s1	missing_0.7	-0.187 [-0.431, +0.043]	-3.4% [-7.9%, +0.8%]	-15.75 [-17.57, -14.03]	1000
rl_pilot_s1	noise_0.1	-0.459 [-0.711, -0.199]	-7.9% [-12.2%, -3.4%]	-11.47 [-13.12, -9.85]	1000
rl_pilot_s1	noise_0.3	-5.797 [-6.193, -5.424]	-46.3% [-49.5%, -43.3%]	-4.80 [-6.39, -3.25]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
rl_pilot_s1	noise_0.5	-7.175 [-7.613, -6.756]	-47.6% [-50.5%, -44.8%]	-3.83 [-5.24, -2.49]	1000
rl_pilot_s1	noise_1	-7.307 [-7.749, -6.853]	-39.4% [-41.8%, -36.9%]	-2.27 [-3.69, -0.85]	1000
rl_pilot_s2	clean	+0.280 [+0.131, +0.429]	+8.0% [+3.7%, +12.2%]	-8.89 [-10.26, -7.56]	1000
rl_pilot_s2	compound_0	-3.764 [-4.108, -3.436]	-37.2% [-40.6%, -34.0%]	-6.52 [-8.18, -4.99]	1000
rl_pilot_s2	missing_0.1	+0.263 [+0.121, +0.402]	+7.4% [+3.4%, +11.3%]	-9.91 [-11.35, -8.54]	1000
rl_pilot_s2	missing_0.3	-0.156 [-0.326, +0.006]	-3.9% [-8.2%, +0.2%]	-12.64 [-14.31, -11.05]	1000
rl_pilot_s2	missing_0.5	-0.340 [-0.522, -0.171]	-7.8% [-11.9%, -3.9%]	-15.18 [-16.94, -13.43]	1000
rl_pilot_s2	missing_0.7	-0.851 [-1.111, -0.607]	-15.5% [-20.2%, -11.0%]	-14.21 [-16.01, -12.44]	1000
rl_pilot_s2	noise_0.1	-1.088 [-1.327, -0.842]	-18.7% [-22.8%, -14.5%]	-10.88 [-12.55, -9.30]	1000
rl_pilot_s2	noise_0.3	-6.550 [-6.936, -6.133]	-52.3% [-55.4%, -49.0%]	-3.81 [-5.59, -2.27]	1000
rl_pilot_s2	noise_0.5	-7.909 [-8.333, -7.484]	-52.5% [-55.3%, -49.7%]	-2.27 [-3.72, -0.84]	1000
rl_pilot_s2	noise_1	-7.689 [-8.153, -7.224]	-41.4% [-43.9%, -38.9%]	-1.69 [-3.08, -0.22]	1000
select_m1_s0	clean	-0.475 [-0.616, -0.345]	-13.5% [-17.5%, -9.8%]	-14.43 [-15.78, -13.06]	1000
select_m1_s0	compound_0	-5.292 [-5.635, -4.956]	-52.4% [-55.7%, -49.0%]	-14.08 [-15.48, -12.75]	1000
select_m1_s0	missing_0.1	-0.517 [-0.649, -0.400]	-14.5% [-18.2%, -11.3%]	-15.35 [-16.82, -13.99]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
select_m1_s0	missing_0.3	-0.933 [-1.083, -0.784]	-23.5% [-27.2%, -19.7%]	-18.25 [-19.88, -16.70]	1000
select_m1_s0	missing_0.5	-1.165 [-1.331, -1.002]	-26.7% [-30.4%, -22.9%]	-20.44 [-22.11, -18.75]	1000
select_m1_s0	missing_0.7	-1.815 [-2.064, -1.584]	-33.0% [-37.6%, -28.8%]	-20.68 [-22.37, -18.96]	1000
select_m1_s0	noise_0.1	-2.259 [-2.485, -2.026]	-38.8% [-42.7%, -34.8%]	-16.38 [-18.04, -14.88]	1000
select_m1_s0	noise_0.3	-8.282 [-8.673, -7.895]	-66.1% [-69.3%, -63.0%]	-12.24 [-13.72, -10.92]	1000
select_m1_s0	noise_0.5	-9.820 [-10.256, -9.391]	-65.2% [-68.1%, -62.3%]	-12.35 [-13.65, -11.13]	1000
select_m1_s0	noise_1	-10.242 [-10.666, -9.809]	-55.2% [-57.5%, -52.9%]	-11.90 [-13.18, -10.66]	1000
select_m1_s1	clean	-0.521 [-0.659, -0.394]	-14.8% [-18.7%, -11.2%]	-13.31 [-14.71, -11.96]	1000
select_m1_s1	compound_0	-5.452 [-5.790, -5.117]	-53.9% [-57.3%, -50.6%]	-13.84 [-15.29, -12.39]	1000
select_m1_s1	missing_0.1	-0.546 [-0.690, -0.410]	-15.4% [-19.4%, -11.5%]	-14.44 [-15.95, -13.10]	1000
select_m1_s1	missing_0.3	-0.932 [-1.095, -0.779]	-23.4% [-27.5%, -19.6%]	-17.22 [-18.85, -15.66]	1000
select_m1_s1	missing_0.5	-1.232 [-1.398, -1.069]	-28.2% [-32.0%, -24.5%]	-19.21 [-20.86, -17.53]	1000
select_m1_s1	missing_0.7	-1.889 [-2.122, -1.662]	-34.4% [-38.6%, -30.3%]	-19.76 [-21.44, -18.09]	1000
select_m1_s1	noise_0.1	-2.281 [-2.516, -2.051]	-39.2% [-43.2%, -35.3%]	-15.75 [-17.37, -14.18]	1000
select_m1_s1	noise_0.3	-8.339 [-8.714, -7.960]	-66.6% [-69.6%, -63.6%]	-12.34 [-13.84, -11.03]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
select_m1_s1	noise_0.5	-9.826 [-10.250, -9.410]	-65.2% [-68.0%, -62.4%]	-12.27 [-13.54, -10.99]	1000
select_m1_s1	noise_1	-10.150 [-10.604, -9.684]	-54.7% [-57.1%, -52.2%]	-11.89 [-13.21, -10.64]	1000
select_m1_s2	clean	-0.471 [-0.611, -0.332]	-13.4% [-17.4%, -9.5%]	-13.51 [-14.84, -12.20]	1000
select_m1_s2	compound_0	-5.414 [-5.760, -5.066]	-53.6% [-57.0%, -50.1%]	-14.16 [-15.64, -12.75]	1000
select_m1_s2	missing_0.1	-0.494 [-0.631, -0.358]	-13.9% [-17.8%, -10.1%]	-14.76 [-16.17, -13.44]	1000
select_m1_s2	missing_0.3	-0.856 [-1.014, -0.693]	-21.5% [-25.5%, -17.4%]	-17.50 [-19.14, -15.90]	1000
select_m1_s2	missing_0.5	-1.152 [-1.330, -0.977]	-26.3% [-30.4%, -22.3%]	-19.79 [-21.45, -18.14]	1000
select_m1_s2	missing_0.7	-1.699 [-1.959, -1.445]	-30.9% [-35.7%, -26.3%]	-20.81 [-22.50, -19.16]	1000
select_m1_s2	noise_0.1	-2.172 [-2.416, -1.911]	-37.3% [-41.5%, -32.8%]	-15.78 [-17.44, -14.19]	1000
select_m1_s2	noise_0.3	-8.383 [-8.784, -7.993]	-66.9% [-70.1%, -63.8%]	-12.45 [-13.93, -11.09]	1000
select_m1_s2	noise_0.5	-10.050 [-10.468, -9.613]	-66.7% [-69.5%, -63.8%]	-12.02 [-13.29, -10.78]	1000
select_m1_s2	noise_1	-10.130 [-10.592, -9.675]	-54.6% [-57.1%, -52.1%]	-11.63 [-12.90, -10.39]	1000
cp	CORE_SUITE	+18.384 [+17.238, +19.613]	+243.4% [+228.3%, +259.7%]	-16.42 [-17.37, -15.51]	1000
cv	CORE_SUITE	+3.958 [+3.491, +4.414]	+52.4% [+46.2%, +58.4%]	+0.93 [-0.80, +2.70]	1000
extra_sft_pilot_s0	CORE_SUITE	-3.615 [-3.752, -3.480]	-47.9% [-49.7%, -46.1%]	+0.30 [-0.53, +1.11]	1000

method	condition	Δ joint minADE@68s	rel. Δ	Δ overlap rate (pp)	n
extra_sft_pilot_s1	CORE_SUITE	-3.676 [-3.816, -3.532]	-48.7% [-50.5%, -46.8%]	+0.97 [-0.00, +1.90]	1000
extra_sft_pilot_s2	CORE_SUITE	-3.679 [-3.833, -3.524]	-48.7% [-50.8%, -46.7%]	+1.23 [+0.27, +2.23]	1000
filtered_sft_pilot_s0	CORE_SUITE	-3.583 [-3.722, -3.448]	-47.4% [-49.3%, -45.7%]	-0.62 [-1.49, +0.21]	1000
filtered_sft_pilot_s1	CORE_SUITE	-3.554 [-3.696, -3.420]	-47.1% [-48.9%, -45.3%]	-0.34 [-1.35, +0.67]	1000
filtered_sft_pilot_s2	CORE_SUITE	-3.565 [-3.719, -3.411]	-47.2% [-49.2%, -45.2%]	-0.00 [-0.97, +0.93]	1000
m0_pilot_s1	CORE_SUITE	-0.371 [-0.502, -0.231]	-4.9% [-6.6%, -3.1%]	+0.09 [-0.69, +0.88]	1000
m0_pilot_s2	CORE_SUITE	-0.372 [-0.523, -0.226]	-4.9% [-6.9%, -3.0%]	+0.98 [+0.09, +1.84]	1000
m1_pilot_s0	CORE_SUITE	-3.602 [-3.741, -3.465]	-47.7% [-49.5%, -45.9%]	-0.26 [-1.16, +0.62]	1000
m1_pilot_s1	CORE_SUITE	-3.662 [-3.804, -3.518]	-48.5% [-50.4%, -46.6%]	+1.82 [+0.87, +2.76]	1000
m1_pilot_s2	CORE_SUITE	-3.642 [-3.806, -3.489]	-48.2% [-50.4%, -46.2%]	+1.04 [+0.10, +2.00]	1000
mind_pilot_s0	CORE_SUITE	+0.316 [+0.191, +0.439]	+4.2% [+2.5%, +5.8%]	+1.11 [+0.36, +1.85]	1000
mind_pilot_s1	CORE_SUITE	+0.237 [+0.092, +0.386]	+3.1% [+1.2%, +5.1%]	+0.96 [+0.20, +1.78]	1000
mind_pilot_s2	CORE_SUITE	-0.335 [-0.480, -0.192]	-4.4% [-6.4%, -2.5%]	+2.83 [+1.97, +3.70]	1000
oracle_log	CORE_SUITE	-7.552 [-7.730, -7.371]	-100.0% [-102.4%, -97.6%]	+nan [+nan, +nan]	1000
rl_pilot_s0	CORE_SUITE	-2.095 [-2.252, -1.944]	-27.7% [-29.8%, -25.7%]	-10.97 [-11.85, -10.11]	1000

method	condition	Δ joint minADE@6 8s	rel. Δ	Δ overlap rate (pp)	n
rl_pilot_s1	CORE_SUITE	-2.115 [-2.274, -1.954]	-28.0% [-30.1%, -25.9%]	-10.04 [-11.01, -9.18]	1000
rl_pilot_s2	CORE_SUITE	-2.630 [-2.791, -2.475]	-34.8% [-37.0%, -32.8%]	-9.12 [-10.07, -8.26]	1000
select_m1_s0	CORE_SUITE	-3.829 [-3.977, -3.684]	-50.7% [-52.7%, -48.8%]	-15.84 [-16.76, -14.96]	1000
select_m1_s1	CORE_SUITE	-3.859 [-4.010, -3.717]	-51.1% [-53.1%, -49.2%]	-15.20 [-16.12, -14.37]	1000
select_m1_s2	CORE_SUITE	-3.851 [-4.011, -3.695]	-51.0% [-53.1%, -48.9%]	-15.39 [-16.28, -14.56]	1000

Denominators: `overlap rate (6, full horizon)` = fraction of (scene, sample) pairs with any box overlap over the full 80-step generated horizon, scenes with initial overlap excluded, NaN for logged futures; `overlap jointly-valid` = same test restricted to sub-steps where both targets have valid logged labels (defined for generated and logged trajectories on identical intervals; the oracle_log row is the reference rate); `offroad frac` = vehicle targets only, full horizon (logged: valid steps only); cp/cv are deterministic single trajectories; learned models use 6 joint samples (32 for the audit columns).

5a. Per-seed final comparison

Development comparison (1000 dev scenes; bootstrap unit = heuristic source-segment group, 95% CI)

Sampling: learned models 6 samples (joint metrics) + 32-sample audit; cp/cv deterministic (1 sample, listed separately).

Per-seed results (joint minADE@6 8 s, m; overlap and offroad rates as fractions)

method	seed	clean jminADE	clean jminFDE	core-suite jminADE	held-out jminADE	clean overlap
cp	det.	25.937 [24.776, 27.178]	50.479 [48.264, 52.878]	25.937 [24.776, 27.178]	25.937 [24.776, 27.178]	0.0000 [0.0000, 0.0000]
cv	det.	11.114 [10.681, 11.562]	28.456 [27.264, 29.601]	11.510 [11.092, 11.948]	13.122 [12.726, 13.531]	0.1725 [0.1494, 0.1966]

method	seed	clean jminADE	clean jminFDE	core-suite jminADE	held-out jminADE	clean overlap
oracle_log	det.	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	—
m0_pilot_s0	0	3.516 [3.357, 3.687]	8.600 [8.150, 9.111]	7.552 [7.371, 7.730]	11.385 [11.149, 11.620]	0.1496 [0.1355, 0.1638]
m0_pilot_s1	1	3.425 [3.262, 3.594]	8.470 [8.029, 8.956]	7.181 [7.007, 7.353]	11.135 [10.871, 11.393]	0.1638 [0.1486, 0.1807]
m0_pilot_s2	2	3.541 [3.372, 3.721]	8.755 [8.261, 9.258]	7.180 [6.999, 7.368]	11.678 [11.414, 11.930]	0.1655 [0.1499, 0.1810]
mind_pilot_s0	0	3.689 [3.527, 3.866]	9.175 [8.706, 9.695]	7.868 [7.681, 8.051]	11.986 [11.742, 12.210]	0.1852 [0.1690, 0.2025]
mind_pilot_s1	1	3.575 [3.417, 3.752]	8.844 [8.370, 9.350]	7.789 [7.599, 7.971]	12.111 [11.866, 12.364]	0.1827 [0.1675, 0.1989]
mind_pilot_s2	2	3.691 [3.525, 3.870]	8.997 [8.512, 9.494]	7.217 [7.029, 7.408]	11.534 [11.268, 11.777]	0.2044 [0.1875, 0.2211]
m1_pilot_s0	0	3.200 [3.055, 3.364]	7.875 [7.433, 8.351]	3.950 [3.821, 4.097]	5.985 [5.826, 6.150]	0.1473 [0.1327, 0.1617]
m1_pilot_s1	1	3.143 [2.991, 3.303]	7.694 [7.273, 8.155]	3.890 [3.751, 4.036]	5.908 [5.741, 6.073]	0.1688 [0.1525, 0.1857]
m1_pilot_s2	2	3.251 [3.088, 3.424]	7.975 [7.517, 8.471]	3.910 [3.773, 4.064]	5.915 [5.739, 6.087]	0.1590 [0.1432, 0.1755]
select_m1_s0	0	3.041 [2.894, 3.204]	7.379 [6.941, 7.838]	3.723 [3.591, 3.871]	5.602 [5.435, 5.763]	0.0053 [0.0020, 0.0094]
select_m1_s1	1	2.994 [2.851, 3.146]	7.307 [6.886, 7.734]	3.693 [3.557, 3.842]	5.555 [5.389, 5.724]	0.0165 [0.0102, 0.0234]
select_m1_s2	2	3.045 [2.892, 3.218]	7.564 [7.099, 8.041]	3.701 [3.561, 3.864]	5.638 [5.467, 5.810]	0.0145 [0.0087, 0.0207]
extra_sft_pilot_s0	0	3.123 [2.976, 3.282]	7.667 [7.258, 8.145]	3.937 [3.808, 4.085]	5.953 [5.789, 6.119]	0.1541 [0.1393, 0.1697]
extra_sft_pilot_s1	1	3.078 [2.925, 3.234]	7.505 [7.073, 7.960]	3.876 [3.732, 4.029]	5.950 [5.790, 6.113]	0.1576 [0.1421, 0.1735]

method	seed	clean jminADE	clean jminFDE	core-suite jminADE	held-out jminADE	clean overlap
extra_sft_pilot_s2	2	3.156 [3.000, 3.322]	7.784 [7.333, 8.258]	3.873 [3.732, 4.033]	5.936 [5.770, 6.110]	0.1561 [0.1393, 0.1729]
filtered_sft_pilot_s0	0	3.178 [3.036, 3.345]	7.724 [7.302, 8.196]	3.969 [3.835, 4.120]	6.154 [5.985, 6.330]	0.1398 [0.1265, 0.1538]
filtered_sft_pilot_s1	1	3.180 [3.032, 3.331]	7.705 [7.283, 8.143]	3.998 [3.857, 4.148]	6.149 [5.977, 6.317]	0.1466 [0.1313, 0.1627]
filtered_sft_pilot_s2	2	3.218 [3.059, 3.396]	7.986 [7.524, 8.463]	3.988 [3.842, 4.153]	6.109 [5.929, 6.291]	0.1546 [0.1388, 0.1705]
rl_pilot_s0	0	4.043 [3.871, 4.232]	10.165 [9.642, 10.706]	5.457 [5.284, 5.645]	7.997 [7.767, 8.214]	0.0388 [0.0315, 0.0467]
rl_pilot_s1	1	4.059 [3.875, 4.246]	10.300 [9.778, 10.848]	5.437 [5.257, 5.630]	7.889 [7.663, 8.118]	0.0455 [0.0367, 0.0546]
rl_pilot_s2	2	3.795 [3.627, 3.977]	9.738 [9.242, 10.237]	4.922 [4.751, 5.103]	7.284 [7.073, 7.482]	0.0607 [0.0508, 0.0720]

Q1 Joint vs independent decoder (m0 − mind; negative = joint better)

seed	clean jminADE Δ	clean jminFDE Δ	clean overlap Δ (pp)	clean overlap32 Δ (pp)
0	-0.174 [-0.301, -0.045] (-4.7%)	-0.575 [-0.979, -0.181] (-6.3%)	-3.56 [-5.00, -2.04]	-3.97 [-5.18, -2.87]
1	-0.150 [-0.284, -0.020] (-4.2%)	-0.375 [-0.761, +0.035] (-4.2%)	-1.89 [-3.51, -0.35]	-2.49 [-3.61, -1.35]
2	-0.149 [-0.274, -0.024] (-4.0%)	-0.242 [-0.629, +0.155] (-2.7%)	-3.90 [-5.52, -2.36]	-2.95 [-4.10, -1.77]
sign agreement	consistent	consistent	consistent	consistent

Q2 Corruption-trained vs clean-trained under degraded inputs (m1 − m0; negative = m1 better)

seed	core-suite jminADE Δ	held-out jminADE Δ	core-suite jminFDE Δ	core-suite overlap Δ (pp)
0	-3.602 [-3.741, -3.465] (-47.7%)	-5.400 [-5.611, -5.182] (-47.4%)	-7.763 [-8.142, -7.422] (-46.2%)	-0.26 [-1.16, +0.62]
1	-3.290 [-3.421, -3.147] (-45.8%)	-5.227 [-5.453, -4.988] (-46.9%)	-7.274 [-7.637, -6.919] (-44.9%)	+1.72 [+0.77, +2.67]
2	-3.270 [-3.424, -3.115] (-45.5%)	-5.763 [-5.988, -5.537] (-49.4%)	-7.288 [-7.703, -6.863] (-44.4%)	+0.06 [-0.91, +1.08]
sign agreement	consistent	consistent	consistent	MIXED

Q3 Clean accuracy and physical consistency (m1 − m0 on clean)

seed	clean jminADE Δ	clean overlap Δ (pp)	clean offroad Δ (pp)	clean implausible Δ (pp)
0	-0.315 [-0.457, -0.185] (-9.0%)	-0.23 [-1.64, +1.21]	-0.59 [-0.98, -0.22]	+0.02 [+0.00, +0.04]
1	-0.282 [-0.403, -0.157] (-8.2%)	+0.50 [-1.01, +2.02]	-1.31 [-1.71, -0.91]	+0.00 [-0.03, +0.02]
2	-0.290 [-0.419, -0.161] (-8.2%)	-0.65 [-2.15, +0.97]	-1.21 [-1.66, -0.77]	-0.01 [-0.03, +0.02]
sign agreement	consistent	MIXED	consistent	MIXED

Q5a Bounded RL vs its M1 initialisation (rl − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+0.843 [+0.719, +0.969] (+26.3%)	+1.507 [+1.390, +1.622] (+38.1%)	-10.85 [-12.09, -9.55]	-10.43 [-11.48, -9.37]	-10.36 [-11.41, -9.29]	-1.32 [-1.65 -0.99
1	+0.916 [+0.785,	+1.547 [+1.423,	-12.34 [-13.84, -10.89]	-11.80 [-12.98, -10.67]	-11.76 [-12.92, -10.65]	-1.46 [-1.78 -1.16

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
	+1.046] (+29.1%)	+1.672] (+39.8%)				
2	+0.545 [+0.428, +0.660] (+16.8%)	+1.012 [+0.916, +1.110] (+25.9%)	-9.83 [-11.14, -8.60]	-9.79 [-10.87, -8.75]	-9.75 [-10.82, -8.71]	-1.40 [-1.7 -1.11
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q5b Extra-SFT control vs M1 (extra_sft − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	clea offr Δ (p
0	-0.077 [-0.167, +0.008] (-2.4%)	-0.013 [-0.058, +0.032] (-0.3%)	+0.69 [-0.38, +1.78]	+0.58 [+0.03, +1.12]	+0.60 [+0.05, +1.13]	+0.25 [+0.0 +0.51
1	-0.065 [-0.147, +0.013] (-2.1%)	-0.015 [-0.055, +0.029] (-0.4%)	-1.12 [-2.20, -0.15]	-1.33 [-1.78, -0.88]	-1.31 [-1.77, -0.86]	+0.00 [-0.21 +0.23
2	-0.094 [-0.166, -0.022] (-2.9%)	-0.038 [-0.081, +0.006] (-1.0%)	-0.28 [-1.39, +0.84]	-0.03 [-0.57, +0.50]	-0.03 [-0.57, +0.50]	+0.40 [+0.1 +0.64
sign agreement	consistent	consistent	MIXED	MIXED	MIXED	consis

Q5c Reward-filtered-SFT control vs M1 (filtered_sft − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	clea offro Δ (p
0	-0.023 [-0.105, +0.060] (-0.7%)	+0.019 [-0.032, +0.068] (+0.5%)	-0.75 [-1.79, +0.22]	-0.57 [-1.04, -0.10]	-0.55 [-1.03, -0.08]	-0.28 [-0.5(-0.06
1	+0.037 [-0.043, +0.119] (+1.2%)	+0.108 [+0.060, +0.158] (+2.8%)	-2.22 [-3.28, -1.17]	-2.32 [-2.86, -1.78]	-2.29 [-2.84, -1.75]	+0.02 [-0.2(+0.2
2	-0.032 [-0.111, +0.044] (-1.0%)	+0.077 [+0.033, +0.123] (+2.0%)	-0.43 [-1.50, +0.64]	-0.79 [-1.34, -0.23]	-0.76 [-1.31, -0.20]	+0.07 [-0.1 +0.31
sign agreement	MIXED	consistent	consistent	consistent	consistent	MIXI

Q5d Reward-based candidate selection vs M1 (select − m1)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	-0.159 [-0.240, -0.080] (-5.0%)	-0.227 [-0.278, -0.179] (-5.8%)	-14.19 [-15.59, -12.82]	+0.06 [-0.40, +0.52]	+0.05 [-0.41, +0.51]	-3.21 [-3.56 -2.87
1	-0.149 [-0.234, -0.062] (-4.7%)	-0.198 [-0.240, -0.151] (-5.1%)	-15.23 [-16.73, -13.82]	+0.10 [-0.34, +0.56]	+0.10 [-0.35, +0.55]	-3.12 [-3.45 -2.81
2	-0.206 [-0.285, -0.126] (-6.3%)	-0.209 [-0.252, -0.165] (-5.4%)	-14.44 [-15.90, -13.07]	-0.13 [-0.57, +0.31]	-0.10 [-0.55, +0.34]	-3.17 [-3.5 -2.86
sign agreement	consistent	consistent	consistent	MIXED	MIXED	consi

Q5e RL vs extra-SFT control (rl − extra_sft)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+0.920 [+0.793, +1.052] (+29.5%)	+1.520 [+1.401, +1.634] (+38.6%)	-11.53 [-12.97, -10.17]	-11.01 [-12.14, -9.92]	-10.95 [-12.06, -9.86]	-1.57 [-1.93 -1.20
1	+0.981 [+0.859, +1.097] (+31.9%)	+1.562 [+1.436, +1.687] (+40.3%)	-11.22 [-12.57, -9.92]	-10.48 [-11.58, -9.44]	-10.44 [-11.53, -9.42]	-1.46 [-1.80 -1.15
2	+0.639 [+0.529, +0.755] (+20.2%)	+1.049 [+0.950, +1.150] (+27.1%)	-9.55 [-10.94, -8.20]	-9.76 [-10.94, -8.69]	-9.73 [-10.86, -8.66]	-1.80 [-2.15 -1.45
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q5f RL vs reward-filtered-SFT control (rl − filtered_sft)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+0.866 [+0.738, +0.997] (+27.2%)	+1.488 [+1.382, +1.596] (+37.5%)	-10.10 [-11.36, -8.89]	-9.86 [-10.91, -8.87]	-9.80 [-10.85, -8.82]	-1.04 [-1.34 -0.72
1	+0.878 [+0.752, +1.009] (+27.6%)	+1.439 [+1.320, +1.555] (+36.0%)	-10.11 [-11.48, -8.95]	-9.48 [-10.51, -8.46]	-9.46 [-10.48, -8.45]	-1.48 [-1.81 -1.19
2	+0.577 [+0.470, +0.683] (+17.9%)	+0.935 [+0.844, +1.024] (+23.4%)	-9.39 [-10.71, -8.02]	-9.00 [-10.07, -7.97]	-8.99 [-10.05, -7.96]	-1.47 [-1.78 -1.16
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q5g RL vs reward-based candidate selection (rl − select_m1; select draws 32 candidates per scene)

seed	clean jminADE Δ	core-suite jminADE Δ	clean overlap6 Δ (pp; the method's own 6 outputs)	clean overlap32 Δ (pp; 32-sample pool — for select_m1 this is the UNSELECTED pool)	clean overlap32 jointly-valid Δ (pp)	cle offr Δ (p
0	+1.002 [+0.879, +1.128] (+33.0%)	+1.734 [+1.623, +1.850] (+46.6%)	+3.34 [+2.64, +4.08]	-10.49 [-11.55, -9.41]	-10.41 [-11.46, -9.32]	+1.89 [+1.5 +2.23
1	+1.064 [+0.937, +1.190] (+35.5%)	+1.745 [+1.618, +1.877] (+47.2%)	+2.89 [+2.04, +3.77]	-11.90 [-13.15, -10.70]	-11.85 [-13.05, -10.68]	+1.66 [+1.3 +1.97
2	+0.750 [+0.645, +0.856] (+24.6%)	+1.221 [+1.125, +1.310] (+33.0%)	+4.61 [+3.73, +5.57]	-9.67 [-10.76, -8.60]	-9.65 [-10.73, -8.57]	+1.77 [+1.5 +2.07
sign agreement	consistent	consistent	consistent	consistent	consistent	consi

Q6a Excessive-stopping diagnostic (descriptive; clean dev; 500 scenes; identical scenes/sampling; eligible = agent with valid logged final step and logged final displacement > 5 m; stopped = predicted final displacement < 1 m; counted per (agent, joint sample); 6 samples/scene (select_m1: best 6 of 32 by reward); torch seed 0; descriptive only, no threshold)

method	eligible (agent, sample) pairs	stopped-while-logged-moved	frequency	seeds
m1_pilot	17082	73	0.43 %	0,1,2
select_m1	17082	75	0.44 %	0,1,2
extra_sft_pilot	17082	74	0.43 %	0,1,2
filtered_sft_pilot	17082	99	0.58 %	0,1,2
rl_pilot	17082	596	3.49 %	0,1,2
m0_pilot	17082	89	0.52 %	0,1,2

Q6b Post-training behaviour summary (clean dev; scene-level means over seeds; 6-sample metrics unless 32 noted)

method	speed_mean (m/s)	GT speed_mean	final_disp_mean (m)	GT final_disp	final / G
m1_pilot	6.41	6.59	49.49	50.95	0.971
select_m1	6.32	6.59	48.84	50.95	0.959
extra_sft_pilot	6.39	6.59	49.23	50.95	0.966
filtered_sft_pilot	6.36	6.59	48.98	50.95	0.961
rl_pilot	5.62	6.59	43.68	50.95	0.857

Sampling cost and wall-clock (clean condition)

Wall-clock seconds per scene are DESCRIPTIVE measurements logged by evaluate.py on a GPU shared with unrelated jobs; they do not establish a reliable relative speed ranking. The deterministic cost driver is the number of joint samples drawn per scene (and, for select_m1, the reward scoring of all 32 candidates).

method	samples drawn per scene	s / scene (descriptive)
cp	1	0.0000
cv	1	0.0000
extra_sft_pilot_s0	6	0.0075
extra_sft_pilot_s1	6	0.0074
extra_sft_pilot_s2	6	0.0076
filtered_sft_pilot_s0	6	0.0073
filtered_sft_pilot_s1	6	0.0075
filtered_sft_pilot_s2	6	0.0075
m0_pilot_s0	6	0.0074
m0_pilot_s1	6	0.0074
m0_pilot_s2	6	0.0076
m1_pilot_s0	6	0.0075
m1_pilot_s1	6	0.0075
m1_pilot_s2	6	0.0077
mind_pilot_s0	6	0.0074
mind_pilot_s1	6	0.0072
mind_pilot_s2	6	0.0074
oracle_log	1	0.0000
rl_pilot_s0	6	0.0076
rl_pilot_s1	6	0.0075
rl_pilot_s2	6	0.0075
select_m1_s0	32 + reward scoring of 32	0.0720

method	samples drawn per scene	s / scene (descriptive)
select_m1_s1	32 + reward scoring of 32	0.0720
select_m1_s2	32 + reward scoring of 32	0.0727

Per-condition joint minADE@6 (mean over seeds; per-seed values in summary.json)

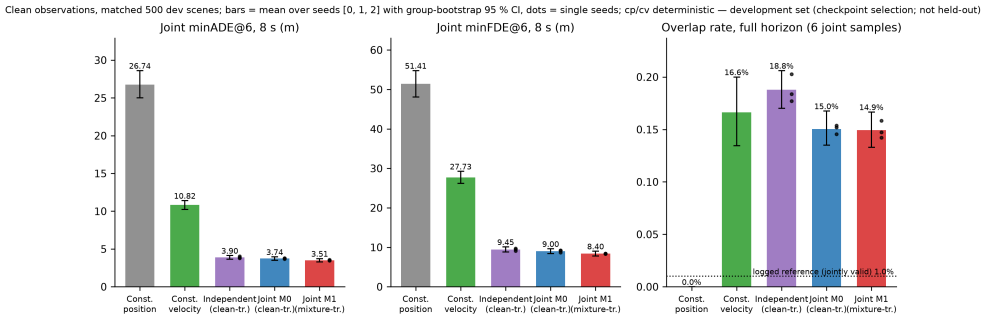
condition	cv	m0_pilot	mind_pilot	m1_pilot	select_m1	rl_pilot
clean	11.114	3.494 (±0.050, n=3)	3.652 (±0.054, n=3)	3.198 (±0.044, n=3)	3.027 (±0.023, n=3)	3.966 (±0.121, n=3)
missing_0.1	11.099	3.548 (±0.046, n=3)	3.706 (±0.040, n=3)	3.166 (±0.023, n=3)	3.037 (±0.021, n=3)	4.009 (±0.135, n=3)
missing_0.3	11.060	3.932 (±0.033, n=3)	4.182 (±0.012, n=3)	3.181 (±0.026, n=3)	3.071 (±0.036, n=3)	4.057 (±0.169, n=3)
missing_0.5	11.045	4.416 (±0.124, n=3)	4.800 (±0.124, n=3)	3.320 (±0.020, n=3)	3.188 (±0.035, n=3)	4.293 (±0.186, n=3)
noise_0.1	11.204	5.797 (±0.063, n=3)	5.985 (±0.359, n=3)	3.779 (±0.018, n=3)	3.580 (±0.047, n=3)	5.098 (±0.268, n=3)
noise_0.3	11.886	11.671 (±0.636, n=3)	11.939 (±0.946, n=3)	4.491 (±0.036, n=3)	4.188 (±0.041, n=3)	6.490 (±0.366, n=3)
noise_0.5	12.769	14.462 (±0.545, n=3)	15.135 (±0.610, n=3)	5.565 (±0.109, n=3)	5.169 (±0.107, n=3)	7.686 (±0.375, n=3)
missing_0.7	11.439	5.720 (±0.288, n=3)	6.165 (±0.354, n=3)	3.741 (±0.035, n=3)	3.691 (±0.078, n=3)	5.058 (±0.296, n=3)
noise_1	15.875	18.258 (±1.170, n=3)	18.851 (±0.412, n=3)	9.125 (±0.071, n=3)	8.382 (±0.049, n=3)	11.225 (±0.283, n=3)
compound_0	12.051	10.220 (±0.260, n=3)	10.616 (±0.680, n=3)	4.942 (±0.045, n=3)	4.721 (±0.068, n=3)	6.887 (±0.387, n=3)

Preregistered thresholds (m1 vs m0, per seed)

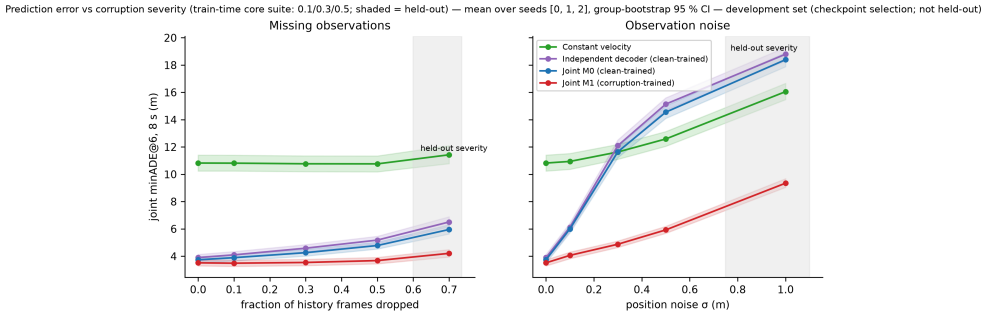
seed	core-suite rel. Δ (need ≤ −5%)	clean rel. Δ (need ≤ +3%)	core-suite overlap Δ pp (need ≤ +0.5)	verdict
0	-47.7% [-49.5, -45.9]	-9.0% [-13.0, -5.3]	-0.26 [-1.16, +0.62]	PASS
1	-45.8% [-47.6, -43.8]	-8.2% [-11.8, -4.6]	+1.72 [+0.77, +2.67]	FAIL

seed	core-suite rel. Δ (need ≤ −5%)	clean rel. Δ (need ≤ +3%)	core-suite overlap Δ pp (need ≤ +0.5)	verdict
2	-45.5% [-47.7, -43.4]	-8.2% [-11.8, -4.6]	+0.06 [-0.91, +1.08]	PASS

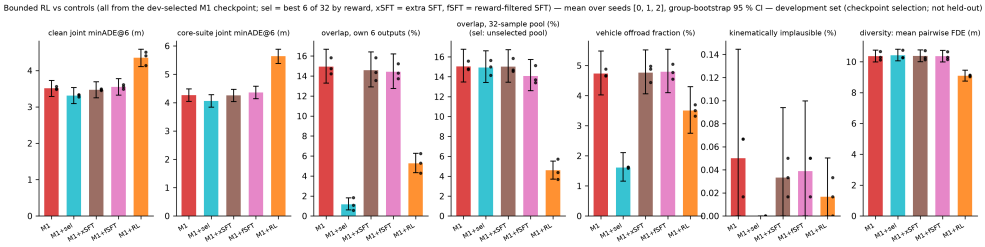
6. Figures



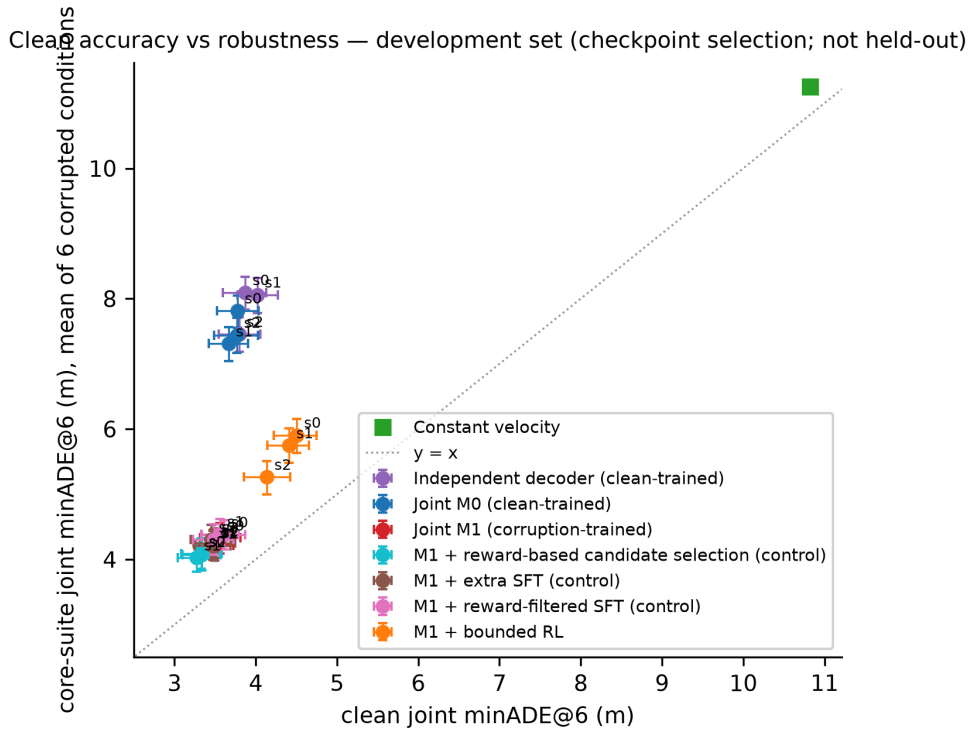
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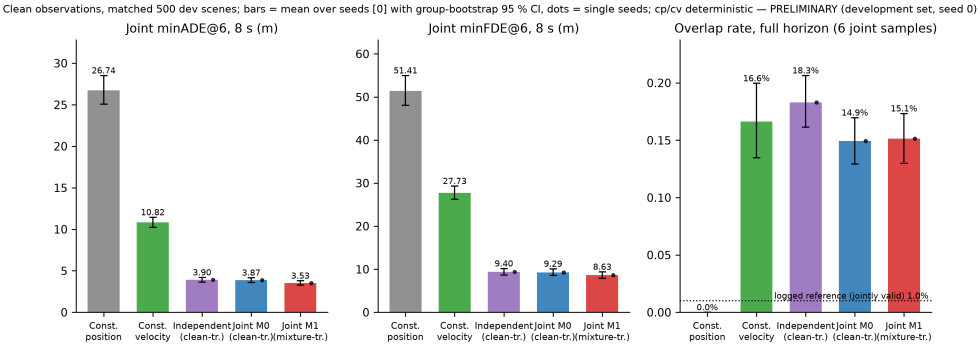
fig_error_vs_severity



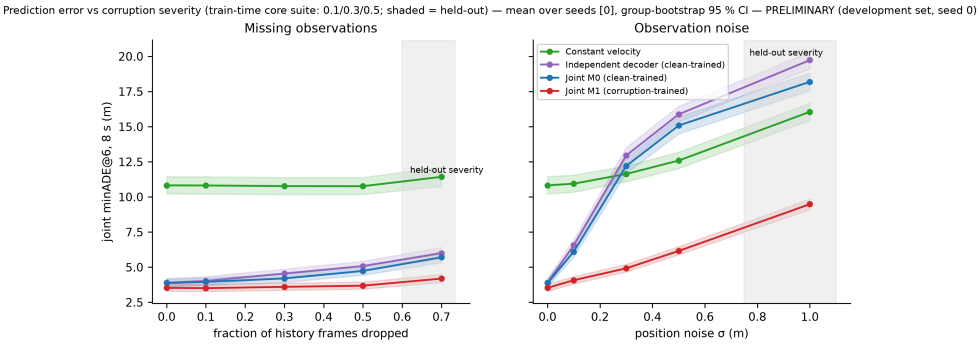
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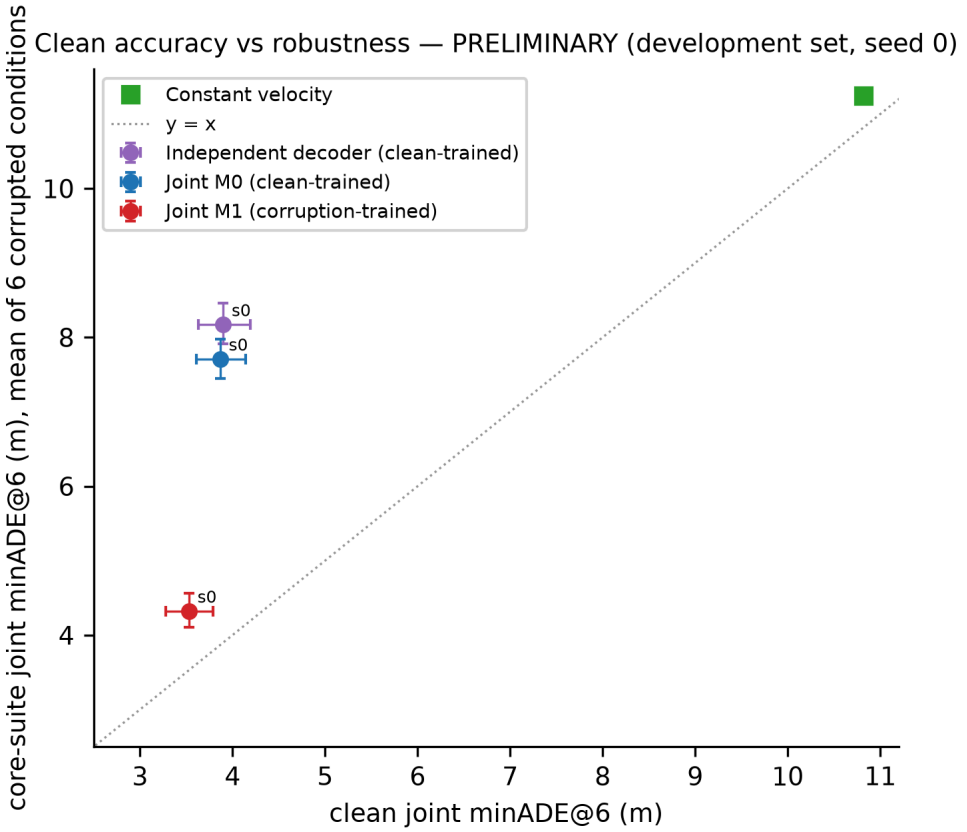
fig_tradeoff



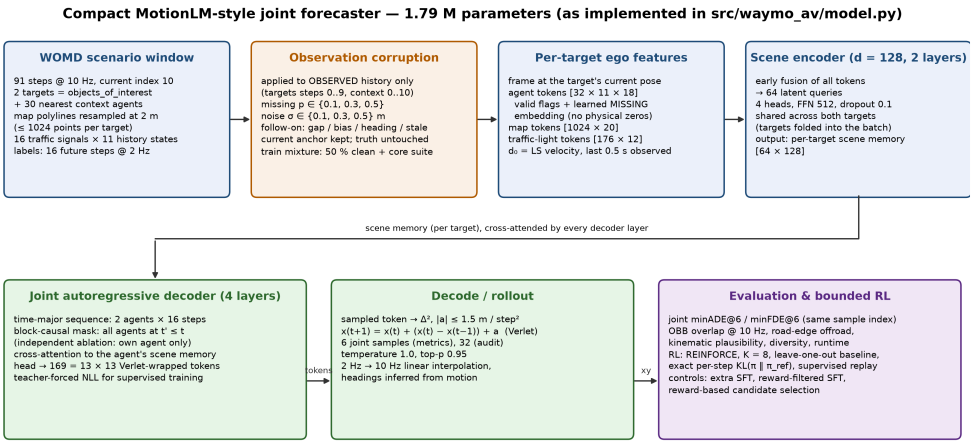
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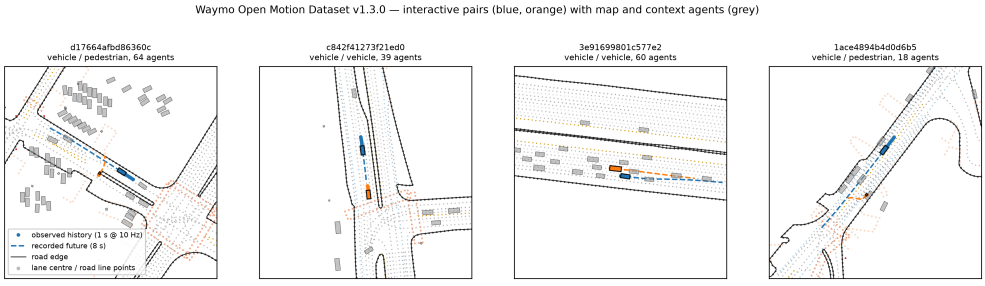
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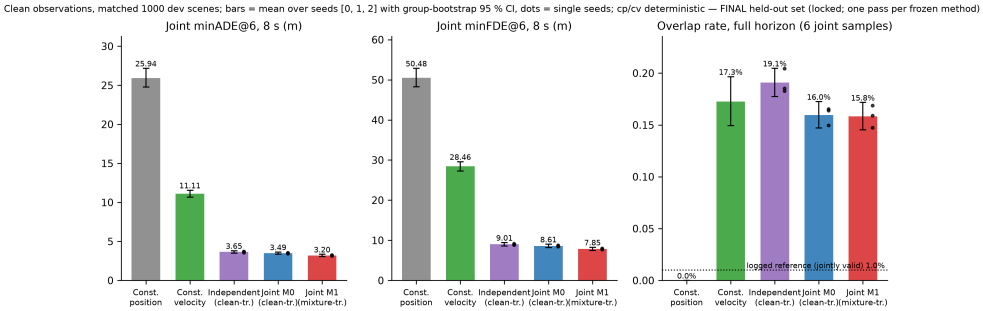
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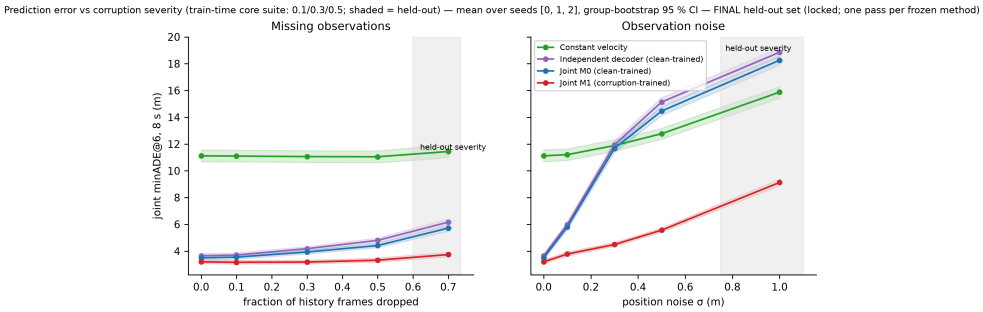
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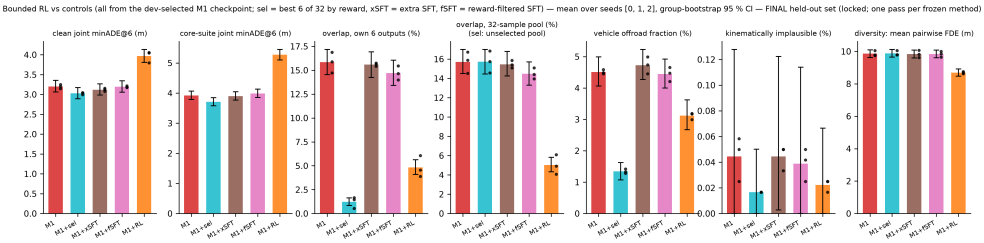
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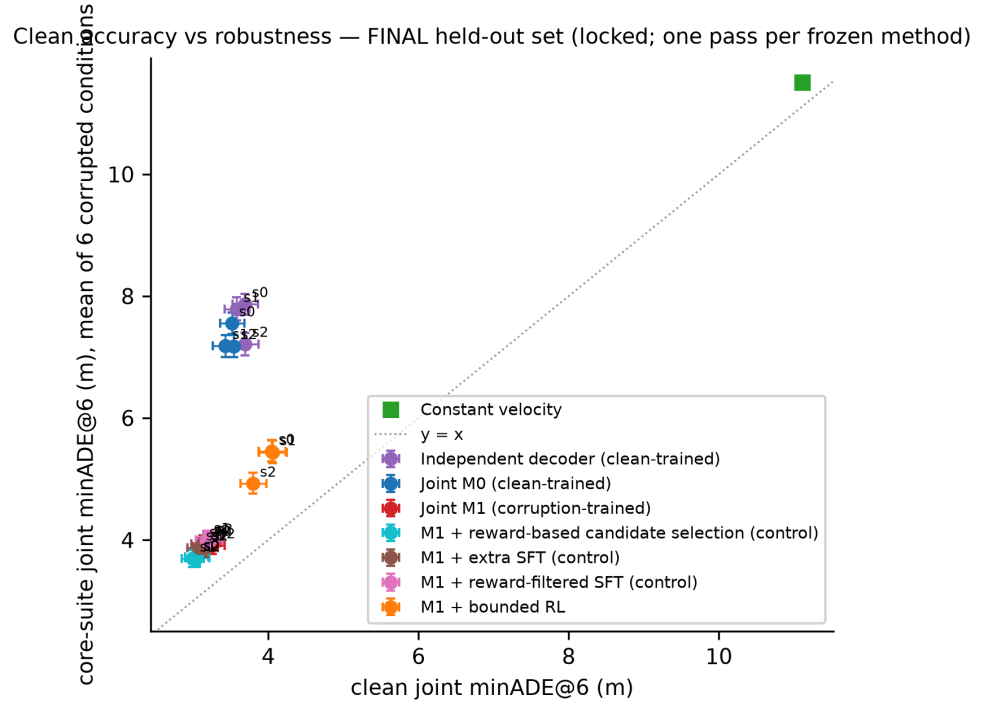
fig_baseline_comparison



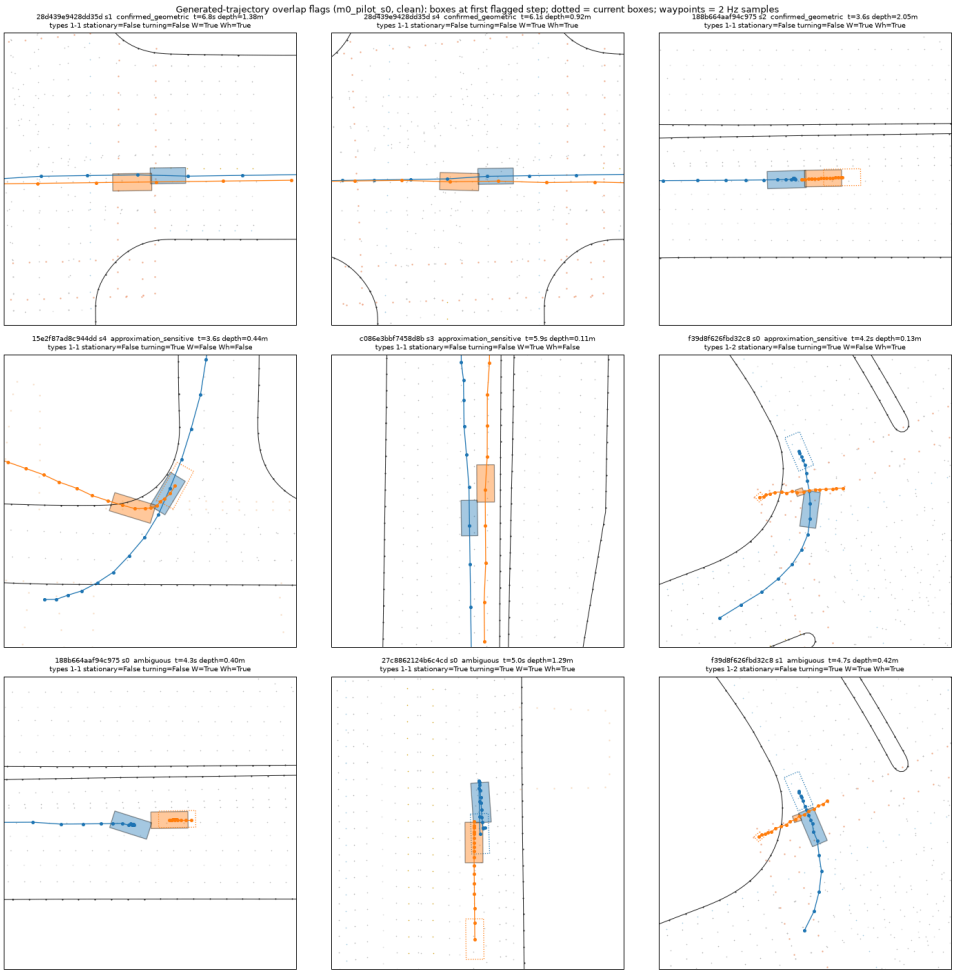
fig_error_vs_severity



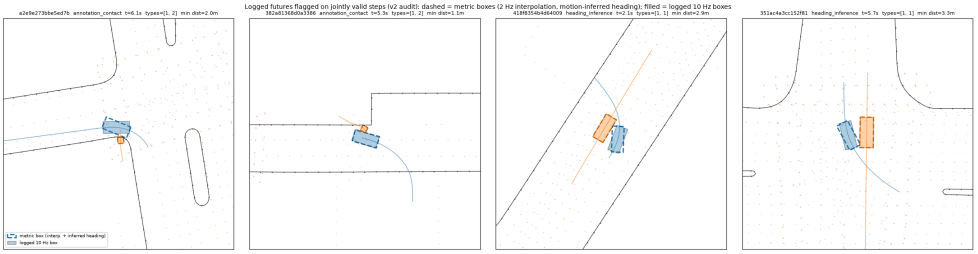
fig_rl_controls



fig_tradeoff



model_overlap_cases_m0_pilot_s0



overlap_floor_cases

7. Animations

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8. Limitations

Limitations (standing; updated as stages complete)

- **No real-data results yet.** Every number produced so far comes from SYNTHETIC fixtures used to verify the implementation; they are labelled as such and are not scientific results.
- **Simplified reproduction.** Compact model (1.8 M params vs MotionLM's 9 M), far fewer training steps, a subset of WOMD shards, no NMS/k-means mode aggregation, six raw samples for local metrics. Not an official challenge submission; the official metrics package could not be installed here (DECISIONS D-019a), so the custom metrics are audited on analytic cases and oracles only.
- **Benchmark-provided targets.** The two targets are the dataset's `objects_of_interest` pair; this is evaluation metadata, not an onboard target-selection algorithm.
- **Open-loop context.** Context agents are not simulated; only the two targets interact (2 Hz joint autoregression). Interpolating 2 Hz outputs to 10 Hz for geometric checks does not make the model a reactive 10 Hz policy.
- **Synthetic corruption.** Missing frames and Gaussian position noise are stress scenarios, not calibrated Waymo sensor-error distributions; they say nothing about raw camera/LiDAR perception.
- **Offroad metric.** Road-edge polylines with a sign convention calibrated by the log-playback oracle; vehicles only; centre-point test (no corner test); invalid maps excluded and counted.
- **Inference unit.** Scenario-level paired bootstrap; WOMD does not expose the originating run segment, so independence between windows is assumed, not verified.
- **Seeds.** Single-seed evidence unless the pilot budget supports three seeds (planned: 3).
- **RL scope.** REINFORCE with LOO baseline/KL/replay over two agents and a 16-step episode; rewards are deterministic generated-trajectory diagnostics. Differs from the all-agent closed-loop setting of the Waymo RL paper.
- **Source-segment overlap.** Identified only through SDC motion n-gram matching (D-025); train scenes sharing a segment with dev were removed, final shares none, but same-segment windows remain inside each split, so the scenario-level bootstrap slightly overstates independence.
- **Overlap-metric reference rate.** On jointly valid intervals the 2 Hz-waypoint OBB test flags 1.0 % of logged joint futures on dev (D-028a: 3 annotation-level vehicle–pedestrian contacts, 2 heading-inference artefacts). The full-horizon rate on logged futures is undefined (invalid labels carry no geometry); model `overlap_rate` (full horizon) and `overlap_rate_jointvalid` (identical intervals as the logged reference) are both reported with their denominators.
- **Mixture training as regulariser.** The pilot over-fits after ≈ 30 epochs (D-027); corruption mixture training also augments the data, so part of any M1 gain may be regularisation rather than corruption awareness. The matched extra-SFT control and clean guardrail bound, but do not remove, this confound.
- **Overlap reference rate is not a lower bound (D-028a).** It is the same approximate test applied to logs; it mixes annotation contacts and heading-inference artefacts, and vehicle–pedestrian pairs dominate it. The RL overlap reward is therefore an approximate geometric-consistency objective, not a verified safety measure.
- **Bootstrap unit.** CIs resample heuristic source-segment groups (D-025a); the grouping is derived from SDC motion matching and cannot guarantee independence.
- **RL outcome (D-031).** Bounded REINFORCE post-training lowered box overlap but worsened clean accuracy by 18–27 % and diversity, with ≈ 8× more excessive stopping; by the frozen rule the RL effect does not count, and the inference-time candidate-selection control dominates it. Reward curves during training are not held-out improvements.

- **Timings.** Seconds-per-scene figures were measured on a GPU shared with unrelated jobs; they are descriptive and do not rank methods by speed. Sampling counts per scene are the cost driver.